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A FLORISTIC STUDY OF SWANTON PACIFIC RANCH, SANTA CRUZ COUNTY, CALIFORNIA

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ABSTRACT

The Swanton Pacific Ranch (SPR) is located north of the town of Davenport, in Santa Cruz County, California, and is owned by California Polytechnic State University, San Luis Obispo. SPR is approximately 1295 ha and contains areas of salt marsh, coastal prairie, coastal bluff scrub, northern coastal scrub, riparian forest, redwood forest, mixed evergreen forest, and chaparral. The property extends over an elevational gradient from 0 m to 420 m. In this study, we conducted a floristic inventory at SPR and documented 636 taxa. Between 2017 and 2019 we vouchered 547 taxa with 974 specimens. We were unable to locate 63 taxa that had been previously reported, 27 represented from historic collections and 36 represented by personal observations, but lacking vouchers. We relocated, but did not collect, 26 taxa known from previous collections. Overall, we documented 93 families at SPR, with Asteraceae, Poaceae, and Fabaceae having the most species, respectively. California rare taxa, listed by the California Native Plant Society, made up 2.5% of the flora (n = 18). Locally rare taxa made up 10% of the flora (n = 65). Non-native taxa made up 26% of the flora (n = 169). We vouchered one new taxon for Santa Cruz County, *Senecio aphanactis* Greene, which has a California Rare Plant Rank of 2B.2; documented a new occurrence of *Hesperocyparis abramsiana* (C.B.Wolf) Bartel var. *abramsiana*, a Federally Threatened conifer; and collected a form of *Sanicula crassicaulis* Poepp. Ex DC that we have published as a distinct variety. Overall, this relatively small coastal ranch contained approximately 10% of the flora of California. The 2020 CZU Lightning Complex Fire burned ca. 90% of SPR within a year of this study being completed. This opens the possibility for this study to be used as a baseline for future studies examining post-fire shifts in plant diversity at a landscape scale.

Key Words: checklist, Santa Cruz, Scott Creek Watershed.

Physical specimens are the gold standard of plant biodiversity research. They create a lasting record that can be reinterpreted in the light of new taxonomy and often can be sampled for genetic material (Staats et al. 2013; Hart et al. 2016). The rate of collecting new plant specimens has fallen drastically in the United States at a time when specimens are becoming more valuable than ever (Prather et al. 2004). The advent of specimen digitization and imaging has exponentially increased the utility and accessibility of specimens, and widespread access to GPS technology makes location data much more accurate (Soberón and Peterson 2004; Page et al. 2015). These data are increasingly used to model current species distributions and predict range shifts under climate change, but more collections and newer collections are needed to ensure model accuracy (Feeley and Silman 2011). The botanical community needs to continue to produce current, high-quality physical specimens to maintain this important resource.

In addition to contributing to a wider pool of knowledge, local floras based on physical specimens have high potential to inform conservation decisions. Many floristic projects take place in underexplored, often remote, areas. These are quite valuable, but this trend has the potential to overlook working lands like ranches and managed forests as important areas for documenting biodiversity. Local floras may also serve as snapshots of plant biodiversity at a certain time. This can be particularly valuable in the case of extreme disturbance events such as flood or wildfire. In 2020, the CZU Lightning Complex Fire burned almost all of the area on which this study took place. This sets the stage for future studies to use the data we recorded as a baseline for comparing plant diversity post fire.

In this study, we explored the vascular plant diversity of the Swanton Pacific Ranch (SPR) in

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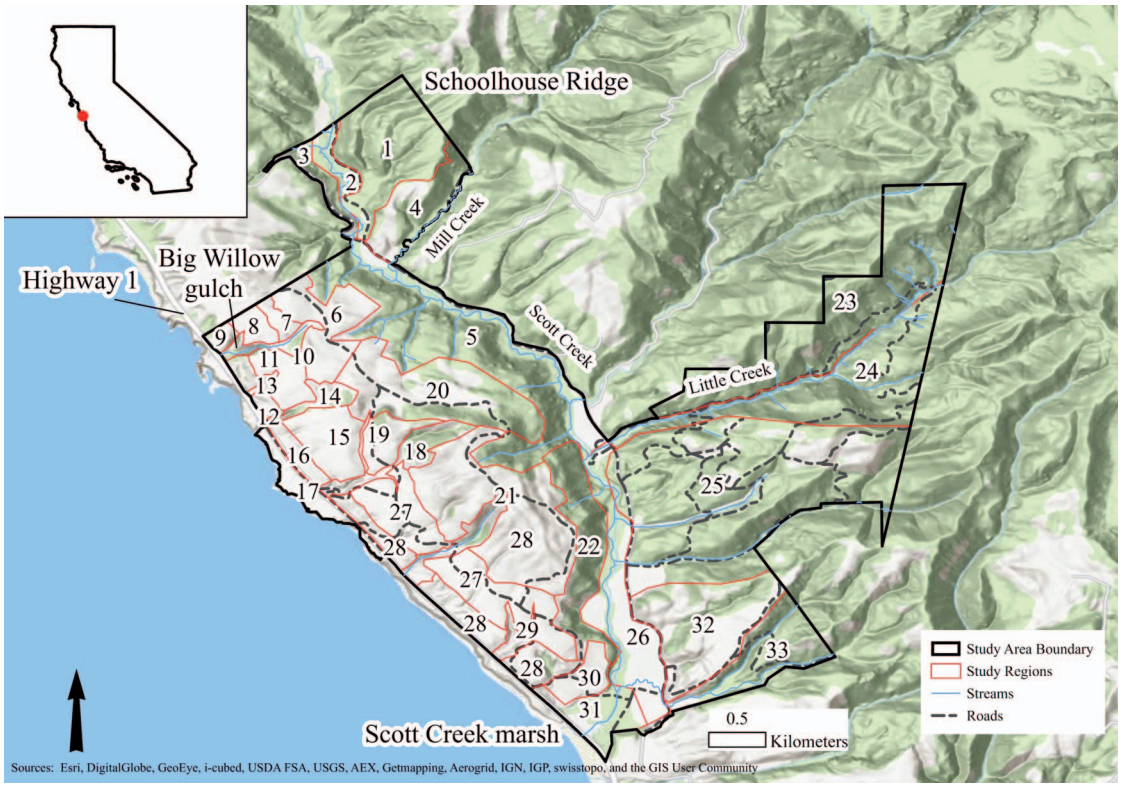

FIG. 1. Swanton Pacific Ranch, Santa Cruz County, CA. Pictured are the study boundary in black, notable features, streams, roads, and study regions (red polygons numbered 1–33). The inset shows the approximate location of SPR in California. A shapefile of the study regions and boundary can be downloaded at https://doi.org/10.25338/B8JP70.

Santa Cruz County, California. Located in the coastal Scott Creek Watershed in the Santa Cruz Mountains, SPR is a working ranch with active cattle grazing and timber operations. We wanted to both document the extraordinary species richness of this area as well as facilitate informed conservation decisions at SPR. To this end, our primary objectives were to: 1) develop a vouchered checklist of vascular plant taxa at SPR, 2) document California rare and locally rare taxa, 3) describe the dominant plant communities across SPR, and 4) collect information on invasive species.

DESCRIPTION OF STUDY AREA

Circumscription

SPR is located in northern Santa Cruz County, just north of the town of Davenport (population ca. 400). SPR's holdings total 1295 ha and include areas of salt marsh, coastal bluff scrub, northern coastal scrub, coastal prairie, riparian forest, redwood forest, mixed evergreen forest, and chaparral in approximate ascending order of elevation. The property extends over an elevational gradient from 0 m to 420 m, is bounded by Highway 1 and the Pacific Ocean to the west, and extends 4.5 km inland. Our floristic study area included all 1295 ha of SPR, including several small inholdings, and a 10 m buffer on either side of Swanton Road where it runs adjacent to SPR (Fig. 1).

Notable Features

Scott Creek flows from north to south for ca. 7 km through SPR. The majority of the Scott Creek riparian corridor on SPR has a steep cliff face to the west and a gently rising slope to the east. The largest sub-watershed within SPR is that of Little Creek, which flows primarily southwest and joins Scott Creek in the middle of SPR (Fig. 1). In contrast to Scott Creek, both sides of the Little Creek riparian corridor are extremely steep. This is predominantly where redwood forest is found at SPR.

The Scott Creek marsh is located where Scott Creek meets the Pacific Ocean (Figs. 1, 2), and for part of the year a sandbar closes it off from the ocean, forming a lagoon. The lagoon is separated into three sections. The northern section is only seasonally inundated. It contains the historic stream channel and has characteristic salt marsh vegetation. The central section consists of a man-made channel with berms on either side and is perennially filled with brackish water. The southern section consists of

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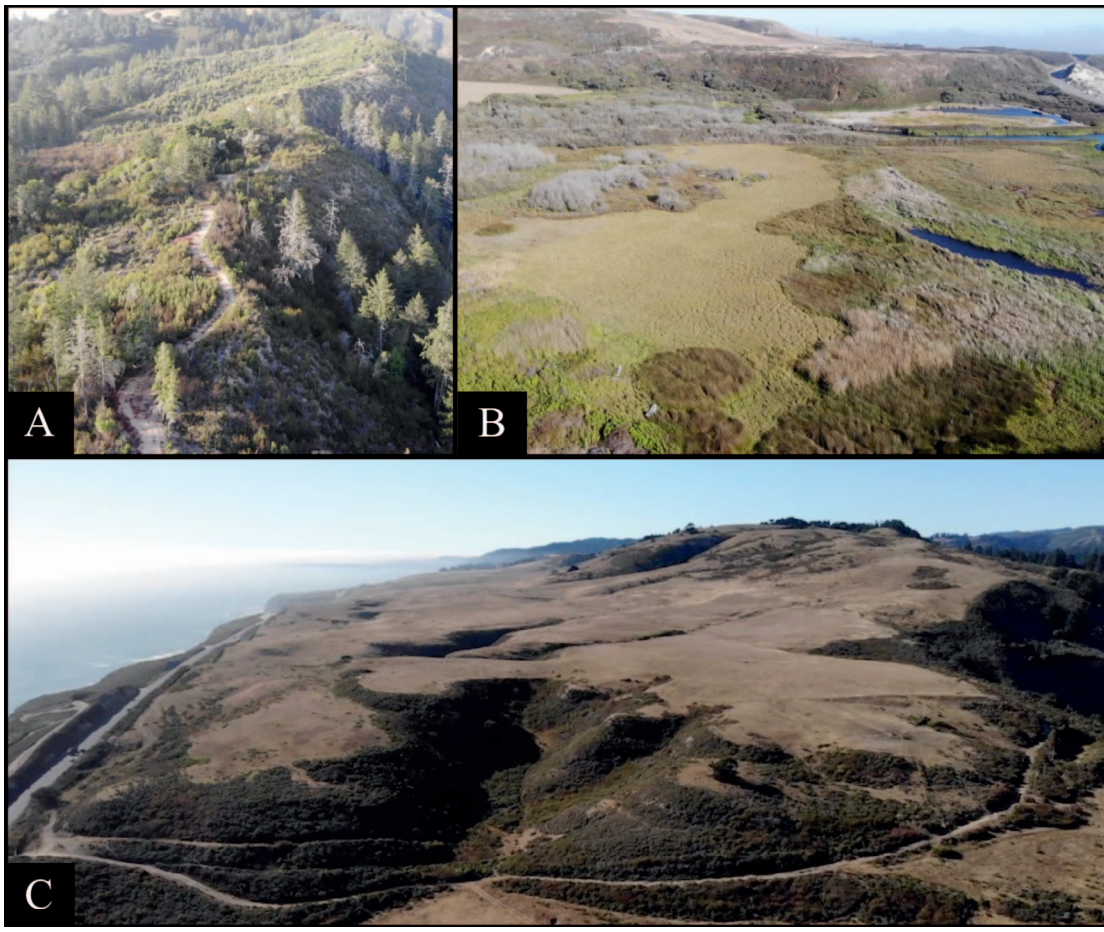

FIG. 2. A. Top of the Schoolhouse Ridge at Swanton Pacific Ranch, Santa Cruz County, showing the white "chalk" soil and hard chaparral community. (CalPhotos ID: 0000 0000 0521 0162). B. Vegetation of the Scott Creek marsh at Swanton Pacific Ranch, Santa Cruz County. View looking south. (CalPhotos ID: 0000 0000 0521 0160). C. The Western Terrace at Swanton Pacific Ranch, Santa Cruz County. The first and second coastal terraces can be seen as well as some of the distinctive gullies. (CalPhotos ID: 0000 0000 0521 0161).

a predominantly freshwater pond that was excavated to provide soil for the berms.

To the north of the Scott Creek marsh is the Western Terrace (Fig. 2C), a large section of rolling coastal prairie that gently slopes towards the ocean. The Western Terrace has deep gullies throughout, each of which acts as a small watershed and drains east to west into the ocean. The largest gully, which supports the greatest variety of plant communities, is called Big Willow gulch (Fig. 1, study region 19), so named for the stand of large willows found at the head of the gulch. Big Willow gulch consists of a marshy area at the head of the gulch, northern coastal scrub on the sides, a steep canyon with riparian vegetation at the bottom, and a relatively flat mouth where it drains under Highway 1. Other notable features are the perched marshes found at the upper rim of the gulch.

The final notable feature of SPR is the Schoolhouse Ridge and adjacent ridges in the northernmost section (Figs. 1, 2A). This complex of more or less parallel ridges is bounded on the east by Mill Creek, and contains the steep, mostly inaccessible, south-facing section of the Mill Creek Watershed. The Schoolhouse Ridge and adjacent ridge complex contain large areas of hard chaparral and interior coastal prairie, as well as unusual vertical grasslands. Many locally rare taxa were found in the hard chaparral, interior coastal prairie and vertical grasslands of the Schoolhouse Ridge complex.

Climate

Swanton Pacific Ranch has a Mediterranean climate. Rainfall is concentrated between December and March, with almost no rain from June–August (Fig. 3). Summer fog is a common occurrence and is likely an important summer water source for some plants at SPR. Summer fog is common on the Western Terrace and diminishes towards the eastern

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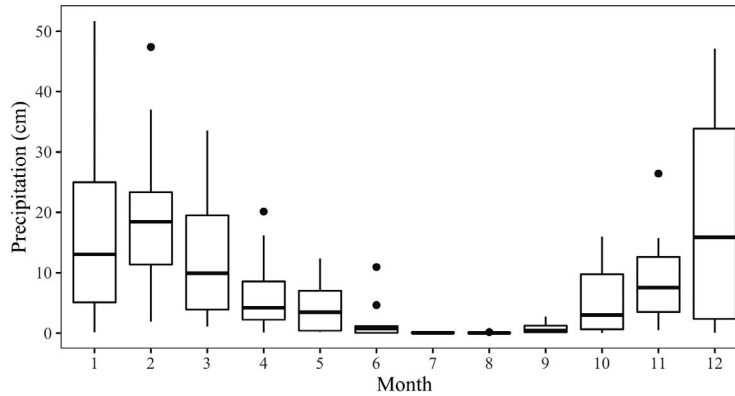

FIG. 3. Average rainfall per month from 2000–2015 at Swanton Pacific Ranch, Santa Cruz County. Measurements are reported from rain gauges in the study area.

ridgetop areas. Between 2000 and 2014, the average annual rainfall recorded was 84 cm (but varied from 23.6–121.4 cm), and February was the wettest month with 19.3 cm of rain on average (data from a tip-bucket rainfall gauge located in study region 25 at ca. 300 m in elevation). The average air temperatures are generally similar (within 2°C) across SPR, with the coastal regions generally being slightly warmer and experiencing less variation in temperature and the inland regions being colder on average with larger fluctuations in temperature. The mean annual air temperature for SPR is 13–14°C, and the annual variation around the mean is 12–13°C. December and January are the coldest months of year with mean temperatures of 8–10°C. August is the hottest month of the year with mean temperatures of 15–17°C. Air temperature data is reported from the CHELSA global climate model and represents the years 1979–2013 (Karger et al. 2017).

Geology

The parent material for the vast majority of SPR is siliceous mudstone, with only the upper half of the Little Creek Watershed underlain by quartz diorite (Cochrane et al. 2015). Soils are variable throughout SPR. Sand and sandy loam is present on the coastal bluffs and extends inland to the edge of the Western Terrace, where it is associated with much greater rates of erosion (Fig. 1, lower portions of study regions 18 and 19). Loam soil is found on the Western Terrace and in interior coastal prairie and forested areas. Weathered siliceous mudstone, often referred to as "chalk," forms an extremely poor soil on ridgetops (Fig. 2A). Distinct from the weathered mudstone on ridgetops are areas of exposed or nearly exposed mudstone only thinly covered by topsoil. These openings harbor unique herbaceous communities in coastal prairie and mixed evergreen forest.

Coastal terraces are a key feature of SPR and are generally located within 1.5 km of the ocean. These unique, step-like formations occur from SPR south to Santa Cruz and are thought to have formed during five separate periods of high sea level when wave action cut a smooth plane into the bedrock, resulting in five terrace "steps." At SPR, there are clear signs of the youngest two terraces, found closest to the ocean. The lower terrace forms the coastal bluffs and is ca. 70–80,000 yr old; the higher terrace, which forms the Western Terrace (Fig. 2C), is ca. 100,000 yr old (White et al. 2008). Gullies cut across the Western Terrace and were presumably formed by the erosive activity of waterfalls running over cliffs formed by the coastal terraces (Bennett 1999). This erosive activity is referred to as a headcut. Over time, the headcuts eroded the terrace until there was no longer a vertical drop. This process created the steep gullies that abruptly flatten out as they reach the downslope edge of the terrace formation. A distinct feature of the Western Terrace is the fold in the bedrock that divides it into two watersheds. The northeastern portion is tilted east and hosts some of the most intact coastal prairie on SPR, including many locally rare taxa. The rest of the Western Terrace is west-facing and drains via the many gullies into the ocean.

SPR has high topographic complexity, characterized by many steep slopes of varying aspects. The two major creeks, Scott Creek and Little Creek, flow roughly perpendicular to each other, meaning SPR has slopes that face all cardinal directions. In addition, the many active gullies present on the Western Terrace create small-scale topographic complexity. High topographic complexity may be a contributing factor to the high species richness at SPR (Yasuhiro et al. 2004).

Land-use History and Ownership

The Swanton area has likely been influenced by human use for thousands of years. Through an analysis of trace plant remains at Año Nuevo State Park, 10 miles north of SPR, Evett and Cuthrell (2013) found strong evidence for widespread burning of vegetation by the indigenous Cotoni people—a practice that created many areas of grassland, which have now undergone succession to northern coastal

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scrub or mixed evergreen forest. The evidence suggests that regular burning had been taking place for hundreds to more than one thousand years, and would only have ceased once the Mission Padres solidified their control over the area in the late 1700s to early 1800s. This archaeological evidence is reinforced by historical accounts recorded by the 1769 Portola expedition that indicate grassland as the dominant vegetation in the region (Cuthrell 2013). In addition to burning, indigenous people commonly harvested the bulbs or corms of local geophytes in this region (Anderson and Rowney 1999) and may have coppiced shrubs such as *Corylus cornuta* Marshall subsp. *californica* (A. DC.) A.E.Murray to increase fruit production and tended beds of *Carex barbarae* Dewey for basketry (Anderson 2005).

When the Mission Padres arrived in 1792, they forbade burning and introduced cattle to the landscape. In 1839, current day SPR and surroundings were formally included in a Mexican land grant to form the Rancho Agua Puerca y Las Trancas (this translates roughly as stagnant pig water and sand bars ranch). The ranch was used primarily for grazing livestock until 1867, when the land grant was sold to James Archibald (Scaramozzino 2015). From 1867 to 1950, the land was used for timber, agriculture, and to generate hydroelectric power. The area was extensively logged—all old-growth redwood as well as the entire mixed evergreen forest was clearcut and tanbark oaks were harvested for their bark. A portion of the coastal prairie was tilled and used for row crops, such as Brussels sprouts and artichokes. Flax was grown during the same period, the feral descendants of which are now a common non-native member of the grassland community. Dams and miles of flume were built to run a hydroelectric power plant that provided electricity to Santa Cruz. The plant operated from 1899 until 1948, when the flume was burnt in a forest fire.

Archibald's holdings were split up over time, but much of the Swanton Valley came to be owned by the settling families. For example, the Gianone family was originally hired by Archibald to manage his dairy operations. Co-author of this study, James A. West is a direct descendant of these settlers and has lived in Swanton Valley since 1946. From ca. 1940 to 1980, the Gianones and other families sold parcels in the Swanton region to Al Smith, who largely kept the area free of development. Smith experimented with different tenants and farming techniques until 1991, when he gifted much of his ranch to California Polytechnic State University, San Luis Obispo, to be used for experiential learning.

Disturbance

Fire has had a large impact on the flora of SPR. Historically, it is likely that anthropogenic fire was common and maintained large areas of grassland (Cuthrell 2013). Recently, fire suppression has kept wildfire to a minimum, while simultaneously rendering the landscape prone to large, catastrophic fires. SPR has experienced two recent wildfires. In 2009, the Lockheed Fire burned at low to high severity across 460 ha of SPR (ca. 1/3 of the ranch), primarily through mixed evergreen forest. In 2020, the CZU Lightning Complex Fire burned almost the entirety of SPR. Post the Lockheed Fire, many areas that were mixed evergreen forest with an open understory pre-burning were converted into a dense soft chaparral, composed primarily of *Ceanothus thyrsiflorus* Eschsch. These all burned again in the CZU Lightning Complex Fire and it will be interesting to observe what regenerates.

Recent anthropogenic disturbances include agriculture, cattle grazing, and timber harvests. Agriculture currently occurs on ca. 30 ha of valley bottom and includes row crops and a small apple orchard. Cattle grazing occurs on ca. 650 ha, primarily on the coastal prairie and on a few small pastures in the southwest corner of SPR. While the impact of cattle grazing on the native flora at SPR is unknown, studies elsewhere have shown both positive and negative effects of cattle grazing on coastal prairie (Dremann and Shaw 2002; Hayes and Holl 2003; Skaer et al. 2013). Rangeland management at SPR includes the removal of encroaching shrubs such as *Baccharis pilularis* = DC. subsp. *consanguinea* (DC.) C.B.Wolf and *Toxicodendron diversilobum* (Torr. & A.Gray) Greene. This may increase local species diversity, as these shrubs are aggressive invaders of coastal prairie and do not support a diverse understory (Zavaleta and Kettley 2006).

Timber harvests are conducted infrequently and selectively at SPR, primarily in the Little Creek Watershed (Fig. 1, study regions 23–25). There have been four harvests in the last 30 yr, one of which was a post-fire salvage operation in 2010. Each harvest was guided by a timber harvest plan that excluded botanical conservation areas from logging (Dietterick et al. 2020). The forested areas of SPR have many, infrequently used logging roads where several infestations of *Genista monspessulana* (L.) L.A.S.-Johnson and non-native *Euphorbia* spp. L. were observed. The installation of skid trails and logging roads is prone to spreading invasive species due to the large degree of soil disturbance involved.

Previous Descriptions of the Flora

The flora of Santa Cruz County was first cataloged by Dr. Charles Lewis Anderson in 1890 and assembled by Ferdinand Clarke (1890) into a pamphlet listing 628 flowering plants, 17 ferns, and 75 grasses. It wasn't until 1961 that an updated flora was produced for Santa Cruz County and the surrounding region—John Hunter Thomas's *Flora of the Santa Cruz Mountains of California* (Thomas 1961). Swanton is listed as a locality in this work, and Thomas made a series of collections on and around SPR, totaling ca. 30 specimens representing 24 taxa. To date, this flora remains the most

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comprehensive account of the phytogeography in Santa Cruz County, as it includes descriptions of the local geology and common plant communities. In 2005, Randall Morgan et al. authored the first edition of *An annotated checklist of the vascular plants of Santa Cruz County, California*. The checklist established 13 floristic regions in the county. The majority of SPR is included in the Swanton floristic region, while the coastal slope that forms the Western Terrace is included in the North Coast floristic region (Morgan et al. 2005). In 2013, Dylan Neubauer published the second updated edition, which included 1594 vascular plant taxa, notes on differentiating closely related taxa, and local distributional and rarity data (Neubauer 2013). In 2016, James A. West authored an essay entitled "Traversing Swanton Road," which documented plant communities and described taxonomic and ecological oddities along Swanton Road (West 2016). An edited summary of this work that focuses on research topics entitled *Research topics for the Scott Creek Watershed and environs* was published online in 2021 (West 2021). "Traversing Swanton Road" and the questions it proposed formed the basis for the present study.

Methods

Our primary goals were to: 1) develop a vouchered checklist of vascular plant taxa at SPR, 2) document California rare and locally rare taxa, 3) describe the dominant plant communities across SPR, and 4) collect information on invasive species. We first conducted a thorough literature and herbarium search for all taxa previously reported within SPR. Literature examined included the following: *Flora of the Santa Cruz Mountains of California, Annotated checklist of the vascular plants of Santa Cruz County, California,* 2nd ed., and *Traversing Swanton Road* (Thomas 1961; Neubauer 2013; West 2016). Seed collections corresponding to the observations documented in *Traversing Swanton Road* and deposited at the UCSC Arboretum seedbank were also used as additional source of record data. The Consortium of California Herbaria (CCH2) was used to identify potential SPR specimens (CCH2 2020). Our CCH2 search included all specimen records georeferenced as occurring inside a 50m buffer around the current SPR property boundary, or with "Swanton Pacific Ranch" included in the locality description. We then filtered out specimens with site descriptions that were clearly outside the SPR property boundary. Many of the specimens housed at the Hoover (OBI) and UC Berkeley/Jepson herbaria (UC/JEPS) were then reviewed in person by R. Kenny.

Field work occurred in 2017 and 2019, with the goal of inventorying species and vouchering specimens across all major plant communities at SPR. To accomplish this, we defined 33 study regions based on geology, topography, and prior knowledge of plant community boundaries (see Fig. 1). Each study region was surveyed at least once, and a species list was generated and vouchers were collected. In total, 63 survey days occurred (July 2–September 21, 2017; March 2–July 26, 2019). All 2017 collection locations were recorded using the My GPS Coordinates app (My GPS Coordinates version 2.12, TappiApps) running on an iPhone 6. Specific location data was lost for many specimens collected in 2017. For these specimens, locality information was reduced to "Swanton Pacific Ranch." In most cases, the general region of SPR where the specimen was collected was known and coordinates were assigned with locational uncertainties that reflected this. Where these vouchers are the only representatives of taxa included in this study, bioregions were assigned if possible. Collection locations in 2019 were recorded using ArcGIS Collector (version 20.1.0, Esri, Redlands, CA) running on an iPhone 8. All specimens collected in 2017 or 2019 were deposited in the Hoover Herbarium (OBI), with any duplicates sent to the Kenneth S. Norris Center for Natural History, University of California Santa Cruz (UCSC). We visited most all accessible portions of SPR. Areas that we did not visit were those with heavy cover of poison oak, impenetrable chaparral, or slopes that were not walkable.

Taxonomy in this study follows The *Jepson eFlora* (as of June 26, 2020) with the exception of one taxon recognized by CNPS as rare (*Silene verecunda* S.Watson subsp. *verecunda*), but not currently recognized in The *Jepson eFlora* (2020). California rare taxa were designated as any taxa having a California Rare Plant Rank (CRPR) assigned by CNPS. CRPR 1B indicates taxa that are rare, threatened, or endangered in California and elsewhere. CRPR 2B indicates taxa that are rare, threatened, or endangered in California, but not elsewhere. CRPR 3 indicates taxa about which more information is needed. CRPR 4 indicates taxa with limited distribution in California. Each rank also includes a threat level. A threat level of 0.1indicates seriously threatened in California (over 80% of occurrences threatened / high degree and immediacy of threat). A threat level of 0.2 indicates moderately threatened in California (20–80% occurrences threatened / moderate degree and immediacy of threat). A value 0.3 indicates not very threatened in California (less than 20% of occurrences threatened / low degree and immediacy of threat or no current threats known). Taxa were determined as locally rare if they had five or fewer extant populations in Santa Cruz County (see Neubauer 2013, p. 10 for a full definition). Occurrence data for all California rare plants was submitted to the California Natural Diversity Database (CNDDB) (California Dept. of Fish and Wildlife 1981). Non-native status was assigned based on The *Jepson eFlora*. A tabulated form of the complete checklist produced is available to download at https://doi.org/10.25338/B8JP70.

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TABLE 1. NUMERIC SUMMARY OF THE FLORA OF SWANTON PACIFIC RANCH, SANTA CRUZ COUNTY.

	Ferns	Gymnosperms	Magnoliids	Eudicots	Monocots	Total
Families	7	2	2	67	15	93
Genera	11	2	2	279	65	359
Non-native taxa	0	0	0	128	41	169
California rare or locally rare taxa	2	0	0	63	18	83
Total taxa	16	4	2	487	127	636

Scott Creek Marsh Vegetation Mapping

To document baseline vegetation patterns in the Scott Creek marsh and facilitate future restoration efforts, Cal Poly undergraduate students Thomas Buchloh, Sam Farrow, and Celeste Dorion, with the supervision of the authors, finely mapped the vegetation in the marsh using a combination of field surveys and drone imagery. Transects were established across the width of the marsh, spaced 5 m apart throughout the survey area. Using a Garmin handheld GPS unit, the coordinates of the start and end points of each transect were recorded in the field. 100 m transect tapes were run between the start and end points of each transect and species presence/absence in a 0.25 m quadrat was collected every 1 m along each transect. Species were recorded as present within the quadrat if any piece of the plant was within the quadrat. Water, bare soil, and dead vegetation were also included as alternatives if there were no species present in the quadrat.

To complement the transect data and to create a visual record of the vegetation of the Scott Creek marsh, aerial photographs were captured using a DJI Phantom 3 Professional drone (DJI Phantom 3 Pro, DJI Corporation, Shenzen, China). The altitude used achieves resolutions of 5–7 cm^2 per pixel. DroneDeploy software was used to create a flight map for the drone. The aerial imagery was used to correct any errors in the species presence-absence data due to inaccurate locations and as a tool to visually interpret where different stands of vegetation occur in the marsh. The combination of the species presence-absence data as well as aerial imagery allowed us to map both individual species distributions in the marsh as well as plant communities. We used ArcGIS Pro for all mapping and analysis.

RESULTS

Flora

In total, we documented 636 taxa at SPR, based off of 1317 collections included in the study, as well as 36 records based off of personal observation (Table 1, Appendix 1, https://doi.org/10.25338/B8JP70). The 1317 collections include all 974 collections made between 2017 and 2019, as well as 343 historic collections that we were able to verify as collected from SPR. During field surveys, we vouchered 547 taxa. We were unable to relocate 63 taxa, 27 represented from historic collections and 36 represented by personal observations, but lacking vouchers. We relocated, but did not collect, 26 taxa known from previous collections. Overall, we documented 93 families at SPR, with Asteraceae, Poaceae, and Fabaceae having the most species, respectively (Table 2). California rare taxa, listed by CNPS, made up ca. 2.5% of the flora (n = 18) (Table 3). Locally rare taxa, following designation in Neubauer (2013), made up 10% of the flora (n = 65). Non-native taxa made up 26% of the flora (n =169).

Rare Taxa

Noteworthy collections. We vouchered one species new to Santa Cruz County. We collected *Senecio*

TABLE 2. NUMERICAL SUMMARY OF THE FLORA OF THE SWANTON PACIFIC RANCH, SANTA CRUZ COUNTY. The top 10 most well-represented families and genera are listed as well as the most common life forms.

	Number of taxa	Percent of total
Family		
Asteraceae	89	14.2%
Poaceae	70	11.1%
Fabaceae	53	8.3%
Apiaceae	24	3.7%
Brassicaceae	23	3.7%
Rosaceae	21	3.3%
Boraginaceae	21	3.2%
Polygonaceae	18	2.9%
Cyperaceae	17	2.7%
Lamiaceae	16	2.5%
Genus		
Trifolium	22	3.5%
Festuca	11	1.8%
Carex	10	1.6%
Juncus	10	1.6%
Rumex	10	1.6%
Erythranthe	8	1.3%
Acmispon	8	1.1%
Lupinus	7	1.1%
Ranunculus	7	1.1%
Vicia	7	1.1%
Life form		
Perennial	331	51.8%
Annual	206	32.6%
Shrub	47	7.5%
Biennial	16	2.5%
Geophyte	14	2.2%
Evergreen tree	12	1.9%
Deciduous tree	4	0.6%
Subshrub	3	0.5%
Caning shrub	2	0.3%

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TABLE 3. CALIFORNIA RARE TAXA PRESENT CURRENTLY OR HISTORICALLY AT SWANTON PACIFIC RANCH, SANTA CRUZ COUNTY. California Rare Plant Rank (CRPR) 1B indicates taxa that are rare, threatened, or endangered in California and elsewhere. CRPR 2B indicates rare, threatened, or endangered in California but not elsewhere. CRPR 3 indicates taxa about which more information is needed. CRPR 4 indicates taxa with limited distribution in California. Each rank also includes a threat level. A 0.1indicates seriously threatened in California (over 80% of occurrences threatened / high degree and immediacy of threat). A 0.2 indicates moderately threatened in California (20–80% occurrences threatened / moderate degree and immediacy of threat). A 0.3 indicates not very threatened in California (less than 20% of occurrences threatened / low degree and immediacy of threat or no current threats known).

Scientific name	CRPR
Agrostis blasdalei	1B.2
Amsinckia lunaris	1B.2
Arctostaphylos glutinosa	1B.2
Collinsia multicolor	1B.2
Elymus californicus	4.3
Hesperocyparis abramsiana var. *abramsiana*	1B.1
Hosackia gracilis	4.2
Micropus amphibolus	3.2
Microseris paludosa	1B.2
Perideridia gairdneri subsp. *gairdneri*	4.2
Pinus radiata	1B.1
Piperia michaelii	4.2
Plagiobothrys chorisianus var. *chorisianus*	1B.2
Sanicula hoffmannii	4.3
Senecio aphanactis	2B.2
Silene verecunda subsp. *verecunda*	1B.2
Stebbinsoseris decipiens	1B.2
Trifolium buckwestiorum	1B.1

aphanactis Greene, which has a CRPR of 2B.2, in two locations. One specimen was growing in the hard chaparral in the upper reaches of the Schoolhouse Ridge. The other was growing in the hard chaparral on the northern ridge of the Little Creek Watershed. The sites are ecologically similar, sharing a degraded mudstone substrate and dominated by *Arctostaphylos crustacea* Eastw. subsp. *crinita* (J.E.Adams) V.T.Parker, M.C.Vasey & J.E.Keeley, and *Adenostoma fasciculatum* Hook. & Arn. var. *fasciculatum*. The closest record formerly was in Santa Clara County, south of San Jose, in the eastern foothills of the Santa Cruz Mountains. Our collection firmly establishes the presence of the species on the western slope of the Santa Cruz Mountains.

County-endemic taxa. One individual of *Hesperocyparis abramsiana* (C.B.Wolf) Bartel var. *abramsiana* was located at SPR. This Federally Threatened and State Endangered taxon is endemic to Santa Cruz County. The individual at SPR is outside of the four main populations in the county, which occur on nutrient-poor, sandstone- or granitic-derived soils in fire-prone areas. The closest population is ca. 6.5 km distant. The SPR tree was growing on a steep, east-facing slope with loamy soil in mixed evergreen forest. This area burned in the 2020 CZU Lightning Complex Fire. The typical response in this genus is that adult trees are killed by fire, but their serotinous cones open and generally seeds sprout vigorously (Adams and Bartel 2009). Subsequent surveys of this area should be conducted to determine the persistence of this taxon at SPR.

Arctostaphylos glutinosa B.Schreib. was collected in 1983 by Roy Buck and Jim West from property that later became part of SPR. This species is endemic to Santa Cruz County and has a CRPR of 1B.2. It occurs predominantly on the ridgetops northeast of SPR in exposed, extremely weathered siliceous mudstone. The collection locality for the 1983 specimen was revisited and searched thoroughly, but no individuals were encountered. The area burned in the 2009 Lockheed Fire and *A. glutinosa,* an obligate-seeder, does not re-sprout after fire. It may be that we missed a seedling in our search, or it is possible that the manzanita was outcompeted by re-sprouting shrubs. This area burned again in the 2020 CZU Lightning Complex Fire.

California rare and locally rare taxa. Ten taxa occurring at SPR have a CRPR of 1B, which indicates that they are rare, threatened, or endangered in California and elsewhere. One of these is *Arctostaphylos glutinosa,* which was addressed above. Here we present habitat information and preliminary population estimates for the other eight taxa. A list of all taxa with a CRPR can be found in Table 3.

Agrostis blasdalei Hitchc. (CRPR 1B.2) is a perennial grass that occurs at SPR in sandy, disturbed areas on coastal bluffs and in wet areas in coastal prairie. There are six occurrences, (which are defined here as plants occurring 0.4 km apart or more), known from collections at SPR. The largest occurs on the coastal bluffs and numbers in the hundreds (Fig. 1, study region 17). The smaller occurrences are found inland on the Western Terrace and vary between 10 and 100 plants. Occurrences are threatened most by the encroachment of non-native species, including *Carduus pycnocephalus* L. subsp. *pycnocephalus* and *Holcus lanatus* L.

Amsinckia lunaris J.F.Macbr. (CRPR 1B.2) is an annual herb with only one occurrence at SPR on a steep, exposed, north-facing slope in mostly intact coastal prairie. Though cows can access the location, the terrain is so steep that they do not appear to graze there. The occurrence consists of ca. 150 individuals.

Collinsia multicolor Lindl. & Paxton (CRPR 1B.2) is an annual herb that has three occurrences at SPR. All of these are roadcuts where a vertical surface of the mudstone parent material has been exposed. The occurrences vary from ca. 10–50 individuals. Road maintenance is the most direct threat to these occurrences. If one of the occurrences were scraped with a tractor, the soil seedbank could be eliminated. Another occurrence is on a quad trail where this is not a threat. All of these occurrences burned in the

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2020 CZU Lightning Complex Fire, but some have reappeared as of 2021.

Microseris paludosa (Greene) J.T.Howell (CRPR 1B.2), a perennial herb, was last observed by James A. West at SPR in the 1970s in a seasonally wet drainage in coastal prairie on the Western Terrace. It was collected from the Arroyo de las Trancas in 2016, which is 1.5 km north of the Western Terrace [see *West s.n.* (UCSC)]. It has also been collected from several other locations in the Scott Creek Watershed including, in 1983, the coastal prairie just north of the northern boundary of SPR [see *Buck & West 269* (JEPS82401)]. However, no individuals were encountered during this study. The habitat may have been impacted by cattle grazing.

Pinus radiata D.Don (CRPR 1B.1) is an evergreen tree that occurs immediately to the north of SPR and has been vouchered from SPR [see *Keil 20657* (OBI 161809)]. This taxon hybridizes with *Pinus attenuata* at SPR and in the surrounding regions. No individuals were observed during this study that did not appear intermediate to some degree between *P. radiata* and *P. attenuata*. The majority of the trees with this intermediate phenotype are found in scattered stands on the Western Terrace.

Plagiobothrys chorisianus (Cham.) I.M.Johnst. var. *chorisianus* (CRPR 1B.2) is an annual herb that has been observed in a small, marshy area between Scott Creek and the northern border of SPR. No individuals were encountered during this study, however the taxon was collected in 2019 within 0.5 km of the study area [see *West s.n.* (OBI 158496)].

Silene verecunda S.Watson subsp. *verecunda* (CRPR 1B.2) is a perennial herb that occurs on siliceous mudstone outcrops on coastal bluffs and in grassland. It is known from three occurrences at SPR, which consist of 2–30 individuals each. There are no readily apparent threats, but the impacts of cattle grazing on this taxon are unknown. The individuals at SPR represent the current known southernmost distribution of this taxon.

Stebbinsoseris decipiens (K.L.Chambers) K.L.Chambers (CRPR 1B.2) is an annual herb known from three occurrences at SPR. Two smaller occurrences are located on thin soils in coastal prairie on the Western Terrace, and the largest is located in interior coastal prairie on the lower reaches of the Schoolhouse Ridge. The two smaller occurrences consist of ca. 150–200 individuals each, and the interior occurrence consists of ca. 500 individuals. There are likely more occurrences of this annual plant at SPR than we documented and documentation of this taxon at SPR should be ongoing. Cattle grazing is currently the largest threat. On the Western Terrace, plants tend to occur in areas that cattle avoid due to poor forage or steep slopes. However, in the currently ungrazed interior coastal prairie, plants are found on less extreme slopes. If grazing were introduced to this area, it seems likely that population numbers would decline. The interior occurrences burned in the 2020 CZU Lightning Complex Fire, and it is unknown whether they have reappeared.

Trifolium buckwestiorum Isely (CRPR 1B.1) is an annual herb that is known from SPR only from the type collection, which was made in 1982 on the lower Schoolhouse Ridge [see *Buck et al. 1* (JEPS 81528)]. No individuals were encountered during this study. However, the species was collected in 2019 within 2 km of the study area [see *West s.n* (OBI)]. It is possible that the habitat was impacted by the 2009 Lockheed Fire. This taxon had previously been found at SPR and in the surrounding areas in mixed evergreen forest with an open understory. Much of this habitat at SPR burned and subsequently experienced heavy *Ceanothus* regeneration, which closed off the understory. Much larger areas of mixed evergreen forest burned in the 2020 CZU Lightning Complex. If these postfire areas experience similar levels of *Ceanothus* regeneration, as seen after the Lockheed Fire, the habitat for this species could be severely impacted.

Seven additional taxa at SPR are ranked as less rare, having a CRPR higher than 1B. All California rare taxa are listed in Table 3 and are denoted in the appendix with an $^{\mathbf{RR}}$. There are 65 taxa vouchered from SPR that do not have a California Rare Plant Rank, but are listed as locally rare in Neubauer (2013). These are denoted in the appendix with an $^{\mathbf{R}}$.

Plant Communities

Below we present a summary of the plant communities at SPR present at the time this study was conducted. The 2020 CZU Lightning Complex Fire burned the majority of SPR, which will likely result in changed vegetation patterns. We have generally followed Holland (1986). These classifications are clearly outdated, but we choose to present them here for two main reasons: 1) No CNPS relevés or rapid assessments were collected during this project, thus our ability to accurately apply the current system set forth in the *Manual of California Vegetation* (MCV) is limited 2) Holland (1986) names are descriptive and likely to be familiar to many readers. In addition to this, many are crosswalked with MCV alliances on the MCV website. Common names are included in this section only, as it may be of interest to a wider audience. We have included the MCV alliances that are potentially represented, but caution that the MCV alliances should be treated as preliminary (California Native Plant Society 2020). The exception to this is the Scott Creek marsh, where the data generated by Buchloh, Doroin, and Farrow was used to identify and map the relevant MCV alliances and associations; see discussion in "Scott Creek Marsh Communities" below (Fig. 4).

Coastal Bluff Scrub. Occurring on the relatively small portion of SPR adjacent to the ocean (Fig. 1, study region 17), this community is floristically distinct from the northern coastal scrub described below. *Erigeron glaucus* Ker Gawl. (Seaside Daisy),

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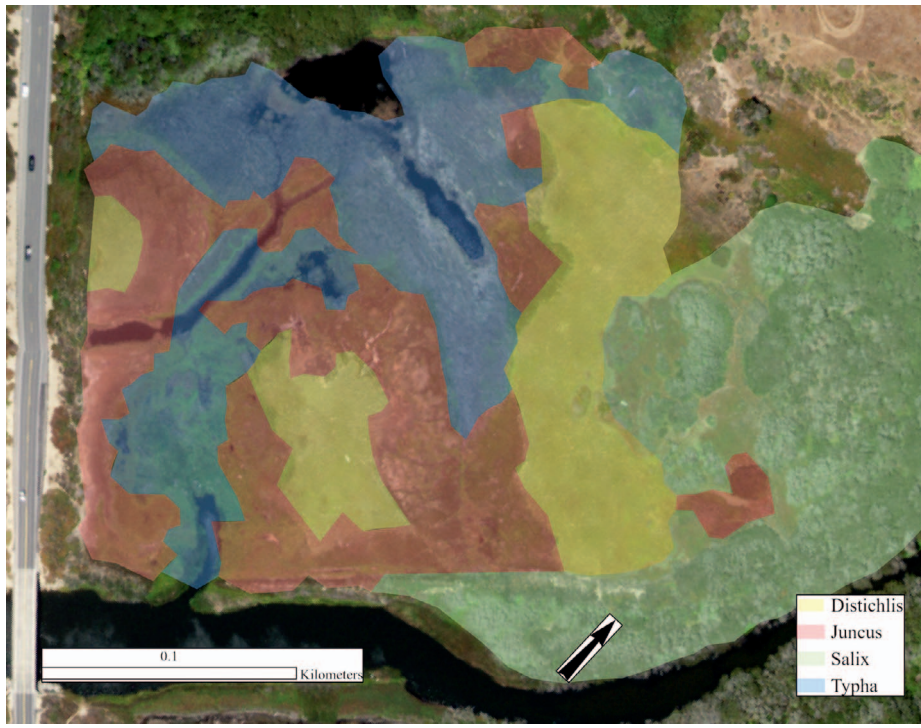

FIG. 4. CNPS MCV vegetation alliances and associations in the Scott Creek marsh at Swanton Pacific Ranch, Santa Cruz County. Distichlis = *Distichlis spicata* alliance; *Distichlis spicata–Sarcocornia pacifica* association. Juncus = *Juncus lescurii* alliance. Salix = *Salix lasiolepis* alliance; provisional *Salix lasiolepis–Persicaria punctata* association. Typha = *Typha* (*angustifolia, domingensis, latifolia*) alliance; *Typha* (*latifolia, angustifolia*) association.

Eriophyllum staechadifolium Lag. (Seaside Woolly Sunflower), and *Eriogonum latifolium* Sm. (Seaside Wild Buckwheat) are co-dominant with *Artemisia californica* (Coastal Sagebrush) to a much greater extent than is found anywhere inland, and are shorter in stature and spaced farther apart. A robust understory is present, which commonly includes *Daucus pusillus* Michx. (Rattlesnake Carrot), *Cryptantha clevelandii* Greene var. *florosa* I.M.Johnst. (Coastal Cryptantha), and *Cryptantha micromeres* (A.Gray) Greene (Minute-flowered Cryptantha). In areas of high erosion and low shrub cover in this community, *Dudleya caespitosa* (Haw.) Britton & Rose (Sea Lettuce), *Fragaria chiloensis* (L.) Mill. (Beach Strawberry), California rare (CRPR 1B.2) *Agrostis blasdalei* (Blasdale's Bent Grass), and *Bromus sitchensis* Trin. var. *maritimus* (Piper) Otting & R.E. Brainerd (Maritime Brome). The MCV alliance that likely describes this community is the *Eriophyllum staechadifolium–Erigeron glaucus–Eriogonum latifolium* Herbaceous Alliance.

Northern Coastal Scrub. Northern coastal scrub occurs primarily on the coastal terraces (Fig. 1, study regions 10–21, 26–31). It is the dominant community on steep slopes that form the sides of coastal gulches and also occurs in large patches interdigitated with non-native grassland on the Western Terrace. Dominant shrubs include *Artemisia californica* Less. (California Sagebrush), *Baccharis pilularis* subsp. *consanguinea* (Coyote Brush) and, *Toxicodendron diversilobum* (Poison Oak), which can form dense, impenetrable stands with a depauperate understory. Noteworthy shrubs include the locally rare *Berberis pinnata* Lag. subsp. *pinnata* (California Barberry) and *Amelanchier utahensis* Koehne (Utah Serviceberry). In areas with lower shrub density the forb understory can be quite diverse and includes noteworthy species such as the locally rare *Angelica tomentosa* S.Watson (Woolly Angelica), *Zeltnera davyi* (Jeps.) G.Mans. (Davy's Centaury), and the California rare *Piperia michaelii* (Greene) Rydb. (Coast Piperia). The MCV alliance that likely describes this community is the *Baccharis pilularis* Shrubland Alliance.

Coastal Prairie. This community is largely restricted to the Western Terrace and the Schoolhouse Ridge (Fig. 1, study regions 1, 4, 6–21, 27–30). It occurs primarily on predominantly north and west-facing slopes within 1.5 km of the ocean in areas that are vernally or perennially moist, or on slopes that are too steep for cattle to easily access. Three grasses are dominant or co-dominant in this community: *Danthonia californica* Bol. (California Oat Grass), *Deschampsia cespitosa* (L.) P.Beauv. (Tufted Hair Grass), and *Hordeum brachyantherum* Nevski subsp. *brachyantherum* (Northern Barley). Noteworthy

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forbs occurring in this community include the California Rare *Amsinckia lunaris* (Bent-flowered Fiddleneck) and *Hosackia gracilis* Benth. (Harlequin Lotus), and the locally rare decorum Fisch. & C.A.Mey (Coast Larkspur). Several smaller and floristically distinct stands of coastal prairie occur on the lower reaches of the Schoolhouse Ridge (Fig. 1, study regions 1, 4), which we refer to as "interior coastal prairie." These areas are drier and are distinguished floristically by a greater abundance of *Danthonia californica, Elymus glaucus* Buckley subsp. *glaucus* (Blue Wild Rye), and *Stipa pulchra* Hitchc. (Purple Needlegrass), and a relative lack of *Deschampsia cespitosa*. The forb species also differ, with *Lomatium caruifolium* (Hook. & Arn.) J.M.Coult. & Rose var. *caruifolium* (Caraway-leaved Lomatium), *Clarkia rubicunda* (Lindl.) F.H.Lewis & M.E.Lewis (Farewell-to-spring), and *Brodiaea terrestris* Kellogg subsp. *terrestris* (Dwarf Brodiaea) being much more common than in the more coastal locations. The MCV alliance that likely describes this community is the *Danthonia californica–Deschampsia cespitosa–Hordeum brachyantherum* Herbaceous Alliance.

Freshwater Marsh. A community somewhat intermediate between Freshwater Seep and Freshwater Marsh can be found at the head of many of the coastal gulches, where subsurface water emerges before flowing down through the gulch (Fig. 1, study regions 10, 14, 18, 21). This community is generally dominated by *Scirpus microcarpus* J.Presl & C.Presl (Small-fruited Bulrush), *Juncus effusus* L. subsp. *pacificus* (Fernald & Wiegand) Piper & Beattie (Pacific Rush), *Juncus hesperius* (Coast Rush), and *Juncus patens* (Spreading Rush), with some having an overstory of *Ribes divaricatum* Douglas var. *pubiflorum* Koehne (Straggle Bush) or *Salix lasiolepis* Benth. (Arroyo Willow). During the 2019 field season, there was evidence of cattle grazing, with prominent hoof punch and grazed tips on the *Juncus*. Despite the grazing, or perhaps because of it, this community tends to be dominated by native species. Noteworthy forbs include *Oenanthe sarmentosa* Presl ex DC. (Pacific Oenanthe), *Hypericum anagalloides* Cham. & Schltdl. (Tinker's Penny), the locally rare *Stachys chamissonis* Benth. (Swamp Stachys), and *Epilobium ciliatum* Raf. (Common Willow Herb). The MCV alliances that likely describe this community are the *Scirpus microcarpus* Herbaceous Alliance and the *Juncus effusus* Herbaceous Alliance.

Freshwater Seep. Numerous freshwater seeps occur in coastal prairie and within large expanses of non-native annual grassland (Fig. 1, study regions 6–21, 26–30). These areas range in size from ca. 100–200 m^2. Dominant taxa include *Carex densa* (L.H.Bailey) L.H.Bailey (Dense Sedge), *Carex harfordii* Mack. (Harford's Sedge), *Carex subbracteata* Mack (Small-bracted Sedge), various putative hybrids of *Carex, Juncus balticus* Willd. subsp. *ater* (Rydb.) Snogerup (Baltic Rush), *Juncus hesperius* (Piper) Lint (Coast Rush), *Juncus patens* E.Mey. (Spreading Rush), and *Juncus phaeocephalus* Engelm var. *phaeocephalus* (Brown-headed Rush). A common forb is *Sisyrinchium bellum* S.Watson (Western Blue-eyed-grass). Non-native species commonly occurring in these areas include *Trifolium repens* L. (White Clover) and *Holcus lanatus* L. (Velvet Grass). The MCV alliances that likely describe this community are the *Carex densa* Provisional Herbaceous Alliance, *Juncus arcticus* (var. *balticus*, var. *mexicanus*) Herbaceous Alliance and the *Juncus* (*oxymeris, xiphioides*) Provisional Herbaceous Alliance.

Wildflower Field. Several areas of nearly exposed mudstone occur primarily in the northern third of the Western Terrace (Fig. 1, study regions 7–13). These areas are characterized by thin, dry soil with few non-native grasses and large displays of native forbs. Though small in extent, this community is floristically distinct from coastal prairie and is dominated by *Lasthenia gracilis* (DC.) Greene (Common Goldfields), *Microseris bigelovii* (A.Gray) Sch. Bip. (Bigelow's Microseris), *Clarkia rubicunda* (Farewell-to-spring), and *Plantago erecta* E.Morris (California Plantain). In addition, this community hosts a large diversity of native *Trifolium* spp. (clovers), as well as the rare *Stebbinsoseris decipiens* (Santa Cruz Microseris). The MCV alliance that likely describes this community is the *Lasthenia californica–Plantago erecta–Vulpia microstachys* Herbaceous Alliance.

Non-native Grassland. Large portions of the coastal terraces support non-native annual grassland dominated by *Avena barbata* Pott ex Link (Slender Wild Oats), *Bromus hordeaceus* L. (Soft Chess), and *Hordeum murinum* L. (Wall Barley) (Fig. 1, study regions 15–21, 26–28). This community occurs primarily on gentle slopes with deep, loamy to sandy soil and it provides much of the forage for SPR's cattle. From the northern to southern edge of the Western Terrace, aeolian sand deposits increase, as does the cover of non-native forbs and grasses. Other common taxa in this community include *Carduus pycnocephalus* subsp. *pycnocephalus* (Italian Thistle), *Silybum marinum* (L.) Gaertn. (Milk Thistle), *Trifolium angustifolium* L. (Narrow-leaved Clover), and *Trifolium subterraneum* L. (Subterranean Clover). The MCV alliance that likely describes this community is the *Avena* spp.–*Bromus* spp. Herbaceous Semi-Natural Alliance.

Central Coast Riparian Forest. This community occurs primarily within 100 m of Scott Creek (Fig. 1, study regions 5, 26). In the upper reaches of Scott Creek, co-dominant overstory trees include *Alnus* rubra Bong. (Red Alder), *Acer macrophyllum* Pursh (Big-leaf Maple), and *Acer negundo* L. (Box Elder). Common midstory trees and shrubs include *Salix* spp. (willows) (*S. lasiandra* Benth., *S. lasiolepis*, and *S. sitchensis* Sanson ex Bong.), *Sambucus nigra* L. subsp. *caerulea* (Raf.) R.Bolli (Blue Elderberry), and *Rubus spectabilis* Pursh (Salmonberry). In the lower reaches of Scott Creek, *Alnus rubra* becomes the sole

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dominant overstory tree with little to no midstory and a thick floodplain understory dominated by *Persicaria punctata* (Elliott) Small (Water Knotweed), *Carex obnupta* L.H.Bailey (Slough Sedge), and *Cyperus eragrostis* Lam. (Umbrella Sedge). The MCV alliances that likely describe this community are the *Alnus rubra* Forest Alliance and the *Acer macrophyllum* Forest & Woodland Alliance.

Mixed Evergreen Forest. This community occurs on most of the slopes in the interior half of SPR (Fig. 1, study regions 1–5, 22–25, 32, 33). *Pseudotsuga menziesii* (Mirb.) Franco subsp. *menziesii* (Douglas Fir) is co-dominant with *Arbutus menziesii* Pursh (Madrone), *Notholithocarpus densiflorus* (Hook. & Arn.) Manos, Cannon & S.H.Oh var. *densiflorus* (Tanoak), and *Quercus parvula* Greene var. *shrevei* (C.H.Mull.) Nixon (Shreve Oak). The midstory is generally open, with scattered *Toxicodendron diversilobum, Frangula californica* (Eschsch.) A.Gray subsp. *californica* (Coffeeberry), and *Heteromeles arbutifolia* (Lindl.) M. Roem (Toyon). The understory ranges from being absent to nearly 100% cover of ferns and forbs such as *Polypodium californicum* Kaulf. (California Polypody), *Osmorhiza berteroi* DC. (Mountain Sweet Cicely), and *Cynoglossum grande* Dougl. ex Lehm (Grand Hound's Tongue). Small stands of mixed evergreen forest also occur on north-facing slopes of the coastal gulches (Fig. 1, study regions 18, 19) and are unique in having *Ligusticum apiifolium* (Nutt.) A.Gray (Celery-leaved Licorice Root) as a component of the understory. Another occasional member of this community is *Torreya californica* Torr. (California Nutmeg), which is co-dominant to dominant in one area of east-facing slopes on the west side of Scott Creek. The MCV alliance that likely describes this community is the *Pseudotsuga menziesii–Notholithocarpus densiflorus* Forest and Woodland Alliance.

Redwood Forest. This community occurs primarily in the Little Creek Watershed, and secondarily in some stands along Scott Creek and in the steep valleys north of the Schoolhouse Ridge (Fig. 1, study regions 1, 5, 23, 24). The overstory is dominated by *Sequoia sempervirens* (Lamb. ex D.Don) Endl. (Coast Redwood), and the midstory is generally sparse with a composition similar to the midstory of mixed evergreen forest. The understory ranges from sparse to dense, with dominant species including *Polystichum munitum* (Kaulf.) C.Presl (Western Sword Fern), *Oxalis oregana* Nutt. (Redwood Sorrel), and *Maianthemum* spp. [*M. racemosum* (L.) Link (False Solomon's Seal) and *M. stellatum* (L.) Link (Slim Solomon's Seal)]. The MCV alliance that likely describes this community is the *Sequoia sempervirens* Forest Alliance.

Northern Maritime Chaparral. This is a soft chaparral community composed almost entirely of *Ceanothus thyrsiflorus* var. *thyrsiflorus* (Blue Blossom), with a minimal understory. This community is almost entirely a result of the heavy recruitment of *C. thyrsiflorus* var. *thyrsiflorus* following the 2009 Lockheed Fire. These areas were previously mixed evergreen forest with a sparse midstory and diverse understory. The MCV alliance that likely describes this community is the *Ceanothus thyrsiflorus* Shrubland Alliance.

Central Maritime Chaparral. This hard chaparral community is primarily composed of the burl-forming *Arctostaphylos crustacea* subsp. *crinita* (Crinite Manzanita) and *Adenostoma fasciculatum* var. *fasciculatum* (Chamise). *Eriodictyon californicum* (Hook. & Arn.) Torr. (California Yerba Santa), and *Dendromecon rigida* Benth. (Bush Poppy) also are frequent. The understory is generally sparse and supports forbs such as *Psilocarphus tenellus* Nutt. (Slender Woolly-marbles), *Senecio aphanactis* Greene (Chaparral Ragwort), and *Acmispon junceus* (Benth.) Brouillet var. *junceus* (Rush Trefoil). The hard chaparral occurs primarily on ridgetops in poor soil with very low moisture retention (see Fig. 2A). There are also isolated patches on the steep, southeast-facing portions of study region 28 (Fig. 1). Portions of the hard chaparral also burned in the 2009 Lockheed Fire, but the shrubs resprouted strongly. The MCV alliance that likely describes this community is the *Arctostaphylos* (*crustacea, tomentosa*) Shrubland Alliance.

Scott Creek Marsh Communities

The Scott Creek marsh and seasonally bar-closed lagoon is located at the far south-western corner of SPR (Fig. 1). California Highway One bridges the outlet of the lagoon. This bridge is due to be replaced in the next decade, a project that will likely involve restoration of the salt marsh and lagoon. The fine-scale mapping of the Scott Creek marsh that was conducted as part of this study allowed us to identify specific MCV alliances and associations (Fig. 4).

Distichlis spicata *alliance* / Distichlis spicata–Sarcocornia pacifica *association*. This association occurs on the highest elevations in the marsh plain. It is dominated by *Distichlis spicata* (L.) Greene (Salt Grass), *Salicornia pacifica* Standl. (Pickleweed), and *Potentilla anserina* L. subsp. *pacifica* (Howell) Rousi (Pacific Silverweed). *Schoenoplectus californicus* (C.A.Mey.) Soják (Southern Bulrush), and *Juncus lescurii* Bol. (San Francisco Rush) are co-dominant in some areas.

Juncus lescurii *alliance*. The mid-elevation transitional regions of the marsh are largely dominated by *Juncus lescurii* with *Bolboschoenus fluviatilis* (Torr.) Soják (River Bulrush) and *Schoenoplectus californicus* co-dominant in some areas. It is possible that this transitional region should be divided up between the *Distichlis* and the *Typha* communities, but we find the *Juncus lescurii* community to be floristically distinct from the other marsh communities.

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Salix lasiolepis *alliance / provisional* Salix lasiolepis–Persicaria punctata *association*. A large area between the marsh plain proper and the upstream alder forest is dominated by *Salix lasiolepis* (Arroyo Willow), with an understory almost entirely dominated by *Persicaria punctata* (Water Knotweed). The soil is generally saturated and small ponds and pools are common. In drier areas, *Ribes divaricatum* var. *pubiflorum* (Straggle Bush) and *Urtica dioica* L. (Stinging Nettle) form a dense midstory.

Typha *(*angustifolia, domingensis, latifolia*) alliance* / Typha *(*latifolia, angustifolia*) association*. This association occurs in the lowest-elevation portions of the marsh, generally adjacent to channels and ponds. *Typha angustifolia* L. (Narrow-leaved Cattail) dominates with *Typha latifolia* L. (Broad-leaved Cattail) occurring in a few small, pure stands.

Non-native and Invasive Taxa

Twenty-six percent of the taxa recorded in this study are non-native. Most non-native taxa are uncommon, but several are dominant in certain communities. In particular, large areas of the coastal and interior grasslands are dominated by European annual grasses, such as *Avena barbata*, *Bromus hordeaceus*, and *Hordeum murinum*. These non-native grasslands seemingly reached equilibrium long ago and are not a priority for management. However, there are several non-native taxa that should be managed, either because they are newly arrived and at low levels or because they appear to be spreading. These are listed below along with observations about their distribution and potential for management.

Silybum marianum *(L.) Gaertn*. This annual or biennial herb is present in high densities in localized areas of the Western Terrace. The most severe infestation is located at the northern edge of the Western Terrace where the soils are deep and loamy (Fig. 1, study region 8). This area tends to be more mesic than the southern portion of the Western Terrace and contains intact coastal prairie. Management is a priority for this species to prevent the degradation of more coastal prairie, as well as to improve the forage quality in the rangelands. Mechanical treatment, such as mowing, is cited as the most effective management. Unfortunately, grazing is cited as relatively ineffective (DiTomaso et al. 2013).

Carduus pycnocephalus *L. subsp*. pycnocephalus. This annual herb is present in high densities in the same northern edge of the Western Terrace as *Silybum marianum*. It poses many of the same risks, but is potentially more manageable. Mowing and grazing can both be effective at limiting the spread and could be implemented at SPR. Measures such as targeted goat grazing could be especially effective (DiTomaso et al. 2013).

Rubus ulmifolius *Schott var*. anoplothyrsus *Sudre*. This shrub is found primarily in one large patch at SPR, growing just off of Old Schoolhouse Road. Though it is not known to be a common invasive in California, its relative *Rubus armeniacus* Focke can be highly invasive in some areas. Given that all the individuals at SPR could be removed in one day of mechanical treatment, it would be prudent to do so in order to prevent further spread.

Holcus lanatus *L*. This perennial grass rapidly colonizes mesic grasslands. It is found in several locations on the Western Terrace and perhaps poses the most extreme threat to the biodiversity of that area. Mesic areas on the Western Terrace tend to harbor a high diversity of native species. If these communities become highly invaded by *Holcus lanatus,* much of the biodiversity in the coastal prairie, including rare plants, could be lost (Hektner and Foin 1977). This troublesome species requires intense levels of grazing to show any reduction, while low levels of grazing may even increase the spread. Additionally, it provides relatively poor forage and is known to invade both grazed and ungrazed grasslands (California Invasive Plant Council 2005). It is extremely important that managers recognize the threat to native biodiversity at SPR that this species poses. A long-term study of different treatment regimens on the Western Terrace could be beneficial to both local ranchers and conservationists.

Delairea odorata *Lem*. This species is present in high density in several patches in the Scott Creek riparian corridor, with the largest patch occurring on the west bank of Scott Creek just below the railroad bridge. On the Central Coast, this species is known to dominate riparian areas and successfully chokes out much of the understory (California Invasive Plant Council 2004a). The potential impacts on the riparian corridor should justify control of this species. If the riparian corridor became extremely invaded, it could pose a threat to the endangered salmonids and Red-Legged frogs that use this habitat (Robison and DiTomaso 2010, Feeley and Silman 2011). Mechanical treatment can be effective, but repeated treatment is necessary as the plant resprouts vigorously from roots and stem fragments. The limited extent of the infestation and the proximity to water make mechanical control the best option for this species. Additionally, biocontrol agents are being developed. A leaf- and stem-mining moth (*Digitivalva delaireae*) is close to being approved for release, and a galling fly (*Parafreutreta regalis*) is already approved (Balciunas et al. 2010; Reddy and Mehelis 2015).

Oxalis pes-caprae *L*. This perennial herb is invading coastal bluffs and exposed mudstone outcrops, both of which harbor many native species. Unfortunately, it is extremely difficult to manage, and there may not currently be a viable strategy for management at SPR. Mechanical control requires

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painstaking removal of all bulblets to prevent resprouting, and grazing is ineffective. Chemical control may be effective, but may not be available as an option (DiTomaso et al. 2013). All efforts should be made to prevent the spread of the bulblets, the primary mechanism for spread of this species.

Euphorbia oblongata *Griseb. and* E. lathyris *L.* These perennial herbs are present at low levels in disturbed areas at SPR. Large patches can be found on Old Schoolhouse Road near the turnoff to the Al Smith House as well as on the trails above the Staub House. As this species spreads, it will begin to have larger impacts on the native vegetation and become significantly more difficult to control. This species is difficult to eradicate given that mechanical removal stimulates the seedbank and can also spread root fragments that will generate new plants. These roots also provide resistance to herbicide. Limiting the spread of these plants at SPR should be the first priority, while an effective control strategy is devised.

Genista monspessulana *(L.) L.A.S.Johnson*. This shrub is present at SPR in mostly localized patches concentrated around the Staub House and on the quad trails above the Staub House, although scattered individuals can be found along access roads. This species should probably be the highest priority for management at SPR. It is known to form very dense stands, outcompete native vegetation, and can pose a major fire risk (Pauchard et al. 2008). The plants also create a long-lived seedbank, making eradication very difficult (Geerts et al. 2013). The longer the infestations go untreated, the longer it will take to eradicate the seedbank. Mechanical removal can be effective for this species and is likely the best option at SPR (Geerts et al. 2013). Removal paired with effective mapping and attempts to avoid soil disturbance in infested or formerly infested areas would likely be effective controls.

Ehrharta erecta *Lam*. This perennial grass is present at low levels at SPR in disturbed pasture and near road and trails through mixed evergreen forest. It is known to spread readily in response to disturbance and form homogenous stands where dominant (California Invasive Plant Council 2004b). Both mechanical removal and herbicide control have been shown to be effective in the Santa Cruz area, with varying effects on non-target species (Ray et al. 2018). Mechanical and chemical control paired with mapping and efforts to reduce spread would likely keep this species at low levels at SPR.

Discussion

About 40% of the 1594 taxa documented in Santa Cruz County occur at SPR, which represents about 1% of the area of the county. About 10% of the California flora can be found at SPR, in an area less than 0.01% of the state. In addition to the sheer species richness of the area, SPR also supports 18 California rare taxa, and 65 locally rare taxa. Despite making the strongest collection effort to date (Fig. 5), totaling about 1000 collections during the course of this study (Fig. 6), it is certain that there are species at SPR that still need to be documented. This is especially true of rare and somewhat ephemeral species like *Plagiobothrys chorisianus* var. *chorisianus*, also true of some common weeds such as Erodium botrys (Cav.) Bertol and Medicago polymorpha L. that were not vouchered despite the fact that they are almost certainly present at SPR. Additionally, SPR only represents a portion of the Scott Creek Watershed, which is estimated to support over 1000 taxa (J.A. West, unpublished data). The fact the CZU Lightning Complex Fire burned large areas in the watershed, and the majority of SPR, highlights the need to document the plant diversity in these areas. Without baseline knowledge of plant diversity in the Scott Creek Watershed, it will be difficult to determine the effect of the fire on the flora. Our hope is that this study will provide enough data to allow for future studies to examine the effect of fire on plant diversity in this coastal watershed.

The rich and unique flora of the Swanton area and the Scott Creek Watershed has long intrigued botanists and has resulted in much speculation about unusual specimens observed and collected in the area. The following sections will explore putative hybridization and taxonomic uncertainty at SPR.

Putative Hybridization

Several taxa documented at SPR during the course of this study appear morphologically intermediate and may represent hybridization events. Here we present the morphological evidence observed and the relevant background literature in hopes that future work will address this putative hybridization.

Lupinus arboreus × latifolius *putative hybrids*. Plants morphologically intermediate between *Lupinus arboreus* Sims and *Lupinus latifolius* J.Agardh occur primarily in the Little Creek Watershed. Notably less woody than *L. arboreus*, these are small shrubs that become somewhat woody at the base but retain flexible and hollow-stemmed branches. The flowers are consistently lilac in color. These plants may also be treated as *L. propinquus* Greene (not treated in The *Jepson eFlora*, but noted as in need of study), though SPR specimens differ from this taxon in having hairs on the upper surface of the leaflets (Sholars and Riggins 2020). Representative specimen: *Kenny 202* (OBI 171060).

Juncus hesperius × patens *putative hybrids*. Plants morphologically intermediate between *J. hesperius* and *J. patens* are somewhat common in wet or seasonally wet areas in coastal prairie. These can be easily distinguished in the field, as they lack the strongly blue-grey culms of *J. patens,* but the flowers have six stamens instead of three. Representative specimens: *Kenny 653* (OBI 173138); *Richardson 148*

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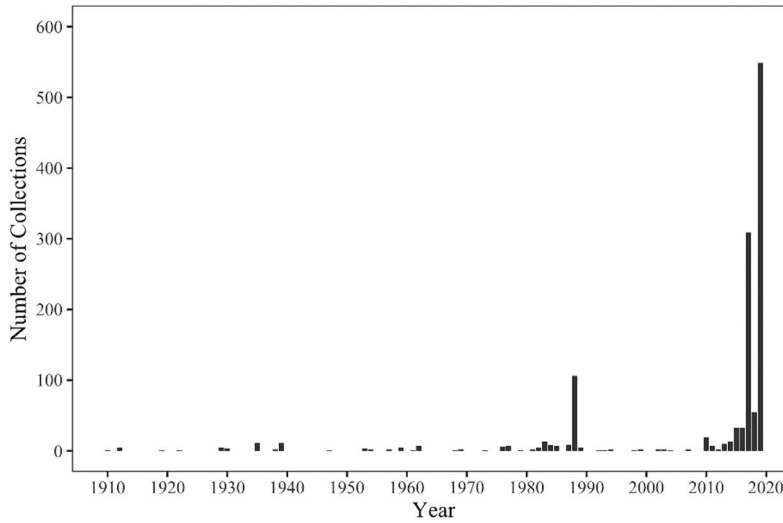

FIG. 5. Collections per year from 1910–2020 at Swanton Pacific Ranch, Santa Cruz County.

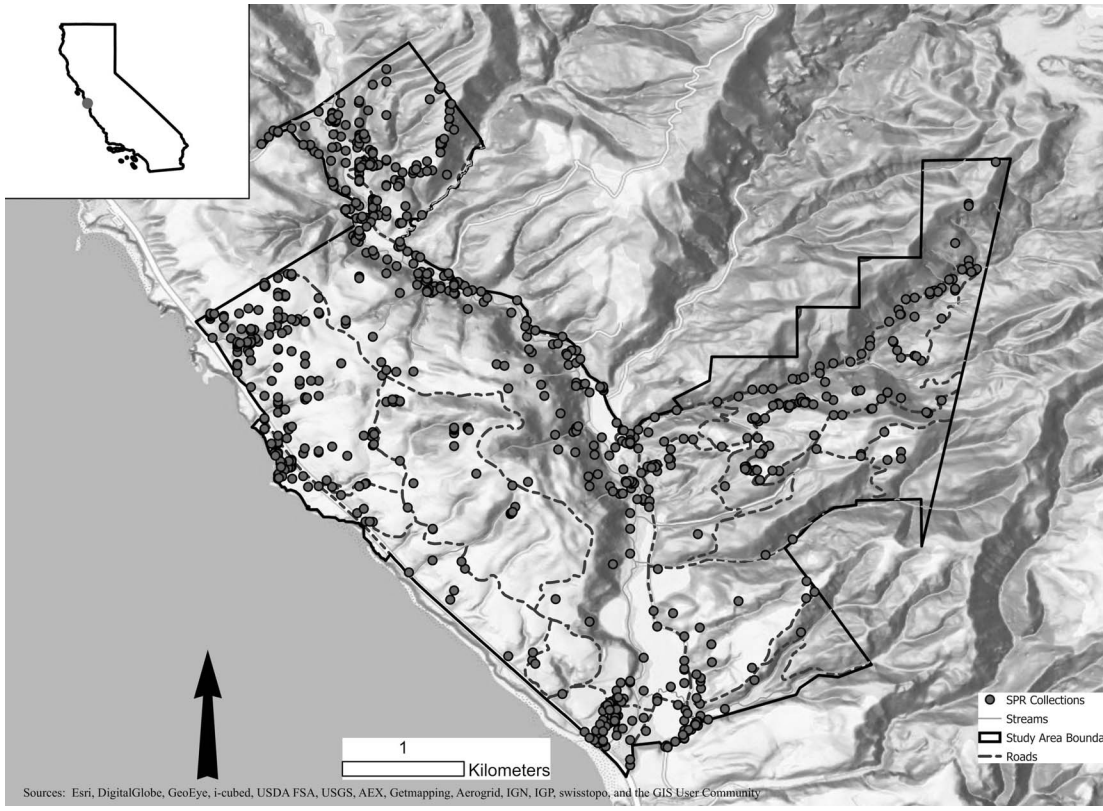

FIG. 6. Collection locations at Swanton Pacific Ranch, Santa Cruz County. The grey dots represent locations of all 1317 collections included in this study, including historic collections as well as collections made during the study. Thirty-six records based off of personal observations were not assigned specific locations and are not represented in the figure. The inset shows the approximate location of SPR in California.

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(OBI 161820); *Richardson 257* (OBI). This putative hybrid has also been documented by Peter Zika (2013).

Carex putative *hybrids*. Many unusual *Carex* individuals occur on the Western Terrace. The oddities primarily seem to be intermediate between *C. harfordii* and *C. subbracteata* or *C. densa* and *C. subbracteata*. Many individuals of these monoecious species appear to be sterile, not forming plump perigynia, although they produce anthers and pollen that may be viable. These plants can have variable inflorescence structures, but are united by their lack of fertile perigynia. Putative hybrids between *C. harfordii* and *C. subbracteata* fit this description and have been noted in the Jepson eFlora treatment (Zika et al. 2015). Other individuals have fertile perigynia, but are intermediate between *C. densa* and *C. subbracteata* in having pistillate flowers above staminate flowers, but also paniculate spikelets and minute teeth on the angles of their stems. Additionally, there are individuals that appear most similar to *C. harfordii*, but have paniculate spikelets. In all cases the paniculate spikelets arise from one or two basal spikelets becoming branched. This is problematic in that sect. *Ovales* Kunth., which contains *C. harfordii* and *C. subbracteata,* as it is currently circumscribed in the *Flora of North America*, specifies plants with racemose inflorescences (Mastrogiuseppe et al. 2002). This led the authors to hypothesize that there may have been a cross-sectional hybridization event between sects. *Ovales* and *Multiflorae* (J.Carey) Kuk., which introduced the paniculate spikelets. The authors' alternative hypothesis is that inflorescence structure is not as strongly regulated as is currently assumed and perhaps occasional branching of basal spikelets is not unusual. This is supported by individuals of another species also occurring at SPR, *C. bolanderi* Olney, which show branched basal spikelets outside of the accepted circumscription. Representative specimens: *Kenny 183* (OBI 161519); *Kenny 184* (OBI 161520); *Kenny 265* (OBI 161270); *Kenny 413* (OBI 174297); *Kenny 528* (OBI 166688); *Kenny 561* (OBI 148055); *Richardson 149* (OBI 144858).

Agrostis *hybrids*. Five species of *Agrostis* occur on SPR. Putative hybrids have been observed between *A. exarata* Trin. and *A. densiflora* Vasey, *A. blasdalei* and *A. densiflora,* and *A. exarata* and *A. blasdalei*. *Richardson 93* (OBI 171115) is an example of a putative *A. exarata* × *A. densiflora*. The *A. blasdalei* × *densiflora* putative hybrids are noted in The *Jepson eFlora* (Peterson and Harvey 2014). This complex needs further study. Many intermediates have also been observed between *A. hallii* Vasey and *A. pallens* Trin. These usually have ligules that are in the 4–7 mm range, which keys to *A. hallii,* but anthers that are in the 0.7–1.8 mm range, which keys to *A. pallens*. *Kenny 724* (OBI 167627) displays these characters.

Taxonomic Uncertainty

Several taxa at SPR hold taxonomic interest as unusual ecotypes or potentially even new species. Notes on these are included to encourage further botanical and evolutionary studies at SPR.

Clarkia davyi *(Jeps.) F.H.Lewis & M.E.Lewis*. There are two distinct forms of what can be keyed to *Clarkia davyi* at SPR. One is prostrate to decumbent and fits the current circumscription of the taxon as described in The *Jepson eFlora* (Lewis 2012). The other form is consistently erect and less robust than the prostrate to decumbent form. These forms could be an interesting illustration of incipient speciation. Representative specimens: *Kenny 518* (OBI 161318); *Kenny 654* (OBI 158552); *Kenny 660* (OBI 158561); *Kenny 666* (OBI 158586); *Kenny 671* (OBI 158547); *Kenny 699* (OBI 158570); *Richardson 110* (OBI 139913); *Richardson 370* (OBI 179982); *Richardson 399a* (OBI 167673).

Isolepis cernua *(Vahl) Roem. & Schult*. There are two forms of *Isolepis cernua* at SPR. One consists of very small-statured annuals, and the other consists of short-lived perennials that form dense clumps or mats. The latest monograph of the genus describes *I. cernua* as an annual to short-lived perennial, so the forms fit within the latest circumscription (Muasya and Simpson 2002). However, the maintenance of two distinct forms that co-occur is of interest. Representative specimens: *Kenny 530* (OBI 161327); *Kenny 559* (OBI 179973).

Sanicula crassicaulis *Poepp. ex DC. var.* nudicaulis *Kenny & West*. An unusual form of *Sanicula crassicaulis* has been collected and observed at SPR for decades. It is distinguishable from *S. crassicaulis* var. *crassicaulis* and var. *tripartita* H.Wolff by the relatively ungraduated nature of the prickles on the schizocarp and the much longer marginal leaf trichomes. It seems to have some ecological differentiation as well, being found much more frequently in riparian, moist, or shaded areas. This form has been collected and observed from northern Santa Barbara County to the San Francisco Bay Area and specimens collected in 1839 were described as *Sanicula nudicaulis* Hook. & Arn. The authors have published this form as variety *nudicaulis*. Representative specimens: *Kenny 187* (OBI 161522); *Kenny 860* (OBI158259). Unfortunately, the marginal leaf trichomes are often lost on herbarium specimens.

Conclusions

This study has highlighted the species richness present at Swanton Pacific Ranch. While we vouchered as many specimens as possible at SPR, there were many taxa that we could not relocate. There is still collecting work to be done, and we encourage more botanical research in the area. In addition, a systematic exploration and mapping of the plant

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communities at SPR using the MCV method would be valuable to inform conservation decisions at SPR and would further our understanding of rare coastal prairie alliances and associations. The Scott Creek Watershed as a whole still has many rare and unusual plant species to be documented, and we hope that this study will serve as a baseline for comparing plant diversity at SPR post the 2020 CZU Lightning Complex Fire as well as a starting point for a larger exploration of the watershed.

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Appendix 1

Annotated Checklist for Swanton Pacific Ranch

* = non-native taxon

[R] = locally rare taxon

[RR] = California rare taxon

Specimens without herbarium accession numbers have not yet been accessioned. These specimens should be locatable using the herbarium code, collector name and collector number.

Ferns and Allies

Athyriaceae

Athyrium filix-femina (L.) Roth var. ***cyclosorum*** Rupr. (Western Lady Fern) – Perennial herb. Common in riparian areas in mixed evergreen forest. *Kenny 639* (OBI 161358). Study Regions: 2, 5, 10, 14, 18, 21, 26, 28.

Blechnaceae

[R]***Struthiopteris spicant*** (L.) Weiss (Deer Fern) – Perennial herb. Uncommon in riparian areas, one individual occurring on wet cliff face adjacent to waterfall in redwood forest. *Kenny 898* (OBI 171055). Study Region: 3.

Woodwardia fimbriata Sm. (Giant Chain Fern) – Perennial herb. Occasional in riparian areas. A patch of this fern also occurs in the southern pond of the Scott Creek marsh growing out of the floating mass of *Typha latifolia* rhizomes. *Richardson 204* (OBI). Study Regions: 10, 13, 21, 23, 26, 31.

Dennstaedtiaceae

Pteridium aquilinum (L.) Kuhn. var. ***pubescens*** Underw. (Bracken Fern) – Perennial herb. Common in northern coastal scrub, coastal prairie, riparian forest, and mixed evergreen forest. *Richardson 45* (OBI 164398). Study Regions: 1, 3, 5, 10, 12, 18, 20, 21, 23–26, 28.

Dryopteridaceae

Dryopteris arguta (Kaulf.) Watt. (Coastal Wood Fern) – Perennial herb. Common in northern coastal scrub and mixed evergreen forest. *Richardson 40* (OBI). Study Regions: 1, 3, 5, 10, 12, 18, 20, 21, 23–26, 28.

Polystichum munitum (Kaulf.) C.Presl (Western Sword Fern) – Perennial herb. Common in mixed evergreen forest. *Richardson 6* (OBI 164279). Study Regions: 1–3, 5, 10, 18, 20, 23–25, 28, 31.

Equisetaceae

Equisetum arvense L. (Common Horsetail) – Perennial herb. Occasional in the Scott Creek riparian corridor. *Kenny 987* (OBI 179967). Study Regions: 2, 5.

Equisetum telmateia Ehrh. subsp. ***braunii*** (J.Milde) Hauke (Giant Horsetail) – Perennial herb. Common in riparian areas. *Richardson 272* (OBI 161822). Study Regions: 2, 5, 10, 13, 17, 18, 21, 23–25, 27, 28, 30.

Polypodiaceae

Polypodium californicum Kaulf. (California Polypody) – Perennial herb. Occasional in mixed evergreen forest. *Kenny 537* (OBI 161331). Study Regions: 1, 2, 5, 10, 13, 20, 23–25, 28.

Polypodium calirhiza S.A.Whitmore & A.R.Sm. (Nested Polypody) – Perennial herb. Occasional in mixed evergreen forest. *Kenny 985* (OBI 179962). Study Regions: 5, 25, 32.

[R]***Polypodium scouleri*** Hook. & Grev. (Leather-leaf Fern) – Perennial herb. Uncommon in mixed evergreen forest. Historically known from only one plant, not encountered during study. *Morgan 1978* (UCSC). Study Region: 2.

Pteridaceae

Adiantum aleuticum (Rupr.) C.A.Paris. (Five-finger Fern) – Perennial herb. Uncommon, occurring in riparian areas in redwood forest. *Richardson 72* (OBI 172597). Study Regions: 5, 23.

Adiantum jordanii Müll. Hal. (California Maidenhair) – Perennial herb. Occasional in riparian areas in mixed evergreen forest. *Richardson 74* (OBI). Study Regions: 1, 5, 10, 21, 24–26, 28, 32.

Pellaea andromedifolia (Kaulf.) Fée (Coffee Fern) – Perennial herb. Occasional on cliffs in mixed evergreen forest. *Kenny 394* (OBI 161467). Study Regions: 1, 10, 23.

Pellaea mucronata (DC. Eaton) DC. Eaton var. ***mucronata*** (Bird's Foot Fern) – Perennial herb. Uncommon in vertical openings in mixed evergreen forest. Historically present, though not encountered during study. *Buck & West 309* (JEPS 82789). Study Region: 4.

Pentagramma triangularis (Kaulf.) Yatsk., Windham & E.Wollenw. (Goldback Fern) – Perennial herb. Common in northern coastal scrub and mixed evergreen forest. *Richardson s.n.* (OBI 159633). Study Regions: 1, 5, 9, 13, 20, 23–25, 28, 32.

Gymnosperms

Cupressaceae

[RR]***Hesperocyparis abramsiana*** (C.B.Wolf) Bartel var. ***abramsiana*** (Santa Cruz Cypress) – Evergreen tree. One individual occurring on steep, east-facing slope above Scott Creek in area dominated by *Torreya californica*. Federally Threatened, State Endangered. *Kenny 196B* (OBI 161496). Study Region: 5.

Sequoia sempervirens (D.Don) Endl. (Coast Redwood) – Evergreen tree. Common in pure stands, as well as intermixed with other conifers and broadleaf trees from near sea level to ridgetops, where it takes on a stunted and yellow growth form. *Kenny 502* (OBI 161307). Study Regions: 2, 3, 5, 23–26.

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Pinaceae

Pinus attenuata Lemmon (Knobcone Pine) – Evergreen tree. Occasional in small stands on coastal headlands. Many individuals appear intermediate between *P. radiata* and *P. attenuata* with ca. 100% of trees on the Western Terrace appearing to be of hybrid origin. Pure forms dominate in the south-eastern portions of SPR. *Havelik 448* (OBI 147353), *Richardson 261* (OBI 161809). Study Regions: 4, 24, 31.

[RR]***Pinus radiata*** D.Don (Monterey Pine) – Evergreen tree. Occasional on coastal headlands. Many individuals appear intermediate between *P. radiata* and *P. attenuata*, and the authors did not collect any specimens that were attributable to a pure form. However, Keil has documented the species as occurring at SPR, so we have included it in the checklist. *Keil 20657* (OBI 161809). Study Region: 2.

Pseudotsuga menziesii (Mirb.) Franco var. ***menziesii*** (Douglas Fir) – Evergreen tree. Common, the dominant tree in mixed evergreen forest. *Keil 20565* (OBI 147619). Study Regions: 1–5, 10, 12, 13, 18–21, 24, 25, 31, 32.

Taxaceae

Torreya californica Torr. (California Nutmeg) – Evergreen tree. Occasional in mixed evergreen forest. There is a large stand on the steep, north-facing slope on south side of Scott Creek. *Keil 20587* (OBI 147725). Study Regions: 5, 24, 25, 31.

Magnoliids

Aristolochiaceae

Asarum caudatum Lindl. (Wild Ginger) – Perennial herb. Occasional in riparian forest. *Kenny 501* (OBI 161306). Study Regions: 2, 5, 23, 24.

Lauraceae

Umbellularia californica (Hook. & Arn.) Nutt. (California Bay Laurel) – Evergreen tree. Common in riparian forest and mixed evergreen forest. *Richardson 279* (OBI 161799). Study Regions: 1–3, 6, 14, 25, 32.

Eudicots

Acanthaceae

****Acanthus mollis*** L. (Bear's Breeches) – Perennial herb. Uncommon near landscaped areas. Not treated in The *Jepson eFlora*. *Richardson 318* (OBI 179966). Study Regions: 5.

Adoxacae

Sambucus nigra L. subsp. ***caerulea*** (Raf.) Bolli (Blue Elderberry) – Shrub to small tree. Common in riparian forest and mixed evergreen forest. *Richardson 59* (OBI 130488). Study Regions: 3, 7, 20, 21, 24, 25, 33.

Sambucus racemosa L. var. ***racemosa*** (Red Elderberry) – Shrub to small tree. Common in riparian forest and mixed evergreen forest. *Keil 20663* (OBI 130425). Study Regions: 1–3, 5, 10, 13, 14, 24–26, 33.

Aizoaceae

****Carpobrotus edulis*** (L.) N.E.Br. (Freeway Iceplant) – Perennial herb. Common in disturbed areas near roadsides on coast. *Richardson 294* (OBI 149487). Study Regions: 17, 31.

Anacardiaceae

Toxicodendron diversilobum (Torr. & A.Gray) Greene (Western Poison Oak) – Shrub to liana. Common. *Richardson 396* (OBI 186156). Study Regions: 1–14, 17–21, 23–28, 30–33.

Apiaceae

[R]***Angelica tomentosa*** S.Watson (Woolly Angelica) – Perennial herb. Uncommon, only two individuals in study area occurring in northern coastal scrub. *Kenny 757* (OBI 166709). Study Region: 33.

****Anthriscus caucalis*** M.Bieb. (Bur-chervil) – Annual herb. Occasional in mixed evergreen forest. Historically present, though not encountered during study. *Keil 20646* (OBI 110442). Study Region: 4.

[R]***Apiastrum angustifolium*** Nutt. (Wild Celery) – Annual herb. Though locally rare in county, common in study area in northern coastal scrub on bluffs and on slopes of coastal gulches and in coastal prairie. *Kenny 406* (OBI 161486). Study Regions: 17, 28.

[R]***Bowlesia incana*** Ruiz & Pav. (Hoary Bowlesia) – Annual herb. Occasional in mixed evergreen forest and soft chaparral. *Kenny 262* (OBI 171077). Study Regions: 20, 25.

****Bupleurum lancifolium*** Hornem. (Lance-leaf Thorow-wax) – Annual herb. Uncommon in the Scott Creek riparian corridor. Historically present, though not encountered during study. *Morgan & West 3363* (UCSC). Study Region: 5.

Cicuta douglasii (DC.) J.M.Coult. & Rose (Douglas's Water Hemlock) – Perennial herb. Occasional on stream banks. *Kenny 740* (OBI 166699). Study Regions: 5, 26.

****Conium maculatum*** L. (Poison Hemlock) – Perennial herb. Common in moist, disturbed areas throughout study area. *Richardson 55* (OBI 110616). Study Regions: 1–3, 6–14, 18–21, 23–28, 30–33.

Daucus pusillus Michx. (Rattlesnake Carrot) – Annual herb. Common on coastal bluffs and slopes of coastal gulches. *Kenny 677* (OBI 161335). Study Regions: 10, 12, 13, 17, 20, 21, 28.

Eryngium armatum (S.Watson) J.M.Coult. & Rose (Coyote Thistle) – Perennial herb. Occasional in wet areas of coastal prairie. Historically present, last observed in 2018, though not encountered during study. Documented in West (2016). Study Region: 7.

****Foeniculum vulgare*** Mill. (Fennel) – Perennial herb. Occasional on roadsides. *Kenny 966B* (OBI 161840). Study Regions: 5, 32.

Heracleum maximum W.Bartram (Cow Parsnip) – Perennial herb. Common in riparian forest and mixed evergreen

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forest. *Richardson 13* (OBI 145008). Study Regions: 2, 3, 5, 9, 10, 17–19, 24, 26, 28.

[R]***Ligusticum apiifolium*** (Nutt.) A.Gray (Celery-leaved Licorice Root) – Perennial herb. Uncommon, occurring in mixed evergreen forest. *Kenny 888* (OBI 166681). Study Regions: 18, 20, 26.

[R]***Lomatium caruifolium*** (Hook. & Arn.) J.M.Coult. & Rose var. ***caruifolium*** (Caraway-leaved Lomatium) – Perennial herb. Locally rare in county. Occasional in interior coastal prairie. *Kenny 113* (OBI 161403). Study Regions: 4, 7.

Oenanthe sarmentosa Presl ex DC. (Pacific Oenanthe) – Perennial herb. Common in freshwater marshes and riparian areas. *Buchloh 23* (OBI). Study Regions: 5, 10, 14, 17, 18, 20, 21, 28, 31.

Osmorhiza berteroi DC. (Mountain Sweet Cicely) – Perennial herb. Common in mixed evergreen forest. *Richardson 355* (OBI 167631). Study Regions: 1, 5, 18, 23–25.

[RR]***Perideridia gairdneri*** (Hook. & Arn.) Mathias subsp. ***gairdneri*** (Gairdner's Yampah) (CRPR 4.2) – Perennial herb. Uncommon, one population in coastal prairie. *Richardson 104* (OBI 175426). Study Region: 11.

Sanicula arctopoides Hook. & Arn. (Footsteps-of-Spring) – Perennial herb. Common in coastal prairie. *Kenny 108* (OBI 161393). Study Regions: 9, 20, 27, 28.

Sanicula bipinnatifida Douglas (Purple Sanicle) – Perennial herb. Occasional in interior coastal prairie. *Kenny 280* (OBI 161281). Study Region: 4.

Sanicula crassicaulis var. ***crassicaulis*** Poepp. ex DC. (Pacific Sanicle) – Perennial herb. Common in mixed evergreen forest and northern coastal scrub. *Keil 20629* (OBI 175592). Study Regions: 1, 5, 9, 10, 12, 17–21, 24, 25, 28, 31, 32.

Sanicula crassicaulis Poepp. ex DC. var. ***nudicaulis*** (Hook & Arn.) Kenny & West (Pacific Fringed Snakeroot) – Perennial herb. Uncommon in mixed evergreen forest. About 50% of *S. crassicaulis* inidviduals found in moist, shaded areas at SPR are assignable to var. *nudicaulis*. This form is distinguished by schizocarp prickles that are less graduated and leaf marginal trichomes that are much longer and thinner than those of the common var. *crassicaulis*. *West 298* (JEPS 83051).

[RR]***Sanicula hoffmannii*** (Munz) R.H.Shan and Constance (Hoffmann's Sanicle) (CRPR 4.3)– Perennial herb. Uncommon in mixed evergreen forest. *Kenny 196* (OBI 161529). Study Region: 5.

****Torilis arvensis*** (Huds.) Link (Tall Sock-destroyer) – Annual herb. Common in disturbed areas. *Richardson 67* (OBI 165013). Study Regions: 23, 25.

****Torilis nodosa*** (L.) Gaertner (Short Sock-destroyer) – Annual herb. Common in disturbed areas. *Kenny 955* (OBI 161831). Study Regions: 1–3, 5, 10, 17, 18, 20, 24, 33.

[R]***Yabea microcarpa*** (Hook. & Arn.) Koso-Pol. (California Hedge Parsley) – Annual herb. Only county records are from the Scott Creek Watershed and on the coastal slope of the Swanton area. Occasional in northern coastal scrub and in grasslands. *Kenny 489* (OBI 161286). Study Regions: 1, 21, 33.

Apocynaceae

****Vinca major*** L. (Greater Periwinkle) – Perennial herb. Occasional in mixed evergreen forest near historically disturbed areas. *Richardson 76* (OBI 165547). Study Regions: 5, 25.

Araliaceae

Aralia californica S.Watson (Elk Clover) – Perennial herb. Common in riparian forest. *Richardson 173* (OBI 140154). Study Regions: 2, 5, 23.

****Hedera helix*** L. (English Ivy) – Perennial herb to liana. Uncommon on roadsides in mixed evergreen forest. *Kenny 850* (OBI 166734). Study Regions: 5, 25.

Hydrocotyle ranunculoides L. f. (Floating Marsh Pennywort) – Perennial herb. Uncommon in ponds. *Kenny 753* (OBI 161865). Study Regions: 5, 20.

Asteraceae

Achillea millefolium L. (Yarrow) – Perennial herb. Common in coastal scrub and grasslands. *Keil 20566* (OBI 141398). Study Regions: 1, 4, 6, 7, 10, 12, 17–21, 24–28, 31, 32.

Adenocaulon bicolor Hook. (Trail Plant) – Perennial herb. Occasional in mixed evergreen forest. *Richardson 134* (OBI 140326). Study Region: 5.

Agoseris grandiflora (Nutt.) Greene var. ***grandiflora*** (Large-flowered Agoseris) – Perennial herb. Occasional in mixed evergreen forest. *Kenny 463* (OBI 161265). Study Regions: 1, 2, 5, 18, 24, 26.

[R]***Agoseris heterophylla*** (Nutt.) Greene var. ***heterophylla*** (Annual Agoseris) – Annual herb. Locally rare in county. Uncommon in interior coastal prairie. *Kenny 448* (OBI 161254). Study Region: 4.

Ambrosia chamissonis (Less.) Greene (Beach Bur-sage) – Shrub. Common in sandy areas around Scott Creek Beach. *Kenny 874* (OBI 166721). Study Region: 31.

Anaphalis margaritacea (L.) Benth. (Pearly Everlasting) – Perennial herb. Common in northern coastal scrub. *Kenny 734* (OBI 166702). Study Regions: 10, 11, 13, 18, 20.

Anisocarpus madioides Nutt. (Woodland Tarweed) – Perennial herb. Common in openings in mixed evergreen forest. *Richardson 130* (OBI 120849). Study Regions: 1, 5, 20, 23–25.

Arnica discoidea Benth. (Rayless Arnica) – Perennial herb. Occasional in mixed evergreen forest. *Kenny 765* (OBI 161861). Study Region: 24.

Artemisia californica Less. (California Sagebrush) – Shrub. Common in northern coastal scrub. *Richardson 43* (OBI 130219). Study Regions: 7, 9, 10, 12, 13, 17–19, 21, 26, 28, 30–32.

Artemisia douglasiana Besser (Mugwort) – Perennial herb. Common in northern coastal scrub, riparian forest, and mixed evergreen forest. *Kenny 845* (OBI 166737). Study Regions: 2, 3, 9, 10, 12–14, 17–21, 25–28, 30, 31.

Baccharis glutinosa Pers. (Marsh Baccharis) – Caning shrub. Occasional in freshwater marshes. *Richardson 264* (OBI 161817). Study Regions: 10, 13, 14, 30, 31.

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Baccharis pilularis DC. subsp. ***consanguinea*** (DC.) C.B. Wolf (Coyote Brush) – Shrub. Common in northern coastal scrub. *Richardson 81* (OBI 175737). Study Region: 33.

****Bellis perennis*** L. (English Daisy) – Perennial herb. Uncommon in disturbed areas. *Kenny 379* (OBI 161483). Study Regions: 1, 25.

****Carduus pycnocephalus*** L. subsp. ***pycnocephalus*** (Italian Thistle) – Annual herb. Common in disturbed areas in coastal prairie. *Richardson 30* (OBI 135718). Study Regions: 1, 2, 5–8, 10–12, 14, 17–21, 23, 25, 27, 28, 30, 32.

****Centaurea melitensis*** L. (Tocalote) – Annual herb. Common in disturbed areas in coastal prairie. *Kenny 954* (OBI 161830). Study Regions: 1, 9, 10, 12, 17, 19, 20, 25, 28, 32.

****Cichorium intybus*** L. (Chicory) – Perennial herb. Uncommon in disturbed areas. *Richardson 329* (OBI 179986). Study Region: 5.

Cirsium brevistylum Cronquist (Indian Thistle) – Perennial herb. Occasional in disturbed areas in northern coastal scrub, mixed evergreen forest, and northern coastal scrub. *Kenny 761* (OBI 148068). Study Regions: 3, 5, 9, 24, 25.

[R]***Cirsium quercetorum*** (A.Gray) Jeps. (Brownie Thistle) – Annual to perennial herb. Uncommon in coastal prairie. Historically present, though not encountered during study. Known from a 1983 collection within 500 m of study area. *Buck & West 284* (JEPS 82811). Documented in West (2016) as occurring in study region 21 as of 2010. Study Region: 21.

****Cirsium vulgare*** (Savi) Ten. (Bull Thistle) – Biennial herb. Common throughout study area. *Buchloh 21* (OBI). Study Regions: 1–8, 10, 11, 13, 14, 17–21, 23–28, 30–33.

Corethrogyne filaginifolia (Hook. & Arn.) Nutt. (California Aster) – Perennial herb to subshrub. Common in northern coastal scrub and grasslands. Prostrate plants occurring near the coast with stems bearing single, large inflorescences have been assigned to *Corethrogyne filaginifolia* var. *californica* (DC.) Saroyan. *Kenny 681B* (OBI). Study Regions: 4, 7, 9, 10, 12, 14, 19–21, 28, 31, 32.

****Cotula coronopifolia*** L. (Brass-buttons) – Perennial herb. Common on coastal bluffs in disturbed areas, or in bare soil. *Kenny 560* (OBI 161372). Study Regions: 17, 19–21, 28.

****Crepis vesicaria*** L. subsp. ***taraxacifolia*** (Thuill.) Thell. (Weedy Hawksbeard) – Annual to perennial herb. Occasional in disturbed areas in coastal prairie. *Kenny 827* (OBI 148249). Study Regions: 20, 23, 28.

Deinandra corymbosa (DC.) B.G.Baldwin (Coastal Tarweed) – Annual herb. Occasional in northern coastal scrub and coastal prairie. *Richardson 298* (OBI 186145). Study Regions: 21, 25.

****Delairea odorata*** Lem. (Cape Ivy) – Perennial herb. Occasional in riparian forest. *Kenny 970* (OBI 161843). Study Regions: 20, 26.

Ericameria arborescens (A.Gray) Greene (Golden-fleece) – Shrub. Uncommon in steep openings in mixed evergreen forest. Historically present, though not encountered during study. Documented in West (2016) as occurring in study regions 23 and 24 as of 8/18/2010. Study Region: 23, 24.

Erigeron canadensis L. (Horseweed) – Annual herb. Common in disturbed areas. *Buchloh 17*. Study Regions: 5, 10, 18, 19, 21, 23, 25, 26, 30–32.

Erigeron foliosus Nutt. var. ***franciscensis*** G.L.Nesom (Leafy Fleabane) – Perennial herb. Uncommon in interface of coastal prairie and mixed evergreen forest. Historically present, though not encountered during study. Personal observation by James A. West in study region 20. Study Region: 20.

Erigeron glaucus Ker Gawl. (Seaside Daisy) – Subshrub. Common on coastal bluffs. *Kenny 266* (OBI 161271). Study Regions: 9, 10, 12, 17, 21, 28.

Eriophyllum confertiflorum (DC.) A.Gray var. ***confertiflorum*** (Golden Yarrow) – Subshrub. Occasional in openings in mixed evergreen forest. *Richardson 21* (OBI 168617). Study Regions: 1, 23–25.

Eriophyllum staechadifolium Lag. (Seaside Woolly Sunflower) – Subshrub. Common in northern coastal scrub. *Richardson 79* (OBI 106191). Study Regions: 10, 12, 13, 17–19, 21, 25, 28, 30–33.

Eurybia radulina (A.Gray) G.L.Nesom (Broad-leaved Aster) – Perennial herb. Occasional in mixed evergreen forest. *Kenny 864* (OBI 158453). Study Regions: 3, 18, 23.

Euthamia occidentalis Nutt. (Western Goldenrod) – Perennial herb. Common in upland portions of the Scott Creek marsh. Uncommon in freshwater marshes throughout study area. *Richardson 250* (OBI 149308). Study Regions: 18, 30, 31.

Gamochaeta ustulata (Nutt.) Holub (Purple Cudweed) – Perennial herb. Common in disturbed areas throughout study area. *Kenny 625* (OBI 161435). Study Regions: 10, 12, 17, 19–21, 23, 24, 27, 28, 30, 31.

****Glebionis coronaria*** (L.) Cass. ex Spach (Crown Daisy) – Annual herb. Uncommon, only occurring in one disturbed area in coastal prairie. *Kenny 977* (OBI 158491). Study Region: 30.

Gnaphalium palustre Nutt. (Lowland Cudweed) – Annual herb. Uncommon in riparian areas and wet pasture. *Kenny 251* (OBI 148281). Study Regions: 2, 25.

Grindelia hirsutula Hook. & Arn. (Hirsute Gumplant) – Perennial herb. Occasional in coastal prairie. *Kenny 892* (OBI 166676). Study Regions: 10, 18, 19.

Helenium puberulum DC. (Sneezeweed Rosilla) – Perennial herb. Common in riparian forest and mixed evergreen forest. *Buchloh 45* (OBI 154050). Study Regions: 5, 14, 17, 18, 23, 24, 31, 33.

****Helminthotheca echioides*** (L.) Holub (Bristly Ox-tongue) – Annual to biennial herb. Common in non-native grasslands and disturbed areas. *Richardson 284* (OBI 174296). Study Regions: 17, 26, 28, 30, 31.

[R]***Heterotheca sessiliflora*** (Nutt.) Shinners subsp. ***bolanderi*** (A.Gray) Semple (Bolander's Golden Aster) – Perennial herb. Uncommon on coastal bluffs and in coastal prairie. Subspecies *bolanderi* and *echioides* both occur in study area, with subsp. *bolanderi* occurring in coastal prairie within 0.4 km of the coast, and subsp. *echioides* occurring in interior grasslands (see below). There are no apparent intergrades in study area. *Kenny 952* (OBI 161828). Study Region: 11.

Heterotheca sessiliflora (Nutt.) Shinners subsp. ***echioides*** (Benth.) Semple (Bristly Golden Aster) – Perennial herb. Uncommon in interior grasslands. *Kenny 264* (OBI 161269). Study Region: 25.

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Hieracium albiflorum Hook. (White Hawkweed) – Perennial herb. Occasional in mixed evergreen forest. *Kenny 758A* (OBI 166710). Study Regions: 3, 23, 24.

****Hypochaeris glabra*** L. (Smooth Cat's-ear) – Annual herb. Common in grasslands and disturbed areas throughout study area. *Kenny 650* (OBI 161367). Study Regions: 1, 4, 6, 7, 9, 11, 12, 17, 19, 21, 28.

****Hypochaeris radicata*** L. (Rough Cat's-ear) – Perennial herb. Common in grasslands and disturbed areas throughout study area. *Richardson 56* (OBI 186134). Study Regions: 1–8, 10–12, 14, 17–21, 23–25, 27, 28, 30, 32, 33.

Jaumea carnosa (Less.) A.Gray (Salt-marsh Jaumea) – Perennial herb. Common in the Scott Creek marsh. *Buchloh 26* (OBI). Study Region: 31.

****Lactuca serriola*** L. (Prickly Lettuce) – Annual herb. Common in disturbed areas in mixed evergreen forest. *Kenny 953* (OBI 161829). Study Regions: 1, 4, 19, 23, 25, 26, 28, 31.

****Lapsana communis*** L. (Nipplewort) – Annual herb. Occasional in disturbed areas in riparian forest and mixed evergreen forest. *Kenny 838* (OBI 166744). Study Regions: 5, 26.

[R]***Lasthenia gracilis*** (DC.) Greene (Common Goldfields) – Annual herb. Occasional in areas of thin soil in coastal prairie. Habitat for this species has been heavily impacted, and it can be used as an indicator species for botanical hotspots in the county (Neubauer 2013). *Kenny 417* (OBI 161481). Study Regions: 7, 9, 10.

[R]***Layia gaillardioides*** (Hook. & Arn.) DC. (Woodland Layia) – Annual herb. Uncommon in grassy openings in mixed evergreen forest. Not encountered during this study, likely due to most of the habitat in study area being converted to *Ceanothus* chaparral following the 2009 Lockheed Fire. *Keil 20612* (OBI 181702). Study Regions: 1, 4, 5, 25.

Layia hieracioides (DC.) Hook. & Arn. (Tall Layia) – Annual herb. Occasional in northern coastal scrub and coastal prairie. *Kenny 460* (OBI 161263). Study Regions: 1, 12, 20, 28, 33.

Layia platyglossa (Fisch. & C.A.Mey.) A.Gray (Tidy-tips) – Annual herb. Uncommon on exposed mudstone in coastal prairie and coastal scrub. *Kenny 281* (OBI 161282). Study Regions: 7, 28, 32.

****Leontodon saxatilis*** Lam. subsp. ***saxatilis*** (Hairy Hawkbit) – Annual to perennial herb. Occasional in disturbed areas in coastal prairie. *Kenny 444* (OBI 161252). Study Regions: 4, 25, 32.

Logfia filaginoides (Hook. & Arn.) Morefield (California Cottonrose) – Annual herb. Uncommon in openings in coastal scrub. Historically present, though not encountered during study. *Buck & West 513* (JEPS 85188). Study Region: 10.

****Logfia gallica*** (L.) Coss. & Germ. (Daggerleaf Cottonrose) – Annual herb. Common in disturbed areas with bare soil throughout study area. *Buchloh 20* (OBI). Study Regions: 7, 9–11, 13, 17, 19, 20, 28, 30–32.

Madia exigua (Sm.) A.Gray (Threadstem Madia) – Annual herb. Uncommon in openings in mixed evergreen forest. *Kenny 763* (OBI 161862). Study Region: 24.

Madia gracilis (Sm.) D.D. Keck & J. Clausen ex Applegate (Gumweed) – Annual herb. Occasional in openings in mixed evergreen forest and on brushy slopes. *Keil 20626* (OBI 157750). Study Regions: 3, 5, 20, 24.

Madia sativa Molina (Coast Tarweed) – Annual herb. Common in coastal scrub, grasslands, and hard chaparral. *Buchloh 38* (OBI 157785). Study Regions: 1, 2, 7, 8, 10, 12, 19–21, 24–26, 28, 30, 31.

[R]***Malacothrix floccifera*** (DC.) S.F.Blake (Woolly Malacothrix) – Annual herb. Uncommon, only occurring on one cliff face in mixed evergreen forest. *Kenny 383* (OBI 161457). Study Region: 1.

Matricaria discoidea DC. (Pineapple Weed) – Annual herb. Uncommon in gravel on roadsides. *Kenny 199* (OBI 171066). Study Regions: 26, 27.

[RR]***Micropus amphibolus*** A.Gray (Mount Diablo Cottonseed) (CRPR 3.2) – Annual herb. Uncommon, only occurring in one area of thin soils in coastal prairie within 0.4 km of coast. *Kenny 521* (OBI 161321). Study Region: 9.

[R]***Micropus californicus*** Fisch. & C.A.Mey. var. ***subvestitus*** A.Gray (Green Cottonweed) – Annual herb. Uncommon in openings in mixed evergreen forest. Historically present, though not encountered during study. Documented in West (2016) as occurring in study region 25 as of 2016. Study Region: 25.

[R]***Microseris bigelovii*** (A.Gray) Sch. Bip. (Bigelow's Microseris) – Annual herb. Occasional in coastal prairie near coast as well as inland. Two forms have been documented in the Scott Creek Watershed that differ in size of heads and fruit; further study needed. *Kenny 399* (OBI 161471). Study Regions: 4, 14, 17, 20.

[RR]***Microseris paludosa*** J.T.Howell (Marsh Microseris) (CRPR 1B.2) – Perennial herb. Uncommon in coastal prairie. Historically present, though not encountered during study. Observed by James A. West in 1970s in a seasonally wet drainage. Habitat may have been impacted by cattle grazing. Study Region: NA.

Petasites frigidus (L.) Fr. var. ***palmatus*** (Aiton) Cronquist (Western Sweet Coltsfoot) – Perennial herb. Common in riparian areas. *Kenny 131* (OBI 161419). Study Regions: 2, 5, 25, 26.

Pseudognaphalium beneolens (Davidson) Anderb. (Fragrant Everlasting) – Annual to perennial herb. Uncommon in mixed evergreen forest. Historically present, though not encountered during study. Documented in West (2016) as occurring in study region 5. Study Region: 5.

Pseudognaphalium biolettii Anderb. (Bioletti's Cudweed) – Perennial herb. Occasional in northern coastal scrub. *Kenny 840* (OBI 166742). Study Regions: 8, 17, 20.

Pseudognaphalium californicum (DC.) Anderb. (California Cudweed) – Perennial herb. Occasional in openings in northern coastal scrub, grasslands, and mixed evergreen forest. *Kenny 936B* (OBI 166666). Study Regions: 5, 11, 21, 24, 31.

****Pseudognaphalium luteoalbum*** (L.) Hilliard & B.L.Burtt (Weedy Cudweed) – Annual herb. Common in northern coastal scrub, grassland, and hard chaparral. *Richardson 291* (OBI 149503). Study Regions: 11, 19, 21, 26, 30, 31.

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Pseudognaphalium ramosissimum (Nutt.) Anderb. (Pink Everlasting) – Perennial herb. Occasional in riparian forest and mixed evergreen forest. *Richardson 194* (OBI 163664). Study Regions: 5, 12, 23.

Pseudognaphalium stramineum (Kunth) Anderb. (Cotton-batting Plant) – Perennial herb. Common in northern coastal scrub. *Kenny 902* (OBI 171052). Study Regions: 1, 2, 10–12, 17, 20, 26, 28, 31, 32.

Psilocarphus tenellus Nutt. (Slender Woolly Marbles) – Annual herb. Occasional in areas of bare soil in grasslands, mixed evergreen forest, and hard chaparral. *Kenny 255* (OBI 171070). Study Regions: 5, 23, 25, 28.

Rafinesquia californica Nutt. (California Chicory) – Annual herb. Uncommon on exposed mudstone cliffs. *Kenny 841* (OBI 166741). Study Regions: 5, 17, 23, 24.

RR***Senecio aphanactis*** Greene (Chaparral Ragwort) (CRPR 2B.2) – Annual herb. Uncommon in hard chaparral. Not in Neubauer (2013). Specimen cited is the first county collection. *Kenny 483* (OBI 161300). Study Region: 23.

****Senecio glomeratus*** Desf. ex Poir. (Cutleaf Burnweed) – Annual to perennial herb. Uncommon in disturbed areas on roadsides. *Kenny 762* (OBI 166712). Study Regions: 8, 24, 25.

****Senecio jacobaea*** L. (Tansy Ragwort) – Perennial herb. Uncommon in disturbed areas on roadside near Scott Creek. *Richardson 275* (OBI 149287). Study Region: 5.

****Senecio minimus*** Poir. (Coastal Burnweed) – Annual to perennial herb. Occasional in disturbed areas on roadsides. *Kenny 934B* (OBI 166668). Study Regions: 1, 5, 10, 24, 31.

****Senecio vulgaris*** L. (Common Groundsel) – Annual herb. Uncommon in disturbed areas on roadsides. *Kenny 190* (OBI 161523). Study Regions: 26, 33.

****Silybum marianum*** (L.) Gaertn. (Milk Thistle) – Annual to perennial herb. Common in disturbed areas in coastal prairie. *Richardson 278* (OBI 149310). Study Regions: 1, 4, 6, 7, 11, 14, 18–21, 25, 27, 28, 30, 32.

Solidago elongata Nutt. (West Coast Canada Goldenrod) – Perennial herb. Occasional in northern coastal scrub and in disturbed areas in riparian forest. *Kenny 895* (OBI 166673). Study Regions: 5, 13, 14, 26, 28.

Solidago velutina DC. subsp. ***californica*** (Nutt.) Semple (California Goldenrod) – Perennial herb. Uncommon in coastal prairie. *Richardson 152* (OBI 145666). Study Region: 7.

****Soliva sessilis*** Ruiz & Pav. (Common Soliva) – Annual herb. Occasional on roadsides. *Kenny 539* (OBI 161373). Study Region: 25.

****Sonchus asper*** (L.) Hill subsp. ***asper*** (Prickly Sow Thistle) – Annual herb. Occasional in disturbed areas. *Richardson 29* (OBI 145760). Study Region: 24.

****Sonchus oleraceus*** L. (Common Sow Thistle) – Annual herb. Common in disturbed areas throughout study area. *Kenny 858* (OBI 166748). Study Regions: 3–5, 8, 17, 19–21, 25, 28, 31, 33.

RR***Stebbinsoseris decipiens*** (K.L.Chambers) K.L.Chambers (Santa Cruz Microseris) (CRPR 1B.2) – Annual herb. Uncommon on thin soils in coastal prairie. This taxon is an allopolyploid hybrid of *Microseris bigelovii* and *Uropappus lindleyi* and is typically found with *Microseris bigelovii* in study area. *Kenny 471* (OBI 174302). Study Regions: 1, 4, 5, 9, 18, 20.

Stephanomeria virgata Benth. subsp. ***pleurocarpa*** (Greene) Gottlieb (Virgate Stephanomeria) – Annual herb. Uncommon in openings in mixed evergreen forest. Specimens collected in course of this study are highly variable. All examined lack a distinct longitudinal groove on the faces of the fruit and have outer phyllaries that are not reflexed, indicating *S. virgata* subsp. *pleurocarpa*. Some individuals observed in the field showed greater reflexion of the outermost phyllaries, a trait that is difficult to see on herbarium specimens. This may indicate potential hybridization with the tetraploid *S. elata* Nutt., a phenomenon that has been noted to occur (Gottlieb, 1972), but normally results in non-reproductive triploid individuals. Caryotyping of several individuals might shed light on the taxonomic status of this interesting population. *Kenny 957* (OBI 149741). Study Region: 1.

Symphyotrichum chilense (Nees) G.L.Nesom (California Aster) – Perennial herb. Common in northern coastal scrub. *Richardson 187* (OBI 176690). Study Regions: 1, 6–8, 10, 13, 14, 17–19, 21, 23, 30–32.

Symphyotrichum subspicatum (Nees) G.L.Nesom (Douglas's Aster) – Perennial herb. Occasional in wet areas in northern coastal scrub. *Kenny 630* (OBI 161352). Study Regions: 26, 31.

****Taraxacum officinale*** F.H.Wigg. (Common Dandelion) – Perennial herb. Uncommon in disturbed areas on roadsides. *Kenny 274* (OBI 161276). Study Regions: 1, 5, 26.

Uropappus lindleyi (DC.) Nutt. (Silverpuffs) – Annual herb. Uncommon in interior coastal prairie. Historically present, though not encountered during study. Documented in West (2016) as occurring in study region 1. Study Region: 1.

R***Wyethia angustifolia*** (DC.) Nutt. (Narrow-leaved Mule's Ears) – Perennial herb. Uncommon in coastal prairie. *Kenny 830* (OBI 148247). Study Regions: 11, 28, 32.

Berberidaceae

R***Berberis pinnata*** Lag. subsp. ***pinnata*** (California Barberry) – Shrub. Uncommon, only observed on north-facing slope of one gully in northern coastal scrub. *Kenny 185A* (OBI 161498). Study Regions: 10, 11.

Betulaceae

Alnus rubra Bong. (Red Alder) – Deciduous tree. Common in riparian forest. *Richardson 7* (OBI 131245). Study Regions: 2, 5, 24–26.

Corylus cornuta Marshall subsp. ***californica*** (A. DC.) A.E. Murray (California Hazel) – Shrub. Common in riparian forest and mixed evergreen forest. *Kenny* 130 (OBI 161413). Study Regions: 1, 3, 5, 18, 23, 25, 32.

Boraginaceae

Amsinckia intermedia Fisch. & C.A.Mey. (Common Fiddleneck) – Annual herb. Occasional in coastal grasslands. *Kenny 549* (OBI 161332). Study Regions: 2, 19, 28.

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[RR]***Amsinckia lunaris*** J.F.Macbr. (Bent-flowered Fiddleneck) (CRPR 1B.2) – Annual herb. Uncommon, only occurring on one steep, exposed north-facing slope in the upper portion of a coastal gully in coastal prairie. *Kenny 285* (OBI 161390). Study Region: 21.

Amsinckia menziesii (Lehm.) A.Nelson. & J.F.Macbr. (Common Fiddleneck, Small-flowered Fiddleneck) – Annual herb. Uncommon in northern coastal scrub. Specimen needs to be examined by expert as the majority of specimens labeled *A. menziesii* from Santa Cruz County have been identified as *A. retrorsa* Suksd. *Richardson 85* (OBI 131569). Study Regions: 1, 33.

Cryptantha clevelandii Greene var. ***florosa*** I.M.Johnst. (Coastal Cryptantha) – Annual herb. Common in northern coastal scrub. *Kenny 456*. Study Regions: 1, 4, 9, 12, 17.

Cryptantha micromeres (A.Gray) Greene (Minute-flowered Cryptantha) – Annual herb. Common in northern coastal scrub and hard chaparral. *Kenny 453* (OBI 161257). Study Regions: 1, 9, 13, 20, 28.

[R]***Cryptantha torreyana*** **(A.Gray) Greene var.** ***pumila*** (A.Gray) Greene (Dwarf Cryptantha) – Annual herb. Uncommon in hard chaparral. Historically present, though not encountered during study. Documented in West (2016) as occurring in study region 23. Study Region: 23.

Cynoglossum grande Dougl. ex Lehm. (Grand Hound's-tongue) – Perennial herb. Occasional in mixed evergreen forest. Synonymous with *Adelinia grande* (Douglas ex Lehm.) J. Cohen. *Kenny 101* (OBI 161392). Study Regions: 5, 23–25.

Emmenanthe penduliflora Benth. (Whispering Bells) var. ***penduliflora*** – Annual herb. Uncommon in hard chaparral. Only observed briefly following the 2009 Lockheed Fire. Historically present, though not encountered during study. Documented in West (2016) as occurring in study regions 1 and 4. Seeds deposited at UCSC Arboretum. Study Regions: 1, 4.

Eriodictyon californicum (Hook. & Arn.) Torr. (California Yerba Santa) – Shrub. Uncommon in hard chaparral. *Kenny 459* (OBI 161262). Study Regions: 1, 20, 23, 25.

****Myosotis discolor*** Pers. (Changing Forget-me-not) – Annual herb. Occasional in northern coastal scrub, disturbed grasslands, and riparian areas. *Kenny 500* (OBI 161305). Study Regions: 9, 10, 14, 20.

****Myosotis latifolia*** Poir. (Broadleaved Forget-me-not) – Perennial herb. Common in riparian areas and mixed evergreen forest. *Monika Richardson 9* (OBI 131068). Study Regions: 1–3, 5, 20, 23–26.

[R]***Nemophila menziesii*** Hook. & Arn. var. ***menziesii*** (Baby Blue-eyes) – Annual herb. Uncommon in openings on steep slopes in mixed evergreen forest. *Kenny 124* (OBI 161408). Study Regions: 1, 2.

Nemophila parviflora Douglas ex Benth. var. ***parviflora*** (Small-flowered Nemophila) – Annual herb. Occasional in mixed evergreen forest. *Kenny 475* (OBI 166690). Study Regions: 1, 3, 5, 23, 25.

[R]***Nemophila pedunculata*** Douglas ex Benth. (Meadow Nemophila) – Annual herb. Uncommon in grassy openings in redwood forest. *Kenny 469* (OBI 174300). Study Regions: 1, 5.

[R]***Nemophila pulchella*** Eastw. var. ***fremontii*** (Elmer) Constance (Fremont's Nemophila) – Annual herb. Uncommon in meadows in redwood forest. The only county collections are in study area. *Kenny 467B* (OBI 174301). Study Regions: 1, 19.

Phacelia californica Cham. (California Phacelia) – Perennial herb. Uncommon on steep, south facing slopes in redwood forest. *Kenny 624* (OBI 161421). Study Regions: 24, 25.

Phacelia distans Benth. (Common Phacelia) – Annual herb. Uncommon in openings on steep slopes in mixed evergreen forest. *Kenny 390* (OBI 161464). Study Region: 1.

Phacelia malvifolia Cham. var. ***malvifolia*** (Stinging Phacelia) – Annual herb. Common in northern coastal scrub and in disturbed areas in mixed evergreen forest. *Kenny 277* (OBI 161278). Study Regions: 3, 20, 21, 25, 26, 28, 31.

[R]***Plagiobothrys bracteatus*** (Howell) I.M.Johnst. (Bracted Popcornflower) – Annual herb. Occasional in seasonally wet areas, pond margins, and seeps. *Kenny 467* (OBI 166692). Study Regions: 1, 2, 5, 6.

[RR]***Plagiobothrys chorisianus*** (Cham.) I.M.Johnst. var. ***chorisianus*** (Choris's Popcornflower) (CRPR 1B.2) – Annual herb. Uncommon, only occurring in a small, marshy area between Scott Creek and the northern border of SPR. Historically present, though not encountered during study. Collected May 2, 1988 by Taylor et al. (UC 1561092). Study Region: 3.

Plagiobothrys hispidulus (Greene) I.M.Johnst. (Harsh Popcornflower) – Annual herb. Uncommon, only occurring in a vernally wet area in mixed evergreen forest. Only known in county and in study area from a single 1988 collection. Though the collection locality is currently out of range for this species (The *Jepson eFlora* 2020) the specimen keys to *P. hispidulus* and may represent a long-distance dispersal event that did not establish a permanent population. *Keil 20636* (OBI 136828). Study Region: 1.

Brassicaceae

[R]***Athysanus pusillus*** (Hook.) Greene (Dwarf Athysanus) – Annual herb. Uncommon, only occurring in one opening in mixed evergreen forest. Historically present, though not encountered during study. *West 54.1* (JEPS 82961). Study Region: 2.

Barbarea orthoceras Ledeb. (American Winter Cress) – Biennial, perennial herb. Occasional in riparian forest. *Kenny 192* (OBI 161525). Study Region: 26

****Brassica nigra*** (L.) W.D.J.Koch (Black Mustard) – Annual herb. Common in disturbed areas on roadsides and adjacent to agricultural fields. *Kenny 380* (OBI 161454). Study Regions: 1, 5, 19, 26, 28, 31, 33.

****Brassica rapa*** L. (Field Mustard) – Annual herb. Occasional in disturbed areas and roadsides. *Buchloh 56*. Study Regions: 5, 31.

****Cakile edentula*** (Bigelow) Hook. (California Sea Rocket) – Annual herb. Uncommon, only occurring on berms bordering the Scott Creek marsh. *Buchloh 31* (OBI 123066). Study Region: 31.

****Cakile maritima*** Scop. (European Sea Rocket) – Annual herb. Uncommon, only occurring on berms bordering the

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Scott Creek marsh. *Kenny 875* (OBI 166720). Study Region: 31.

Cardamine californica (Nutt.) Greene (Milk Maids) – Perennial herb. Occasional in disturbed areas in northern coastal scrub and in mixed evergreen forest. *Kenny 409* (OBI 161478). Study Regions: 3, 5, 18, 20, 23, 25.

****Cardamine hirsuta*** L. (Hairy Bitter Cress) – Annual herb. Occasional in mixed evergreen forest and moist, disturbed areas. Not listed in Neubauer (2013) but noted as potentially occurring. It may be displacing the native *Cardamine oligosperma*, which was formerly common in study area but was not recorded during study. *Kenny 191*. Study Regions: 1, 23, 33.

Cardamine oligosperma Nutt. (Popweed) – Annual herb. Occasional in mixed evergreen forest. Historically present, though not encountered during study. *West s.n.* (OBI 100037). Study Region: 1.

Caulanthus lasiophyllus (Hook. and Arn.) Payson (California Mustard) – Annual herb. Uncommon in northern coastal scrub and coastal prairie. Historically present, though not encountered during study. Known from a 1912 collection (CAS:BOT-BC 39432) in the Swanton area and documented in West (2016) as occurring in study region 10, as of 2006. Study Region: 10.

****Hirschfeldia incana*** (L.) Lagr.-Foss. (Summer Mustard) – Perennial herb. Common in disturbed areas throughout study area. *Richardson 87* (OBI 149300). Study Regions: 19, 26–28, 30–33.

****Lepidium didymum*** L. (Lesser Swine Cress) – Annual herb. Occasional in vernally moist, disturbed areas. *Buchloh 53* (OBI 123064). Study Regions: 25, 26, 31.

Lepidium nitidum Nutt. (Shining Peppergrass) – Annual herb. Occasional in disturbed areas in coastal prairie. *Kenny 279* (OBI 161280). Study Region: 28.

Lepidium virginicum L. subsp. ***menziesii*** (DC.) Thell. (Virginia Peppergrass) – Annual herb. Occasional in disturbed areas in coastal prairie. *Kenny 465* (OBI 148251). Study Regions: 27, 28.

****Lunaria annua*** L. (Money Plant) – Annual herb. Uncommon, known only from one historic collection. Historically present, though not encountered during study. *Keil 20640*. Study Region: 4.

Nasturtium officinale W.T.Aiton (Water Cress) – Perennial herb. Common in riparian areas. *Kenny 658* (OBI 161504). Study Regions: 2, 5, 10, 14, 18, 20, 26, 28.

****Raphanus raphanistrum*** L. (Jointed Charlock) – Annual to perennial herb. Occasional in disturbed areas on roadsides. *Kenny 946* (OBI 166658). Study Regions: 19, 26, 30, 31.

****Raphanus sativus*** L. (Radish) – Annual to perennial herb. Common in disturbed areas in coastal prairie. *Kenny 945* (OBI 166657). Study Regions: 3, 19, 25, 26, 28, 30, 31.

Rorippa curvisiliqua (Hook.) Bessey ex Britton (Western Yellow Cress) – Annual to perennial herb. Occasional in Scott Creek. *Kenny 873* (OBI 166719). Study Regions: 5, 26, 31.

****Sisymbrium officinale*** (L.) Scop. (Hedge Mustard) – Annual herb. Occasional in disturbed areas in mixed evergreen forest. *Kenny 966A* (OBI 161839). Study Regions: 6, 19, 32.

Thysanocarpus curvipes Hook. subsp. ***curvipes*** (Hairy Fringepod) – Annual herb. Uncommon in openings in mixed evergreen forest. *Kenny 432* (OBI 148266). Study Regions: 1, 2, 5, 21.

[R]***Thysanocarpus laciniatus*** Nutt. (Narrow Leaved Fringe Pod) – Annual herb. Uncommon, only occurring in openings in mixed evergreen forest. *Kenny 393* (OBI 161484). Study Region: 1.

Turritis glabra L. (Tower Mustard) – Annual herb. Uncommon in steep mixed evergreen forest with dense midstory. Historically present, though not encountered during study. Seeds deposited at UCSC Arboretum. Study Region: 1.

Campanulaceae

Asyneuma prenanthoides (Durand) McVaugh (California Harebell) – Perennial herb. Occasional in mixed evergreen forest. Synonym to *Smithiastrum prenanthoides* (Durand) Moirin. *Richardson 175* (OBI 102556). Study Regions: 23, 24.

Triodanis biflora (Ruiz & Pav.) Greene (Venus's Looking Glass) – Annual herb. Uncommon in mixed evergreen forest. *Kenny 395* (OBI 161468). Study Region: 1.

Caprifoliaceae

Lonicera hispidula (Lindl.) Dougl. ex Torr. & A.Gray. (Hairy Honeysuckle) – Perennial vining herb to liana, Common in mixed evergreen forest and hard chaparral. *Richardson 53* (OBI 102873). Study Regions: 1, 3, 5, 9, 10, 13, 17, 20, 23–25, 32.

Symphoricarpos albus (L.) S.F.Blake var. ***laevigatus*** (Fernald) S.F.Blake (Snowberry) – Shrub. Occasional in mixed evergreen forest. *Kenny 962* (OBI 161835). Study Region: 2.

Symphoricarpos mollis Nutt. ex Torr. & Gray (Creeping Snowberry) – Shrub. Occasional in mixed evergreen forest. *Kenny 886* (OBI 166716). Study Regions: 18, 20, 24, 28.

Caryophyllaceae

****Herniaria hirsuta*** L. var. ***cinerea*** (DC.) Loret & Barrandon (Gray Hernaria) – Annual herb. Uncommon, only occurring in coastal prairie. Historically present, though not encountered during study. Observed in 1990s. Study Region: NA.

Minuartia douglasii (Torr. & A.Gray) Mattf. (Douglas's Sandwort) – Annual herb. Uncommon in vertical openings in mixed evergreen forest. Historically present, though not encountered during study. Collected by Roy Buck and James A. West on April 22, 1983 in study region 3. Study Region: 3.

****Polycarpon tetraphyllum*** L. var. ***tetraphyllum*** (Four-leaved Allseed) – Annual herb. Occasional on coastal bluffs and in coastal prairie. Though habitat exists for the native *P. depressum* Nutt., no individuals were encountered during study. *Kenny 856*. Study Regions: 17, 19.

****Sagina apetala*** Ard. (Dwarf Pearlwort) – Annual herb. Uncommon in wet, disturbed areas on coastal bluffs. *Kenny 689* (OBI 124217). Study Region: 17.

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Sagina decumbens Torr. & A.Gray subsp. ***occidentalis*** (S.Wats.) G.E.Crow (Western Pearlwort) – Annual herb. Uncommon in wet, disturbed areas on coastal bluffs. *Kenny 718* (OBI 124216). Study Region: 17.

Silene antirrhina L. (Sleepy Catchfly) – Annual herb. Uncommon, only occurring in open northern coastal scrub. Historically present, though not encountered during study. *Buck & West 517* (JEPS 85191). Study Region: 11.

****Silene gallica*** L. (Small-flower Catchfly) – Annual herb. Common in disturbed areas in coastal prairie. *Kenny 391* (OBI 161465). Study Regions: 1, 7, 9–12, 17, 19–21, 24, 27, 28, 30–32.

Silene verecunda S.Watson subsp. ***verecunda*** (San Francisco campion) (CRPR 1B.2) – Perennial herb. Uncommon, occurring primarily on siliceous mudstone outcrops in coastal prairie. Study area is at southern distributional limit for species. *The Jepson eFlora* does not recognize any subspecies in their treatment (Hartman et al. 2012). *Richardson 97* (OBI 140747). Study Region: 7.

****Spergula arvensis*** L. (Stickwort) – Annual herb. Common in disturbed areas in coastal prairie near roadsides. *Kenny 106* (OBI 161395). Study Regions: 7, 8, 17, 19, 27, 28.

****Spergularia bocconi*** (Scheele) Graebn. (Boccone's Sandspurrey) – Annual herb. Occasional in disturbed areas in coastal prairie. *Richardson 297* (OBI 186143). Study Region: 25.

****Spergularia rubra*** (L.) J.Presl & C.Presl (Red Sandspurrey) – Annual herb. Occasional in disturbed areas in coastal prairie. *Kenny 890* (OBI 166678). Study Regions: 17–19, 25, 26, 32.

****Stellaria media*** (L.) Vill. (Common Chickweed) – Annual herb. Common in disturbed areas in coastal prairie. *Kenny 129* (OBI 161412). Study Regions: 1, 5, 10, 11, 18, 20, 23, 25, 26, 28.

Stellaria nitens Nutt. (Shining Chickweed) – Annual herb. Uncommon. Historically present, not encountered during study. Documented in West (2016) as occurring in study regions 2 and 3 as of 2017. Seed collections deposited at the UCSC seedbank. Study Regions: 2, 3.

CELASTRACEAE

Euonymus occidentalis Nutt. ex Torr. var. ***occidentalis*** (Western Burning Bush) – Shrub. Uncommon in riparian forest. *Kenny 504* (OBI 161308). Study Region: 5.

CHENOPODIACEAE

****Atriplex prostrata*** Boucher ex DC. (Fat-hen) – Annual herb. Occasional in *Distichlis*- and *Salicornia*-dominated areas of the Scott Creek marsh. *Buchloh 43* (OBI 123065). Study Region: 31.

Chenopodium californicum (S.Watson) S.Watson (California Goosefoot) – Perennial herb. Occasional in disturbed areas near coast. *Kenny 189* (OBI 161492). Study Regions: 13, 14, 26, 33.

****Chenopodium murale*** L. (Nettle Leaf Goosefoot) – Annual herb. Occasional in the Scott Creek marsh. *Kenny 848* (OBI 158550). Study Region: 31.

****Dysphania ambrosioides*** (L.) Mosyakin & Clemants (Mexican Tea) – Perennial herb. Occasional in riparian forest. *Kenny 835* (OBI 166746). Study Regions: 5, 26.

Salicornia pacifica Standl. (Pickleweed) – Perennial herb. Common in portions of the Scott Creek marsh. *Kenny 868* (OBI 166717). Study Region: 31.

CISTACEAE

****Cistus incanus*** L. (Rock-rose) – Perennial herb. Uncommon, only occurring in ecotone of disturbed area in coastal prairie and mixed evergreen forest. *Kenny 828* (OBI 158506). Study Region: 32.

Crocanthemum scoparium Millsp. var. ***vulgare*** (Jeps.) Sorrie (Peak Rush-rose) – Shrub. Uncommon in hard chaparral. *Kenny 481* (OBI 161298). Study Region: 1.

CONVOLVULACEAE

Calystegia purpurata (Greene) Brummitt subsp. ***purpurata*** (Western Morning Glory) – Perennial vining herb. Occasional in hard chaparral. *Richardson s.n.* (OBI 125805). Study Regions: 1, 25.

****Convolvulus arvensis*** L. (Bindweed) – Perennial vining herb. Common in disturbed areas in coastal prairie. *Richardson 351* (OBI 186115). Study Regions: 2, 6, 8, 19–21, 25, 27, 32.

CORNACEAE

Cornus sericea L. subsp. ***sericea*** (American Dogwood) – Shrub. Common in riparian areas. *Richardson 214* (OBI 104519). Study Regions: 2, 5, 10, 25, 26, 31.

CRASSULACEAE

Crassula aquatica (L.) Schönl. (Water Pygmy-weed) – Annual herb. Uncommon in vernally wet areas. Historically present, though not encountered during study. *Keil 20560*. Study Region: 31.

Crassula connata (Ruiz & Pav.) A.Berger (Pygmy-weed) – Annual herb. Occasional in coastal prairie. *Kenny 123* (OBI 161415). Study Regions: 1, 9, 17, 28.

****Crassula tillaea*** Lest.-Garl. (Moss Pygmy-weed) – Annual herb. Uncommon, only occurring in moist gravel in parking lot. *Kenny 200* (OBI 161490). Study Region: 26.

Dudleya caespitosa (Haw.) Britton & Rose (Sea Lettuce) – Perennial herb, common on coastal bluffs and on exposed mudstone faces. Plants on the coast tend to be smaller, have more glaucous and red-tinged leaves, and are potentially more variable than plants inland. *Richardson 361* (OBI 186136). Study Regions: 7, 9, 10, 12, 13, 17, 20, 21, 23, 28, 32.

CUCURBITACEAE

Marah fabacea (Naudin) Greene (California Man-root) – Perennial vining herb. Common in northern coastal scrub and hard chaparral. *Richardson 323* (OBI 186127). Study Regions: 1–3, 5, 9, 10, 12, 13, 17–19, 21, 23–26, 28, 31, 33.

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DIPSACACEAE

****Dipsacus fullonum*** L. (Wild Teasel) – Perennial herb. Uncommon in wet, disturbed areas near roadsides. *Kenny 944* (OBI 166660). Study Regions: 5, 26.

ERICACEAE

Arbutus menziesii Pursh (Pacific Madrone) – Evergreen tree. Common in mixed evergreen forest. *Richardson 301* (OBI 149500). Study Regions: 1, 4, 5, 17, 23–25, 32.

Arctostaphylos crustacea Eastw. subsp. ***crinita*** (J.E.Adams) V.T.Parker, M.C.Vasey & J.E.Keeley (Crinite Manzanita) – Shrub. Uncommon, only occurring in hard chaparral. *Kenny 114* (OBI 161404). Study Regions: 4, 23.

Arctostaphylos crustacea Eastw. subsp. ***crustacea*** (Brittle-leaved Manzanita) – Shrub. Uncommon, only occurring in hard chaparral. *Richardson 112* (OBI 142315). Study Region: 1.

[RR]***Arctostaphylos glutinosa*** B.Schreib. (Schreiber's Manzanita) (CRPR 1B.2) – Shrub. Uncommon, only occurring in hard chaparral. Historically present near border of study area, though not encountered during study. This occurrence appears not to have survived the 2009 Lockheed Fire. *Buck & West 152* (JEPS 81980). Study Region: 1.

Gaultheria shallon Pursh (Salal) – Shrub. Uncommon, only occurring in one area in mixed evergreen forest. *Kenny 767* (OBI 161860). Study Region: 24.

Rhododendron occidentale (Torr. & A.Gray) A.Gray (California Azalea) – Shrub. Uncommon, only occurring in one area in mixed evergreen forest. *Kenny 643* (OBI 161362). Study Region: 5.

Vaccinium ovatum Pursh (California Huckleberry) – Shrub. Occasional in mixed evergreen forest and hard chaparral. *Kenny 480* (OBI 161284). Study Regions: 1, 5, 23, 24.

EUPHORBIACEAE

Euphorbia crenulata Engelm. (Chinese Caps) – Annual herb. Uncommon on steep slopes in mixed evergreen forest with dense midstory. Historically present, though not encountered during study. Seeds deposited at UCSC Arboretum. Study Region: 2.

****Euphorbia lathyris*** L. (Caper Spurge) – Perennial herb. Occasional in disturbed areas and on roadsides. *Buchloh 18* (OBI 123062). Study Regions: 5, 18, 20, 30–32.

****Euphorbia oblongata*** Griseb. (Eggleaf Spurge) – Perennial herb. Occasional on roadsides. One of the most invasive weeds in the county (Neubauer 2013), this species appears to be spreading in the study area and should be a high priority for weed management. *Richardson 26*. Study Regions: 24, 25.

****Euphorbia peplus*** L. (Petty Spurge) – Annual herb. Common in disturbed areas in mixed evergreen forest. *Kenny 967* (OBI 161841). Study Regions: 1, 3, 5, 17, 23, 25, 26, 31.

Euphorbia serpillifolia Pers. subsp. ***serpillifolia*** (Thyme-leaved Spurge) – Annual herb. Uncommon, only found in one weedy lawn. *Kenny 940* (OBI 158475). Study Region: 26.

Euphorbia spathulata Lam. (Warty Spurge) – Annual herb. Uncommon in mixed evergreen forest. *Kenny 707* (OBI). Study Region: 4.

FABACEAE

Acmispon americanus (Nutt.) Rydb. var. ***americanus*** (American Trefoil) – Annual herb. Uncommon in hard chaparral. *Kenny 392* (OBI 161466). Study Region: 1.

Acmispon glaber (Vogel) Brouillet var. ***glaber*** (Deerweed) – Perennial herb. Common in northern coastal scrub and hard chaparral. *Richardson* 331 (OBI 149511). Study Regions: 1, 7, 10, 12, 17, 21, 23, 24, 28, 32.

Acmispon heermannii (Durand & Hilg.) Brouillet var. ***orbicularis*** (A.Gray) Brouillet (Woolly Trefoil) – Perennial herb. Uncommon, only occurring on roadside in mixed evergreen forest in Little Creek Watershed. Historically present, though not encountered during study. Documented in West (2016). Study Region: 23.

Acmispon junceus (Benth.) Brouillet var. ***junceus*** (Rush Trefoil) – Perennial herb. Uncommon in open areas in hard chaparral. *Kenny 613* (OBI 161429). Study Regions: 1, 23.

[R]***Acmispon maritimus*** (Nutt.) D.D.Sokoloff var. ***maritimus*** (Coastal Trefoil) – Annual herb. Uncommon, only occurring on the lower reaches of the Schoolhouse Ridge. Historically present, though not encountered during study. Documented in West (2016). Study Region: 4.

Acmispon parviflorus (Benth.) D.D.Sokoloff (Small-flowered Trefoil) – Annual herb. Occasional in openings in northern coastal scrub and hard chaparral. *Kenny 454* (OBI 148058). Study Regions: 1, 4, 8, 20, 23, 32.

Acmispon strigosus (Nutt.) Brouillet (Strigose Trefoil) – Annual herb. Uncommon, only occurring near trail in hard chaparral. *Kenny 482* (OBI 161299). Study Region: 1.

Acmispon wrangelianus (Fisch. & C.A.Mey.) D.D.Sokoloff (Chilean Trefoil) – Annual herb. Uncommon in coastal prairie. *Kenny 388* (OBI 161462). Study Regions: 1, 5, 9, 17, 25, 28.

[R]***Astragalus gambelianus*** E.Sheld. (Gambel's Milkvetch, Little Blue Loco) – Annual herb. Uncommon in areas of thin soil in coastal prairie. *Kenny 416* (OBI 161480). Study Region: 7.

****Genista monspessulana*** (L.) L.A.S.Johnson (French Broom) – Shrub. Occasional in mixed evergreen forest. One of the most invasive weeds in the county (Neubauer 2013). Populations in the study area are relatively small and should be managed to prevent spread. *Richardson 54* (OBI 117314). Study Regions: 2, 3, 24, 25.

[RR]***Hosackia gracilis*** Benth. (Harlequin Lotus) (CRPR 4.2) – Perennial herb. Uncommon, only occurring in wet areas of intact coastal prairie. *Kenny 414* (OBI 161488). Study Region: 11.

****Lathyrus latifolius*** L. (Perennial Sweet Pea) – Perennial vining herb. Uncommon on roadsides. *Kenny 948* (OBI 161825). Study Regions: 3, 17, 33.

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Lathyrus vestitus Nutt. var. ***vestitus*** (Hillside Pea) – Perennial vining herb. Occasional in mixed evergreen forest. *Keil 20575* (OBI 117635). Study Region: 5.

****Lotus corniculatus*** L. (Bird's-foot Trefoil) – Perennial herb. Occasional in disturbed areas and on roadsides. *Buchloh 44* (OBI). Study Regions: 14, 20, 31, 32.

Lupinus arboreus Sims (Yellow Bush Lupine) – Shrub. Common in northern coastal scrub and coastal prairie. *Kenny 729* (OBI). Study Regions: 7, 10, 18–21, 25, 26, 28, 30–32.

Lupinus bicolor Lindl. (Miniature Lupine) – Annual herb. Occasional in coastal prairie. *Keil 20625* (OBI 108026). Study Regions: 5, 6, 28.

[R]***Lupinus formosus*** Greene var. ***formosus*** (Summer Lupine) – Perennial herb. Uncommon in coastal prairie. *Richardson 109* (OBI 108196). Study Region: 20.

Lupinus latifolius J. Agardh var. ***latifolius*** (Broad-leaved Lupine) – Perennial herb. Uncommon in openings in mixed evergreen forest. *Kenny 377* (OBI 161452). Study Regions: 1, 2, 4, 5, 23.

Lupinus nanus Douglas ex Benth. (Sky Lupine) – Annual herb. Occasional in coastal prairie. *Keil 20610* (OBI 122067). Study Regions: 1, 4–6, 20, 28.

Lupinus succulentus Douglas ex K.Koch (Arroyo Lupine) – Annual herb. Uncommon in coastal prairie. *Kenny 404* (OBI 161475). Study Region: 11.

Lupinus variicolor Steud. (Lindley's Varied Lupine) – Perennial herb. Occasional in coastal prairie. *Kenny 647* (OBI 161365). Study Regions: 6, 7, 9, 11, 17, 19, 20, 27.

****Medicago arabica*** (L.) Huds. (Spotted Burclover) – Annual herb. Uncommon on roadsides. *Kenny 963* (OBI 161836). Study Regions: 1, 2.

****Medicago lupulina*** L. (Black Medick) – Annual herb. Uncommon on roadsides. *Kenny 933* (OBI 153830). Study Regions: 1, 24.

Rupertia physodes (Douglas ex Hook.) J.W.Grimes (California Tea) – Perennial herb. Uncommon in mixed evergreen forest. *Kenny 859* (OBI 166747). Study Region: 5.

****Trifolium angustifolium*** L. (Narrow-leaved Clover) – Annual herb. Common in disturbed areas in coastal prairie. *Richardson 47* (OBI 115130). Study Regions: 1, 4, 6–12, 14, 17–21, 23, 25, 27, 28, 30–32.

Trifolium bifidum A.Gray var. ***decipiens*** Greene (Pinole Clover) – Annual herb. Occasional in openings in mixed evergreen forest. *Kenny 553* (OBI 161383). Study Regions: 4, 20.

[RR]***Trifolium buckwestiorum*** Isely (Santa Cruz Clover) (CRPR 1B.1) – Perennial herb. Uncommon, only occurring in small swale in interior coastal prairie. Historically present in study region 4, though not encountered during study. *Buck, West, Hawke, & Vigno 1* (JEPS 81528). Study Region: 4.

****Trifolium campestre*** Schreb. (Hop Clover) – Annual herb. Common in disturbed areas in coastal prairie. *Kenny 620* (OBI 161437). Study Regions: 1, 4, 9, 20, 21, 23–25, 28, 30.

Trifolium ciliolatum Benth. (Foothill Clover) – Annual herb. Occasional in coastal prairie and in openings in mixed evergreen forest. *Kenny 431* (OBI 148265). Study Regions: 1, 3, 5, 25, 28.

Trifolium depauperatum Desv. var. ***truncatum*** (Greene) J.S.Martin ex Isely (Dwarf Sack Clover) – Annual herb. Uncommon in areas of thin soil in coastal prairie. *Kenny 402* (OBI 161473). Study Regions: 14, 20.

****Trifolium dubium*** Sibth. (Little Hop Clover) – Annual herb. Common in disturbed areas in coastal prairie and on roadsides. *Kenny 619* (OBI 161434). Study Regions: 17, 23, 24, 28, 31, 32.

****Trifolium fragiferum*** L. (Strawberry Clover) – Annual herb. Occasional in disturbed areas in coastal prairie. *Kenny 964* (OBI 161837). Study Regions: 21, 27, 28, 30, 32.

****Trifolium glomeratum*** L. (Clustered Clover) – Annual herb. Uncommon in disturbed areas in coastal prairie. *Kenny 499* (OBI 161304). Study Regions: 20, 28.

Trifolium gracilentum Torr. & A.Gray (Pinpoint Clover) – Annual herb. Occasional in coastal prairie, and on roadcut in redwood forest. *Kenny 443* (OBI 161251). Study Regions: 4, 17, 20, 28, 32.

****Trifolium hirtum*** All. (Rose Clover) – Annual herb. Uncommon in disturbed areas in coastal prairie. *Kenny 541* (OBI 161375). Study Region: 25.

Trifolium macraei Hook. & Arn. (Macrae's Clover) – Annual herb. Uncommon in coastal prairie and on roadside in mixed evergreen forest. Individuals with extremely small inflorescences that primarily occur directly adjacent to the coastal bluffs may be taxonomically distinct. *Kenny 538B* (OBI 166866). Study Regions: 4, 6, 9, 11, 14, 20, 21, 25.

Trifolium microcephalum Pursh (Small-headed Clover) – Annual herb. Occasional in coastal prairie and in openings in mixed evergreen forest. *Kenny 442* (OBI). Study Regions: 1, 4, 9, 24, 26.

Trifolium microdon Hook. & Arn. (Thimble Clover) – Annual herb. Uncommon in coastal prairie. *Kenny 554* (OBI 161384). Study Regions: 4, 20.

[R]***Trifolium obtusiflorum*** Hook. (Clammy Clover) – Annual herb. Uncommon in dense riparian forest. Historically present in study region 5, though not encountered during study. *Buck & West 415* (JEPS 82876). Study Region: 5.

Trifolium oliganthum Steud. (Few-flowered Clover) – Annual herb. Uncommon on roadsides in mixed evergreen forest. Historically present in study region 5, though not encountered during study. *Keil 20637* (OBI 115654). Study Region: 5.

****Trifolium repens*** L. (White Clover) – Perennial herb. Common in wet areas in coastal prairie. *Kenny 381* (OBI 161455). Study Regions: 1, 5, 19, 27, 30, 31.

****Trifolium striatum*** L. (Knotted Clover) – Annual herb. Occasional in disturbed areas in coastal prairie. *Kenny 696* (OBI 166684). Study Regions: 1, 4, 19, 20, 25, 30.

****Trifolium subterraneum*** L. (Subterranean Clover) – Annual herb. Occasional in disturbed areas in coastal prairie. *Kenny 259* (OBI 171074). Study Regions: 9, 25, 28.

Trifolium variegatum Nutt. (Variegated Clover) – Annual herb. Uncommon, only occurring in opening in mixed evergreen forest. *Kenny 832* (OBI 166695). Study Region: 2.

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Trifolium willdenovii Spreng. (Tomcat Clover) – Annual herb. Occasional in openings in mixed evergreen forest. *Kenny 375* (OBI 161482). Study Regions: 1, 2, 5, 20, 24, 25.

[R]***Trifolium wormskioldii*** Lehm. (Cow Clover) – Perennial herb. Uncommon in wet areas in coastal prairie. Historically present, though not encountered during study. Documented in West (2016) as occurring in a perennially moist swale in study region 27. Study Region: 27.

Vicia americana Muhl. Ex Willd. Subsp. ***americana*** (American Vetch) – Perennial herb. Occasional in wet openings in mixed evergreen forest. *Kenny 253* (OBI 171068). Study Regions: 23, 25, 32.

Vicia gigantea Hook. (Giant Vetch) – Perennial herb. Occasional on wet slopes in mixed evergreen forest. *Richardson 193* (OBI 115982). Study Regions: 5, 10, 23–25, 28, 31.

[R]***Vicia hassei*** S.Watson (Slender Vetch) – Annual herb. Uncommon in matrix of mixed evergreen forest and coastal prairie. Historically present in study region 25, though not encountered during study. *Keil 30758* (OBI 128994). Study Region: 25.

****Vicia hirsuta*** (L.) Gray (Hairy Vetch) – Annual herb. Uncommon in mosaic of mixed evergreen forest and coastal prairie. Historically present in study region 25, though not encountered during study. *Keil 30757* (OBI 128992). Study Region: 25.

****Vicia lutea*** L. (Yellow Vetch) – Annual herb. Uncommon, only occurring on roadside. *Keil 20635* (OBI 107004). Study Regions: 1, 2, 5.

****Vicia sativa*** L. subsp. ***nigra*** (L.) Ehrh. (Narrow-leaved Vetch) – Annual herb. Occasional on margins of openings in mixed evergreen forest. *Kenny 261* (OBI 171076). Study Region: 25.

****Vicia tetrasperma*** (L.) Schreb. (Sparrow Vetch) – Annual herb. Common in disturbed areas throughout study area. *Kenny 492* (OBI 161287). Study Regions: 2, 5, 7, 8, 10, 11, 13, 17, 24, 26, 30.

FAGACEAE

Notholithocarpus densiflorus (Hook. & Arn.) Manos, Cannon & S.H.Oh var. ***densiflorus*** (Tanoak) – Evergreen tree. Common. Occurring in almost pure stands or occasional in mixed evergreen forest. *Richardson 281* (OBI 149298). Study Regions: 2, 3, 5, 23–25.

Quercus agrifolia Née var. ***agrifolia*** (Coast Live Oak) – Evergreen tree. Common on south- and west-facing slopes and in mixed evergreen forest. *Keil 20658* (RSA 659249). Study Regions: 1, 5–7, 20, 23, 25, 30–32.

Quercus chrysolepis Liebm. (Canyon Live Oak) – Evergreen tree. Uncommon in hard chaparral, occurring in shrub form. *Kenny 117* (OBI 161407). Study Region: 1.

Quercus parvula Greene var. ***shrevei*** (C.H.Mull) Nixon (Shreve Oak) – Evergreen tree. Common in mixed evergreen forest. *Keil 20641* (OBI 112618). Study Regions: 1, 3–5, 10, 13, 18–20, 26, 32.

FRANKENIACEAE

Frankenia salina (Molina) I.M.Johnst. (Alkali Heath) – Perennial herb. Common in the Scott Creek marsh. *Buchloh 30*. Study Region: 31.

GENTIANACEAE

Cicendia quadrangularis (Lam.) Griseb. (Timwort) – Annual herb. Uncommon on areas of bare soil near seeps. *Kenny 258* (OBI 171073). Study Region: 25.

Zeltnera davyi (Jeps.) G.Mans. (Davy's Centaury) – Annual herb. Common in northern coastal scrub and coastal prairie. *Kenny 670* (OBI 161508). Study Regions: 10, 11, 13, 17, 19, 27, 31.

Zeltnera muehlenbergii (Griseb.) G.Mans. (Monterey Centaury) – Annual herb. Common in northern coastal scrub and coastal prairie. Specimens have been mistaken for *Centaurium tenuiflorum* (Hoffmanns. & Link) Fritsch. *Kenny 773* (OBI 166698). Study Regions: 11, 17, 20, 21, 30.

GERANIACEAE

****Erodium cicutarium*** (L.) L'Hér. (Redstem Filaree) – Annual herb. Common in disturbed areas in coastal prairie. *Kenny 975* (OBI 166659). Study Regions: 6, 8, 9, 17, 19, 28.

****Erodium moschatum*** L'Hér. (Greenstem Filaree) – Annual herb. Common in disturbed areas in coastal prairie. *Kenny 854* (OBI 166752). Study Regions: 1, 6, 7, 19, 21, 26, 27, 30, 32.

****Geranium core-core*** Steud. (New Zealand Geranium) – Perennial herb. Uncommon, only occurring on roadside. *Kenny 743* (OBI 161855). Study Region: 3.

****Geranium dissectum*** L. (Common Cranesbill) – Annual herb. Common in disturbed areas throughout study area. *Buchloh 16*. Study Regions: 2, 4, 6, 9, 10, 17, 20, 23, 25, 28, 30–32.

****Geranium molle*** L. (Soft Cranesbill) – Annual herb. Uncommon on roadside. *Kenny 826* (OBI 158471). Study Region: 25.

****Geranium robertianum*** L. (Herb Robert) – Annual herb. Uncommon on roadside in redwood forest. *Kenny 825* (OBI 173144). Study Regions: 5, 25.

****Geranium rotundifolium*** L. (Round-leaved Geranium) – Annual herb. Uncommon on train tracks in riparian forest. *Kenny 881* (OBI 173060). Study Regions: 25, 26.

GROSSULARIACEAE

Ribes divaricatum Douglas var. ***pubiflorum*** Koehne (Straggle Bush) – Caning shrub. Occasional in riparian and marshy areas. *Kenny 752* (OBI 166706). Study Regions: 2, 5, 10, 11, 13, 15.

Ribes menziesii Pursh var. ***menziesii*** (Canyon Gooseberry) – Shrub. Occasional in mixed evergreen forest. *Kenny 103* (OBI 161400). Study Regions: 5, 23–25, 31, 32.

HYPERICACEAE

Hypericum anagalloides Cham. & Schltdl. (Tinker's Penny) – Annual to perennial herb. Occasional in freshwater marshes. *Kenny 659* (OBI 161443). Study Regions: 9, 10.

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****Hypericum perforatum*** L. subsp. ***perforatum*** (Klamath-weed) – Perennial herb. Uncommon in disturbed northern coastal scrub. *Richardson 89* (OBI 152226). Study Region: 26.

Lamiaceae

Clinopodium douglasii (Benth.) Kuntze (Yerba Buena) – Perennial herb. Common in northern coastal scrub, mixed evergreen forest, and hard chaparral. *Richardson 65* (OBI 107443). Study Regions: 5, 9, 10, 13, 17, 18, 20, 21, 23–25, 28, 30, 32.

****Glechoma hederacea*** L. (Ground Ivy) – Perennial herb. Uncommon in riparian forest understory. *Kenny 837* (OBI 166745). Study Region: 5.

****Lamium purpureum*** L. (Dead Nettle) – Annual herb. Uncommon in disturbed areas in riparian forest. *Kenny 132* (OBI 161414). Study Regions: 2, 5, 25.

Lepechinia calycina (Benth.) Epling (Pitcher Sage) – Shrub. Uncommon in hard chaparral. *Kenny 455*. Study Region: 1.

****Melissa officinalis*** L. (Lemon Balm) – Perennial herb. Uncommon in disturbed riparian areas. *Kenny 983* (OBI 167657). Study Regions: 2, 3.

****Mentha*** L. (Mint) – Perennial herb. Occasional in disturbed areas throughout study area. Likely a hybrid, as it does not key to species. *Kenny 998* (OBI 158577). Study Regions: 25, 26.

Monardella villosa Benth. subsp. ***franciscana*** (Elmer) Jokerst (Franciscan Coyote Mint) – Perennial herb. Uncommon in northern coastal scrub near coast. *Kenny 903* (OBI 171051). Study Regions: 1, 4, 11, 12, 14, 17, 21, 33.

Monardella villosa Benth. subsp. ***villosa*** (Coyote-mint) – Perennial herb. Occasional in northern coastal scrub and soft chaparral. *Richardson 77* (OBI 149288). Study Regions: 1, 2, 5, 10, 23.

[R]***Pogogyne serpylloides*** (Torr.) A.Gray (Thymeleaf Beard-style) – Annual herb. Uncommon, occurring only in one vernally moist, sandy area in coastal prairie. Historically present, though not encountered during study. *West 139* (JEPS 82600). Study Region: 8.

Prunella vulgaris L. var. ***vulgaris*** (Self-heal) – Perennial herb. Occasional in moist, disturbed areas. *Richardson 123* (OBI 119079). Study Regions: 2, 5, 26.

Salvia columbariae Benth. (Chia) – Annual herb. Uncommon, occurring only on one south-facing cliff in coastal gully. *Kenny 524* (OBI 161323). Study Region: 9.

Scutellaria tuberosa Benth. (Dannie's Skullcap) – Perennial herb from tuber. Uncommon in openings in mixed evergreen forest. *Kenny 389* (OBI 161463). Study Region: 1.

Stachys ajugoides Benth. (Bugle Hedge Nettle) – Perennial herb. Uncommon in wet, marshy area of coastal prairie. *Kenny 531* (OBI 161328). Study Region: 6.

Stachys bullata Benth. (Hedge Nettle) – Perennial herb. Common in northern coastal scrub, riparian forest, and mixed evergreen forest. *Richardson 18* (OBI 111132). Study Regions: 1–3, 5, 7, 10, 11, 13, 17, 18, 20, 21, 23–26, 28, 31, 32.

[R]***Stachys chamissonis*** Benth. (Swamp Stachys) – Perennial herb. Uncommon, occurring only in one marshy area in coastal prairie. *Reed Kenny 950* (OBI 174310). Study Region: 14.

Stachys rigida **Nutt. ex Benth.** var. ***quercetorum*** (A.Heller) G.A.Mulligan & D.B.Munro (Rigid Hedge Nettle) – Perennial herb. Uncommon in redwood forest. *Kenny 615* (OBI 161430). Study Region: 23.

Linaceae

****Linum bienne*** Mill. (Narrow-leaved flax) – Annual herb. Common in coastal prairie. Cultivated in study area during the mid-20th century. *Richardson 44* (OBI 186124). Study Regions: 4–12, 14, 17, 19–21, 25, 27, 28, 30–32.

Lythraceae

****Lythrum hyssopifolia*** L. (Hyssop Loosestrife) – Annual to perennial herb. Common in wet areas of coastal prairie *Buchloh 27* (OBI 104536). Study Regions: 6–8, 10, 11, 13, 14, 17–21, 25, 27, 28, 30, 31.

Malvaceae

****Malva parviflora*** L. (Cheeseweed) – Annual herb. Uncommon in disturbed lawn area. *Kenny 941* (OBI 166715). Study Region: 26.

****Modiola caroliniana*** (L.) G.Don (Modiola) – Perennial herb. Uncommon on roadside in oak woodland. *Kenny 506* (OBI 161309). Study Region: 5.

Sidalcea malviflora (DC.) A.Gray subsp. ***malviflora*** (Checkerbloom) – Perennial herb. Occasional in coastal prairie. *Kenny 270* (OBI 161275). Study Regions: 9–12, 17, 21, 28.

Montiaceae

Calandrinia menziesii (Hook.) Torr. & A.Gray (Red Maids) – Annual herb. Uncommon in coastal prairie. *Kenny 542* (OBI 161377). Study Regions: 17, 25.

[R]***Claytonia exigua*** Torr. & A.Gray subsp. ***exigua*** (Little Spring Beauty) – Annual herb. Uncommon in areas of thin soil in coastal prairie. Historically present, though not encountered during study. Documented in West (2016) as occurring in study region 20 in 2013. Study Region: 20.

Claytonia perfoliata Donn ex Willd. (Miner's Lettuce) – Annual herb. Occasional in mixed evergreen forest. *Kenny 887* (OBI 166723). Study Regions: 1, 3, 9, 18, 20, 23–25, 28.

[R]***Claytonia sibirica*** L. (Candy Flower) – Perennial herb. Uncommon in redwood forest, locally abundant on roadside in study region 25. *Kenny 206A* (OBI 148051). Study Regions: 23, 25, 26.

Montia parvifolia (Moc. ex DC.) Greene (Small-leaved Montia) – Perennial herb. Uncommon, occurring only on cliff face near riparian area in mixed evergreen forest. *Kenny 863* (OBI 147809). Study Region: 5.

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Myricaceae

Morella californica (Cham. & Schltdl.) Wilbur (Wax Myrtle) – Shrub. Uncommon in wet areas in coastal prairie. *Kenny 656* (OBI 161502). Study Regions: 10, 13.

Myrsinaceae

****Lysimachia arvensis*** (L.) U.Manns & Anderb. (Scarlet Pimpernel) – Annual herb. Common in coastal prairie. *Kenny 437* (OBI 148243). Study Regions: 1, 2, 4–6, 8–12, 14, 17, 19–21, 23, 25, 27, 28, 30–32.

Lysimachia latifolia (Hook.) Cholewa (Pacific Starflower) – Perennial herb. Occasional in mixed evergreen forest. *Kenny 477* (OBI 174303). Study Regions: 1, 23–25.

Lysimachia minima (L.) U.Manns & Anderb. (Chaff-weed) – Annual herb. Uncommon in vernally wet, sandy area in coastal prairie. *Kenny 527* (OBI 161326). Study Region: 6.

Oleaceae

****Ligustrum lucidum*** W.T.Aiton (Chinese Privet) – Shrub. Uncommon, a few individuals naturalized in one drainage. *Kenny 989* (OBI). Study Region: 24.

Onagraceae

[R]***Clarkia davyi*** (Jeps.) F.H.Lewis & M.E.Lewis (Davy's Clarkia) – Annual herb. Uncommon in coastal prairie, primarily in areas of thin soil or on steep sides of coastal gullies. Two distinct forms occur in study area, one strongly prostrate, one strongly erect. *Kenny 699* (OBI 158570). Study Regions: 6, 9, 11, 19.

Clarkia purpurea subsp. ***quadrivulnera*** (Lindl.) H.Lewis & M.Lewis (Four-spot) – Annual herb. Uncommon in coastal prairie. *Kenny 704* (OBI). Study Region: 4.

Clarkia rubicunda (Lindl.) F.H.Lewis & M.E.Lewis (Farewell-to-spring) – Annual herb. Common in coastal prairie. *Kenny 703* (OBI). Study Regions: 1, 2, 4, 7, 10, 13, 19, 21, 24, 32, 33.

Epilobium brachycarpum C.Presl (Panicled Willow Herb) – Annual herb. Occasional in disturbed areas. *Kenny 972* (OBI 161845). Study Regions: 26, 32.

Epilobium canum (Greene) P.H.Raven subsp. ***canum*** (California Fuchsia) – Perennial herb. Occasional in northern coastal scrub and coastal prairie. *Kenny 976* (OBI 161848). Study Regions: 7, 10, 28, 30, 31.

Epilobium ciliatum Raf. subsp. ***ciliatum*** (Common Willow Herb) – Perennial herb. Common in wet areas in coastal prairie. *Kenny 631* (OBI 161353). Study Regions: 1, 2, 5, 10, 13, 17, 20, 21, 24–26, 28, 31.

Epilobium ciliatum Raf. subsp. ***watsonii*** (Barbey) Hoch & P.H.Raven (Watson's Willow Herb) – Perennial herb. Uncommon in wet areas in coastal prairie. *Richardson 24* (OBI 129328). Study Regions: 18, 24.

Taraxia ovata (Nutt.) Small (Sun Cup) – Perennial herb. Occasional in coastal prairie. *Kenny 107* (OBI 161396). Study Regions: 4, 9, 27.

Orobanchaceae

Aphyllon fasciculatum (Nutt.) Torr. & A.Gray (Clustered Broomrape) – Perennial herb. Uncommon in northern coastal scrub, parasitic on *Eriogonum latifolium*. *Kenny 1000* (OBI 149749). Study Region: 7.

****Bellardia trixago*** (L.) All. (Mediterranean Linseed) – Annual herb. Occasional in disturbed areas in coastal prairie. *Kenny 721* (OBI 158442). Study Regions: 17, 32.

Castilleja affinis Hook. & Arn. subsp. ***affinis*** (Indian Paintbrush) – Perennial herb. Uncommon in northern coastal scrub. *Kenny 517* (OBI 161317). Study Regions: 5, 9, 10, 12, 13, 17, 20, 21, 28, 33.

Castilleja densiflora (Benth.) T.I.Chuang & Heckard subsp. ***densiflora*** (Owl's Clover) – Annual herb. Uncommon in coastal prairie. Individuals found on coastal headlands with cream colored bract tips and a vanilla scent may fit into the former taxon *Orthocarpus densiflorus* Benth. var. *noctuinus* (Eastw.) J.T.Howell, while populations in study regions 1 and 4 are typical of *C. densiflora* subsp. *densiflora*. *Kenny 458* (OBI 161261). Study Regions: 4, 19, 25.

[R]***Castilleja exserta*** (A.Heller) T.I.Chuang & Heckard subsp. ***latifolia*** (S.Watson) T.I.Chuang & Heckard (Purple Owl's Clover) – Annual herb. Uncommon in area of thin soil in coastal prairie. Historically present in Study Region 9, though not encountered during study. *West 165* (JEPS 82594). Study Region: 9.

****Parentucellia viscosa*** (L.) Caruel (Yellow Parentucellia) – Annual herb. Uncommon in disturbed areas in coastal prairie. *Kenny 889* (OBI 166731). Study Region: 18.

Triphysaria eriantha (Benth.) T.I.Chuang & Heckard subsp. ***rosea*** (A.Gray) T.I.Chuang & Heckard (Pelican Flower) – Annual herb. Uncommon in coastal prairie. Historically present, though not encountered during study. Documented in West (2016) as occurring in Study Region 9. Study Region: 9.

Triphysaria pusilla (Benth.) T.I.Chuang & Heckard (Dwarf Owl's Clover) – Annual herb. Uncommon in coastal prairie. *Kenny 180* (OBI 161494). Study Region: 20.

Oxalidaceae

Oxalis oregana Nutt. (Redwood Sorrel) – Perennial herb. Common in redwood forest. *Richardson 17* (OBI 103455). Study Regions: 2, 3, 5, 23, 24.

****Oxalis pes-caprae*** L. (Bermuda Buttercup) – Perennial herb. Common in disturbed areas in coastal prairie. This species can invade cliff faces and other areas that often harbor rare natives. Efforts should be made to contain its spread. *Kenny 683* (OBI 161340). Study Regions: 17, 23, 28.

Oxalis pilosa Nutt. (Hairy Wood-sorrel) – Perennial herb. Common in disturbed areas in mixed evergreen forest. *Kenny 188* (OBI 161493). Study Regions: 4, 5, 9, 19, 21, 23, 27, 32.

****Oxalis purpurea*** L. (Purple Wood-sorrel) – Perennial herb. Uncommon, only occurring on roadside in redwood forest. *Kenny 994* (OBI 153832). Study Region: 3.

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PAPAVERACEAE

Dendromecon rigida Benth. (Bush Poppy) – Shrub. Uncommon in hard chaparral. *Kenny 116* (OBI 161406). Study Regions: 1, 23.

Dicentra formosa (Haw.) Walp. (Pacific Bleeding Heart) – Perennial herb. Occasional in mixed evergreen forest. *Kenny 497* (OBI 161290). Study Regions: 5, 25.

Eschscholzia californica Cham. (California Poppy) – Perennial herb. Common in coastal prairie. *Richardson 51* (OBI 103893). Study Regions: 1, 2, 4, 6, 7, 10, 12, 17, 19–21, 25–28, 31–33.

****Fumaria capreolata*** L. (White Ramping Fumitory) – Perennial herb. Uncommon, only occurring on roadside in mixed evergreen forest. *Kenny 370* (OBI 161446). Study Region: 1.

Platystemon californicus Benth. (Cream Cups) – Annual herb. Uncommon on exposed mudstone surfaces in coastal prairie. *Kenny 278* (OBI 161279). Study Regions: 21, 25.

PHRYMACEAE

Diplacus aurantiacus (Curtis) Jeps. (Orange Bush Monkeyflower) – Shrub. Common in northern coastal scrub. Individuals occurring on coastal bluffs with varying red to orange corollas appearing intermediate with *D. longiflorus* Nutt. Presumably, pollen of *D. longiflorus* was introduced at some point though its origin is unknown. *Kenny 839* (OBI 166743). Study Regions: 1, 3–5, 7, 8, 10, 12, 13, 17–21, 23–25, 28, 31, 32.

Erythranthe arvensis (Greene) G.L.Nesom (Villous-bracted Monkeyflower) – Annual herb. uncommon in seeps in coastal prairie and in wet areas in mixed evergreen forest. *Kenny 701* (OBI 174299). Study Region: 26.

Erythranthe cardinalis (Douglas ex Benth.) Spach (Scarlet Monkeyflower) – Perennial herb. Occasional in the Scott Creek riparian corridor. *Kenny 896* (OBI 166672). Study Regions: 5, 26.

Erythranthe floribunda (Lindl.) G.L.Nesom (Many-flowered Monkeyflower) – Annual herb. Uncommon on gravel bars in the Scott Creek riparian corridor. *Kenny 878* (OBI 166724). Study Regions: 5, 20, 25, 31.

Erythranthe grandis (Greene) G.L.Nesom (Magnificent Seep Monkeyflower) – Perennial herb. Occasional in wet areas in coastal prairie. *Kenny 693* (OBI 171064). Study Regions: 13, 14, 17, 18.

Erythranthe guttata? (Greene) G.L.Nesom (Common Monkeyflower) – Perennial herb. Uncommon in the Scott Creek riparian corridor. Plants stoloniferous with more or less glabrous leaves, but corolla exserted much more than 5 mm beyond calyx. Specimens in county and study area need further study. *Richardson 140* (OBI 138488). Study Region: 5.

Erythranthe microphylla (Benth.) G.L.Nesom (Small-leaved Monkeyflower) – Annual herb. Uncommon in wet areas in mixed evergreen forest and growing in willows. Not in Neubauer (2013). *Kenny 493* (OBI 161288). Study Region: 26.

Erythranthe nasuta (Greene) G.L.Nesom (Snout-nosed Monkeyflower) – Annual herb. Uncommon, only occurring in the Scott Creek riparian corridor. *Kenny 880* (OBI 166726). Study Region: 26

Erythranthe ptilota G.L.Nesom (Wing-leaf Monkeyflower) – Perennial herb. Occasional, occurring in the Scott Creek riparian corridor. Historic specimens synonymous to *E. moschata* (Douglas ex Lindl.) G.L.Nesom are likely to be *E. ptilota* and should be examined by an expert. *Kenny 834* (OBI 1666697). Study Region: 26.

PLANTAGINACEAE

Antirrhinum kelloggii Greene (Twining Snapdragon) – Annual herb. Uncommon, only occurring on a steep slope in opening in mixed evergreen forest. *Kenny 385* (OBI 161459). Study Regions: 1, 8.

Callitriche heterophylla Pursh var. ***bolanderi*** (Hegelm.) Fassett (Bolander's Woolly Starwort) – Perennial herb. Uncommon in stock ponds. Historically present, though not encountered during study. Documented in West (2016). Study Region: 10.

Callitriche marginata Torr. (California Water-starwort) – Annual herb. Occasional in ponds and marshy areas. *Kenny 466* (OBI 1612670. Study Regions: 1, 7.

Collinsia heterophylla Buist ex Graham var. ***heterophylla*** (Chinese Houses) – Annual herb. Uncommon on steep, exposed mudstone slopes. *Kenny 831* (OBI 166696). Study Regions: 1–3, 26.

[RR]***Collinsia multicolor*** Lindl. & Paxton (San Francisco Collinsia) (CRPR 1B.2) – Annual herb. Uncommon, only occurring on one exposed mudstone cliff face in mixed evergreen forest. *Kenny 495* (OBI 161289). Study Regions: 1, 3, 5, 33.

****Cymbalaria muralis*** G.Gaertn., B.Mey. & Scherb. (Kenilworth Ivy) – Perennial herb. Uncommon in disturbed areas in coastal prairie. *Kenny 711* (OBI 161416). Study Region: 4.

****Kickxia elatine*** (L.) Dumort. (Sharp-leaved Fluellin) – Annual to perennial herb. Uncommon in disturbed meadows and coastal prairie. *Kenny 995* (OBI 158581). Study Regions: 1, 2.

Nuttallanthus texanus (Scheele) D.A.Sutton (Blue Toadflax) – Annual herb. Uncommon on exposed mudstone slopes in coastal prairie and in openings in mixed evergreen forest. *Kenny 387* (OBI 161461). Study Regions: 1, 9.

****Plantago coronopus*** L. (Cut-leaved Plantain) – Perennial herb. Common in disturbed areas in coastal prairie. *Kenny 691* (OBI 161343). Study Regions: 1, 17, 19, 26–28, 30, 31.

Plantago erecta E.Morris (California Plantain) – Annual herb. Occaisional in coastal prairie. Plants near the coast tend to be larger with more succulent leaves than plants inland. *Kenny 755* (OBI 166707). Study Regions: 12, 13, 28.

****Plantago lanceolata*** L. (English Plantain) – Perennial herb. Common in coastal prairie. *Richardson 52* (OBI 6032). Study Regions: 1, 2, 4, 6–8, 10–12, 14, 18–21, 24, 25, 27, 28, 30–32.

****Plantago major*** L. (Common Plantain) – Perennial herb. Uncommon in wet, disturbed areas. *Kenny 687* (OBI 124214). Study Regions: 5, 26.

Plantago maritima L. (Common Seaside Plantain) – Perennial herb. Uncommon, found only in sandy, eroding

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soil on coastal bluffs. *Kenny 637* (OBI 161356). Study Region: 17.

Plantago subnuda Pilg. (Mexican Plantain) – Perennial herb. Occasional in wet areas in coastal prairie. *Kenny 937* (OBI 166665). Study Regions: 24, 27, 28.

Veronica americana Schwein. ex Benth. (American Brooklime) – Perennial herb. Occasional in marshy and riparian areas. *Kenny 636* (OBI 161355). Study Regions: 2, 5, 10, 13, 14, 18, 20.

Veronica serpyllifolia L. (Bright-blue Speedwell) – Perennial herb. Uncommon, only occurring on SPR side of Mill Creek, historically present, though not encountered during study. *Morgan 3748* (UCSC 3115). Study Region: 5.

Plumbaginaceae

Armeria maritima (Mill.) Willd. subsp. ***californica*** (Boiss.) A.E.Porsild (Sea Pink) – Perennial herb. Uncommon in northernmost coastal prairie. *Kenny 663* (OBI 161505). Study Regions: 5, 11.

Polemoniaceae

[R]***Allophyllum divaricatum*** (Nutt.) A.D.Grant & V.E.Grant (Purple False Gilia) – Annual herb. Uncommon, only occurring on sandbank in Scott Creek. Historically present in Study Region 26, though not encountered during study. *Buck & West 108* (JEPS81555). Study Region: 26.

Collomia heterophylla Douglas ex Hook. (Variable Leaf Collomia) – Annual herb. Occasional in openings in mixed evergreen forest. *Kenny 436* (OBI 148254). Study Regions: 1, 2, 23, 25.

Gilia achilleifolia Benth. subsp. ***achilleifolia*** (California Gilia) – Annual herb. Occasional in openings in mixed evergreen forest and hard chaparral. *Kenny 386* (OBI 161460). Study Regions: 1, 2, 8, 25.

[R]***Gilia achilleifolia*** Benth. subsp. ***multicaulis*** (Benth.) V.E.Grant & A.D.Grant (Many-stemmed Gilia) – Annual herb. Uncommon in mixed evergreen forest. *Kenny 717* (OBI). Study Regions: 4, 25.

[R]***Gilia clivorum*** (Jeps.) V.E.Grant (Grassland Gilia) – Annual herb. Uncommon, only occurring in area of thin soil in coastal prairie. Historically present in Study Region 2, though not encountered during study. *Buck & West 210* (JEPS 82571). Study Region: 2.

Leptosiphon androsaceus Benth. (Common Leptosiphon) – Annual herb. Uncommon in openings in mixed evergreen forest. *Kenny 510* (OBI 161312). Study Region: 1.

Navarretia squarrosa (Eschsch.) Hook. & Arn. (Skunkweed) – Annual herb. Common in northern coastal scrub. *Richardson 31* (OBI 127078). Study Regions: 1, 2, 10, 17, 19, 20, 23, 24, 28, 30–32.

Polygalaceae

Polygala californica Nutt. (California Milkwort) – Perennial herb. Occasional in mixed evergreen forest. *Kenny 479* (OBI 161297). Study Regions: 1, 23, 24.

Polygonaceae

Eriogonum latifolium Sm. (Seaside Wild Buckwheat) – Perennial herb. Common in northern coastal scrub. *Richardson 95* (OBI 127840). Study Regions: 7, 9–12, 17, 20, 21, 28, 32.

Eriogonum nudum Benth. var. ***auriculatum*** (Benth.) Jeps. (Naked-stemmed Buckwheat) – Perennial herb. Uncommon, only occurring on slide area in mixed evergreen forest. *Kenny 958* (OBI 161833). Study Region: 1.

Persicaria amphibia (L.) Delarbre (Water Smartweed) – Perennial herb. Common in the Scott Creek marsh. *Kenny 866* (OBI 166716). Study Region: 31.

****Persicaria capitata*** (Buch.-Ham. ex D.Don) H.Gross (Himalayan Smartweed) – Perennial herb. Uncommon in riparian forest understory. Not in Neubauer (2013). *Kenny 971* (OBI 161844). Study Region: 26.

****Persicaria maculosa*** Gray (Lady's Thumb) – Perennial herb. Uncommon in riparian forest. *Kenny 969* (OBI 161842). Study Region: 26.

Persicaria punctata (Elliott) Small (Water Knotweed) – Perennial herb. Common in the Scott Creek riparian corridor. *Richardson 158* (OBI 150677). Study Regions: 5, 13, 14, 26, 31.

****Polygonum aviculare*** L. subsp. ***depressum*** (Meisn.) Arcang. (Common Knotweed) – Perennial herb. Common in disturbed areas on roadsides. *Kenny 942* (OBI 166662). Study Regions: 18, 19, 25–27, 31.

Pterostegia drymarioides Fisch. & C.A.Mey. (Woodland Threadstem) – Annual herb. Uncommon in openings in mixed evergreen forest. *Kenny 472* (OBI 161292). Study Regions: 1, 17, 28.

****Rumex acetosella*** L. (Sheep Sorrel) – Perennial herb. Common in disturbed areas in coastal prairie. *Richardson 57* (OBI 150883). Study Regions: 1, 2, 4, 6–14, 17, 19–21, 24, 25, 27, 28, 30–33.

[R]***Rumex californicus*** Rech. f. (California Dock) – Perennial herb. Occasional in wet areas in northern coastal scrub and coastal prairie. *Richardson 82* (OBI 150846). Study Regions: 5, 25, 26, 33.

****Rumex conglomeratus*** Murray (Whorled Dock) – Perennial herb. Common in disturbed areas in coastal prairie and on roadsides throughout study area. *Kenny 982* (OBI 167668). Study Regions: 1–3, 5, 14, 18, 25, 31.

Rumex crassus Rech. f. (Leatherleaf Dock) – Perennial herb. Uncommon on coastal bluffs and in the Scott Creek marsh. *Kenny 1003* (OBI 158260). Study Regions: 17, 31.

****Rumex crispus*** L. (Curly Dock) – Perennial herb. Common in disturbed areas in coastal prairie. *Buchloh 42* (OBI). Study Regions: 1, 2, 7, 8, 11, 20, 21, 27, 28, 30, 31, 33.

****Rumex obtusifolius*** L. (Bitter Dock) – Perennial herb. Uncommon in the Scott Creek riparian corridor. *Kenny 986* (OBI 179970). Study Region: 5.

[R]***Rumex occidentalis*** S.Watson (Western Dock) – Perennial herb. Uncommon on coastal bluffs. *Kenny 900* (OBI 171053). Study Regions: 2, 6, 17.

****Rumex pulcher*** L. (Fiddle Dock) – Perennial herb. Common in disturbed areas in coastal prairie. *Buchloh 37*

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(OBI 170138). Study Regions: 5, 6, 8, 11, 19–21, 25, 27, 28, 30–33.

Rumex salicifolius Weinm. (Willow-leaved Dock) – Perennial herb. Occasional in wet areas on coastal bluffs and in coastal prairie. *Kenny 745* (OBI 161856). Study Regions: 17, 26, 28, 31.

[R]***Rumex transitorius*** Rech. f. (Willow Dock) – Perennial herb. Occasional in riparian forest. *Kenny 725* (OBI). Study Regions: 2, 4, 5, 8, 10, 26, 31, 33.

PRIMULACEAE

Primula clevelandii var. ***gracilis*** (Greene) A.R.Mast & Reveal (Padre's Shooting Star) – Perennial herb. Uncommon in interior coastal prairie. *Kenny 115* (OBI 161405). Study Region: 4.

Primula hendersonii (A.Gray) Mast & Reveal (Mosquito Bills) – Perennial herb. Uncommon on eroded mudstone. Historically present in study region 20, though not encountered during study. *West 347* (JEPS 83088). Study Region: 20.

RANUNCULACEAE

Actaea rubra (Aiton) Willd. (Baneberry) – Perennial herb. Occasional in riparian forest. *Kenny 498* (OBI 161291). Study Regions: 5, 25.

Anemone grayi Behr & Kellogg (Windflower) – Perennial herb. Uncommon, one occurrence of ca. 100 plants on a steep, north-facing slope in mixed evergreen forest. *Kenny 766* (OBI 166713). Study Region: 24.

Aquilegia formosa Fisch. ex DC. (Western Columbine) – Perennial herb. Uncommon in mixed evergreen forest. *Kenny 204* (OBI 148054). Study Region: 23.

Clematis lasiantha Nutt. (Chaparral Clematis) – Perennial herb. Uncommon in mixed evergreen forest. *Kenny 760* (OBI 166711). Study Region: 24.

[R]***Delphinium californicum*** Torr. & A.Gray subsp. ***californicum*** (California Larkspur) – Perennial herb. Uncommon on roadsides in northern coastal scrub and mixed evergreen forest *Kenny 829* (OBI 148282). Study Regions: 26, 33.

[R]***Delphinium decorum*** Fisch. & C.A.Mey. subsp. ***decorum*** (Coast Larkspur) – Perennial herb. Uncommon, two occurences of ca. 100 plants found in areas of thin soil on north-facing slopes of gullies in coastal prairie. *Kenny 286* (OBI 161389). Study Region: 21.

Delphinium patens Benth. subsp. ***patens*** (Woodland Larkspur) – Perennial herb. Uncommon in slide area in mixed evergreen forest. *Kenny 374* (OBI 161450). Study Regions: 1, 2.

[R]***Myosurus minimus*** L. (Common Mousetail) – Annual herb. Uncommon on banks of the Scott Creek marsh. *Kenny 980* (OBI 167641). Study Regions: 24, 31.

Ranunculus aquatilis L. var. ***diffusus*** With. (Water Buttercup) – Perennial herb. Uncommon in the Scott Creek riparian corridor. *Kenny 667* (OBI 161507). Study Region: 5.

Ranunculus californicus Benth. var. ***californicus*** (Western Buttercup) – Perennial herb. Uncommon in coastal prairie. *Kenny 288* (OBI 161418). Study Region: 27.

Ranunculus hebecarpus Hook. & Arn. (Downy Buttercup) – Annual herb. Uncommon in wet areas in mixed evergreen forest. *Kenny 252* (OBI 171067). Study Regions: 4, 25.

****Ranunculus muricatus*** L. (Prickle-fruited Buttercup) – Annual to biennial herb. Uncommon, only occurring in wet, disturbed soil in trail. *Kenny 439* (OBI). Study Regions: 1, 23.

Ranunculus parviflorus L. (Few-flowered Buttercup) – Annual herb. Uncommon, only occurring on roadside in mixed evergreen forest. Historically present, though not encountered during study. Personal observation by James A. West. Study Region: NA.

****Ranunculus repens*** L. (Crowfoot) – Perennial herb. Occasional in disturbed riparian areas. *Kenny 640* (OBI 161359). Study Regions: 2, 5, 26.

[R]***Ranunculus uncinatus*** D.Don ex G.Don (Barbed Buttercup) – Perennial herb. Uncommon in wet areas in mixed evergreen forest. *Kenny 503* (OBI 166687). Study Regions: 5, 23, 25.

Thalictrum fendleri Engelm. ex A.Gray var. ***polycarpum*** Torr. (Meadow Rue) – Perennial herb. Occasional in riparian forest. *Kenny 193* (OBI 161526). Study Regions: 5, 26.

RHAMNACEAE

Ceanothus thyrsiflorus Eschsch. var. ***thyrsiflorus*** (Blue Blossom) – Shrub. Common in mixed evergreen forest. *Richardson 34* (OBI 124425). Study Regions: 1, 3, 23–25, 32, 33.

Frangula californica (Eschsch.) A.Gray subsp. ***californica*** (California Coffee Berry) – Shrub. Common in northern coastal scrub and mixed evergreen forest. *Richardson 306* (OBI 149501). Study Regions: 1–3, 5, 10, 12, 18–21, 25, 27, 28, 32.

ROSACEAE

Acaena pinnatifida Ruiz & Pav. var. ***californica*** (Bitter) Jeps. (California Acaena) – Perennial herb. Uncommon in coastal prairie. *Kenny 672* (OBI 161512). Study Regions: 10, 11, 27.

Adenostoma fasciculatum Hook. & Arn. var. ***fasciculatum*** (Chamise) – Shrub. Uncommon, only found in hard chaparral. *Richardson 340* (OBI 186118). Study Region: 4.

[R]***Amelanchier utahensis*** Koehne (Utah Service-berry) – Shrub. Uncommon on slopes of coastal gulches. *Kenny 682* (OBI 161339). Study Regions: 10, 21.

Aphanes occidentalis (Nutt.) Rydb. (Western Lady's Mantle) – Annual herb. Occasional in coastal prairie and on roadsides. *Kenny 545* (OBI 161333). Study Regions: 20, 25, 28, 33.

Drymocallis glandulosa (Lindl.) Rydb var. ***wrangelliana*** (Fisch. & Avé-Lall.) Ertter (Sticky Cinquefoil) – Perennial herb. Uncommon, only occurring in a riparian area in mixed evergreen forest understory. *Richardson 174* (OBI 133088). Study Region: 23.

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Fragaria chiloensis (L.) Mill. (Beach Strawberry) – Perennial herb. Common on coastal bluffs and in northern coastal scrub. *Kenny 268* (OBI 161273). Study Regions: 5, 17, 32.

Fragaria vesca L. (Wood Strawberry) – Perennial herb. Common in mixed evergreen forest. *Kenny 371* (OBI 161447). Study Regions: 1, 4, 5, 10, 13, 18, 20, 21, 23–25, 32.

Heteromeles arbutifolia (Lindl.) M.Roem. (Toyon) – Shrub. Occasional in mixed evergreen forest. *Richardson 28* (OBI 1331910. Study Regions: 3, 23–25.

Holodiscus discolor (Pursh) Maxim. var. ***discolor*** (Ocean-spray) – Shrub. Occasional in openings in mixed evergreen forest. *Richardson 190* (OBI 133278). Study Regions: 1, 5, 13, 20, 23–25, 28.

Horkelia cuneata Lindl. var. ***cuneata*** (Wedge-leaved Horkelia) – Perennial herb. Occasional in coastal prairie. *Kenny 526* (OBI 161325). Study Regions: 10, 11, 19.

Oemleria cerasiformis (Hook. & Arn.) J.W.Landon (Oso Berry) – Shrub. Occasional in northern coastal scrub. *Kenny 505* (OBI 161250). Study Regions: 1–5, 10, 12, 18, 20, 21, 25, 26, 28.

Potentilla anserina L. subsp. ***pacifica*** (Howell) Rousi (Pacific Silverweed) – Perennial herb. Common in the Scott Creek marsh. *Richardson 245* (OBI 174306). Study Region: 31.

[R]***Prunus emarginata*** (Douglas) Eaton (Bitter Cherry) – Shrub. Uncommon in northern coastal scrub. *Kenny 488* (OBI 161303). Study Region: 33.

Rosa californica Cham. & Schltdl. (California Rose) – Shrub. Occasional in riparian areas in coastal prairie and mixed evergreen forest. *Richardson 102* (OBI 133958). Study Regions: 8, 10, 31, 32.

Rosa gymnocarpa Nutt. var. ***gymnocarpa*** (Wood Rose) – Shrub. Occasional in mixed evergreen forest. *Richardson 179* (OBI 133984). Study Regions: 10, 17, 18, 23, 24.

Rosa spithamea S.Watson (Coast Ground Rose) – Shrub. Occasional in mixed evergreen forest. *Richardson 327* (OBI 186106). Study Regions: 3, 5, 23.

Rubus leucodermis Douglas ex Torr. & A.Gray (Whitebark Raspberry) – Perennial herb. Occasional in mixed evergreen forest. *Kenny 616* (OBI 161431). Study Regions: 23, 24.

Rubus parviflorus Nutt. (Thimbleberry) – Perennial herb. Common in riparian forest. *Richardson 10* (OBI 109097). Study Regions: 2, 3, 5, 25, 26.

[R]***Rubus spectabilis*** Pursh (Salmonberry) – Shrub. Locally rare in county, but common in the Scott Creek riparian corridor. *Kenny 664* (OBI 161506). Study Regions: 1, 5, 26.

****Rubus ulmifolius*** Schott var. ***anoplothyrsus*** Sudre (Thornless Blackberry) – Perennial herb. Uncommon, primarily occurring on one roadside in mixed evergreen forest. Plants could likely be extirpated if treated. *Kenny 771* (OBI 148071). Study Region: 25.

Rubus ursinus Cham. & Schltdl. (California Blackberry) – Perennial herb. Common in northern coastal scrub and mixed evergreen forest. *Richardson 63* (OBI 109149). Study Regions: 1–14, 17–21, 23–28, 30–33.

Rubiaceae

Galium aparine L. (Goose Grass) – Annual herb. Common in northern coastal scrub and mixed evergreen forest. *Kenny 706* (OBI). Study Regions: 1–5, 9, 10, 13, 18, 20, 23–25, 28.

Galium californicum Hook. & Arn. subsp. ***californicum*** (California Bedstraw) – Perennial herb. Common in mixed evergreen forest. *Kenny 452* (OBI 161256). Study Regions: 1, 5, 10, 18, 20, 23–25.

Galium porrigens Dempster var. ***porrigens*** (Climbing Bedstraw) – Perennial herb. Common in northern coastal scrub and mixed evergreen forest. *Richardson 78* (OBI 109685). Study Regions: 1, 5, 12, 19, 21, 23–25, 31–34.

Galium triflorum Michx. (Sweet-scented Bedstraw) – Perennial herb. Occasional in mixed evergreen forest. *Kenny 478* (OBI 161296). Study Regions: 1, 5, 23–25.

****Sherardia arvensis*** L. (Field Madder) – Annual herb. Common in disturbed areas in coastal prairie. *Kenny 373* (OBI 161449). Study Regions: 1, 4, 7, 9, 17, 20, 21.

Salicaceae

Salix lasiandra Benth. var. ***lasiandra*** (Pacific Willow) – Shrub. Common in riparian areas. *Keil 20662* (OBI 134300). Study Regions: 2, 5, 10, 13, 26.

Salix lasiolepis Benth. (Arroyo Willow) – Shrub. Common in riparian areas. *Kenny 844* (OBI 166738). Study Regions: 2, 5, 10, 11, 13, 14, 17, 19, 21, 26–28, 30–33.

Salix sitchensis Sanson ex Bong. (Sitka Willow) – Shrub. Occasional in riparian areas. *Richardson 165* (OBI 134555). Study Regions: 5, 25, 26.

Sapindaceae

Acer macrophyllum Pursh (Big-leaf Maple) – Deciduous tree. Common in riparian forest. *Keil 20585* (OBI 134636). Study Regions: 3, 5, 24–26.

Acer negundo L. (Box Elder) – Deciduous tree. Occasional in riparian forest. *Keil 20659* (OBI 134699). Study Regions: 2, 5, 25, 26, 31.

Aesculus californica (Spach) Nutt. (California Buckeye) – Deciduous tree. Occasional in gulch bottoms and in riparian forest. *Keil 20633* (OBI 134796). Study Regions: 1–3, 5, 10, 13, 14, 20, 25, 26, 31, 32.

Saxifragaceae

Heuchera micrantha Douglas ex Lindl. (Small-flowered Alum-root) – Perennial herb. Occasional in mixed evergreen forest. *Kenny 372* (OBI 161448). Study Regions: 1–3, 5, 24, 25.

Lithophragma affine A.Gray (Woodland Star) – Perennial herb. Uncommon in mixed evergreen forest. *Kenny 490* (OBI 161285). Study Region: 5.

Lithophragma heterophyllum (Hook. & Arn.) Hook. & Arn. (Hill Star) – Perennial herb. Uncommon in mixed evergreen forest. *Kenny 430* (OBI 158507). Study Region: 3.

Micranthes californica (Greene) Small (California Saxifrage) – Perennial herb. Uncommon, only occurring on a cliff in

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mixed evergreen forest. *Kenny 111* (OBI 161401). Study Region: 1.

Tellima grandiflora (Pursh) Douglas ex Lindl. (Fringe Cups) – Perennial herb. Uncommon in wet areas in mixed evergreen forest. *Kenny 626* (OBI 161436). Study Region: 24.

Tiarella trifoliata L. var. ***unifoliata*** (Hook.) Kurtz (Sugar-scoop) – Perennial herb. Uncommon in wet areas in mixed evergreen forest. Study area near southern distributional limit for taxon. *Kenny 938* (OBI 166664). Study Region: 24.

Scrophulariaceae

Scrophularia californica Cham. & Schltdl. (California Figwort) – Perennial herb. Common in northern coastal scrub. *Kenny 947* (OBI 161824). Study Regions: 1–5, 7, 9–13, 17–21, 23–26, 28, 31, 32.

Solanaceae

Solanum douglasii Dunal (Douglas's Nightshade) – Perennial herb. Common in northern coastal scrub and mixed evergreen forest. *Kenny 100* (OBI 161391). Study Regions: 2, 3, 5, 11, 23–26.

Solanum umbelliferum Eschsch. (Blue Witch) – Perennial herb. Common in northern coastal scrub and mixed evergreen forest. *Kenny 112* (OBI 161402). Study Regions: 1, 4, 5, 20, 23, 24, 27.

Tropaeolaceae

****Tropaeolum majus*** L. (Garden Nasturtium) – Perennial herb. Uncommon, only occurring near buildings. *Kenny 997* (OBI 158579). Study Region: 26.

Urticaceae

Hesperocnide tenella Torr. (Western Nettle) – Annual herb. Occasional on cliff faces in mixed evergreen forest. *Kenny 474* (OBI 161294). Study Regions: 1, 3, 5, 18, 20, 24, 25, 32.

****Parietaria judaica*** L. (Asthma-weed Pellitory) – Perennial herb. Occasional on roadsides. *Kenny 894* (OBI 166674). Study Regions: 5, 18.

Urtica dioica L. (Hoary Nettle) – Perennial herb. Occasional in riparian areas. *Richardson 358* (OBI 167637). Study Regions: 2, 3, 5, 10, 14, 18, 20, 24–26, 28, 31, 33.

Valerianaceae

[R]***Plectritis ciliosa*** (Greene) Jeps. (Long-spurred Plectritis) – Annual herb. Uncommon, only occurring on roadside in riparian forest. Historically present in Study Region 5, though not encountered during study. Documented in West (2016) and seeds deposited at UCSC Arboretum. Study Region: 5.

Plectritis congesta (Lindl.) DC. subsp. ***brachystemon*** (Fisch. & C.A.Mey.) Morey (Pale Plectritis) – Annual herb. Occasional in coastal prairie and in openings in mixed evergreen forest. *Kenny 181B* (OBI 161500). Study Regions: 1, 4, 13, 20.

Verbenaceae

Verbena lasiostachys Link var. ***lasiostachys*** (California Vervain) – Perennial herb. Common in mixed evergreen forest. *Richardson 33* (OBI 114841). Study Regions: 2, 10, 17, 19, 21, 24, 25, 28, 30.

Violaceae

Viola ocellata Torr. & A.Gray (Western Heart's Ease) – Perennial herb. Uncommon in mixed evergreen forest. *Kenny 610A* (OBI 161427). Study Regions: 23.

****Viola odorata*** L. (English Violet) – Perennial herb. Uncommon in mixed evergreen forest. *Kenny 988* (OBI 179963). Study Region: 5.

Viola pedunculata Torr. & A.Gray (Johnny-jump-up) – Perennial herb. Occasional in coastal prairie. *Kenny 451* (OBI 174304). Study Regions: 4, 19.

Viola sempervirens Greene (Evergreen Violet) – Perennial herb. Occasional in redwood forest. *Kenny 104* (OBI 161399). Study Region: 23.

Vitaceae

****Parthenocissus inserta*** (A.Kern.) Fritsch (Woodbine) – Perennial herb. Uncommon, only occurring on *Salix lasiolepis* near Scott Creek. Historically present, though not encountered during study. Personal observation by James A. West. Study Region: NA.

Monocots

Agavaceae

Chlorogalum pomeridianum (DC.) Kunth var. ***divaricatum*** (Lindl.) Hoover (Divaricate Soap Plant) – Geophyte. Occasional in coastal prairie. *Kenny 685* (OBI 161342). Study Region: 10.

Chlorogalum pomeridianum (DC.) Kunth var. ***pomeridianum*** (Soap Plant) – Geophyte. Occasional in northern coastal scrub and hard chaparral. Observed in 2019, but not collected. Study Regions: 4, 9, 10, 12, 13, 17, 18, 20, 21, 23, 28, 32, 33.

Alliaceae

****Allium triquetrum*** L. (Three-angled Onion) – Geophyte. Uncommon, only occurring in one area on Scott Creek in riparian forest. *Kenny 990* (OBI 179977). Study Region: 5.

Allium unifolium Kellogg (Single-leaved Onion) – Geophyte. Uncommon, only occurring in perched marsh above Big Willow Gulch in generally saturated soils. *Kenny 732B* (OBI 158439). Study Region: 10.

Amaryllidaceae

****Amaryllis belladonna*** L. (Naked Ladies) – Geophyte. Uncommon in disturbed areas in coastal prairie. Potentially waifs from historic plantings. *Kenny 1002* (OBI 179984). Study Region: 18.

****Leucojum aestivum*** L. (Snowflake) – Geophyte. Waif. Observed in 2019, but not collected. Study Region: 26.

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****Narcissus tazetta*** L. (Paper White) – Geophyte. Waif. Observed in 2019, but not collected. Study Region: 26.

Araceae

****Arum italicum*** Mill. (Italian Arum) – Perennial herb. Occasional in coastal scrub and grasslands. *Kenny 275* (OBI 161277). Study Regions: 26, 28, 30, 31.

****Arum palaestinum*** Boiss. (Black Calla, Solomon's Lily) – Perennial herb. Uncommon near roadsides in mixed evergreen forest. Historically present in study region 25, though not encountered during study. *Keil 30753* (OBI 144112). Study Region: 25.

Lemna minor L. (Smaller Duckweed) – Perennial herb. Occasional in riparian areas. Observed in 2019, but not collected. Study Region: 28.

Lemna minuta Kunth (Least Duckweed) – Perennial herb. Occasional in riparian areas. Observed in 2019, but not collected. Study Region: 28.

****Zantedeschia aethiopica*** (L.) Spreng. (Calla Lily) – Perennial herb. Occasional in wet areas and as a waif. *Buchloh 40* (OBI). Study Region: 26.

Commelinaceae

****Tradescantia fluminensis*** Vell. (Small-leaf Spiderwort) – Perennial herb. Occasional along Scott Creek. *Kenny 984* (OBI 179987). Study Region: 5.

Cyperaceae

[R]***Bolboschoenus fluviatilis*** (Torr.) Soják (River Bulrush) – Perennial herb. Uncommon, only found in several dense patches in the Scott Creek marsh. Outside of study area, only known in county from one 1909 collection from the Pajaro River. *Kenny 872* (OBI 174312). Study Region: 31.

Carex amplifolia Boott (Big-leaf Sedge) – Perennial herb. Common in riparian areas. *Kenny 836* (OBI 158458). Study Regions: 2, 5, 26.

Carex barbarae Dewey (Whiteroot) – Perennial herb. Occasional, found in freshwater portions of the Scott Creek marsh as well as in a perched freshwater marsh upslope from the main marsh. *Kenny 904* (OBI 171057). Study Regions: 30, 31, 33.

Carex bolanderi Olney (Bolander's Sedge) – Perennial herb. Occasional along the Scott Creek riparian corridor. Some individuals collected have compound spikelets, which is outside of the current species circumscription and merits further study. *Kenny 770* (OBI 161858). Study Regions: 2, 5, 11, 24, 26.

Carex brevicaulis Mack. (Short-stemmed Sedge) – Perennial herb. Occasional in coastal prairie. Synonym for *Carex zikae* E.H.Roalson & M.J.Waterway. *Kenny 842* (OBI 166740). Study Regions: 6, 12, 17, 32.

Carex densa (L.H.Bailey) L.H.Bailey (dense sedge) – Perennial herb. Occasional in wet areas in coastal prairie. *Kenny 289* (OBI 167676). Study Regions: 5, 6, 8, 17, 27, 28.

Carex globosa Boott (Round-fruited Sedge) – Perennial herb. Occasional in mixed evergreen forest. *Richardson 178* (OBI 144629). Study Regions: 5, 23.

[R]***Carex gracilior*** Mack. (Slender Sedge) – Perennial herb. Uncommon in coastal prairie. *Kenny 933* (OBI 153830). Study Regions: 2, 3, 17.

Carex harfordii Mack. (Harford's Sedge) – Perennial herb. Common in coastal prairie. May hybridize with *Carex subbracteata*. *Richardson 126* (OBI 144725)). Study Regions: 5, 6, 8, 10–12, 17, 18, 21, 23, 24, 32.

Carex obnupta L.H.Bailey (Slough Sedge) – Perennial herb. Common in riparian areas. *Kenny 684* (OBI 161341). Study Regions: 6, 10, 13, 14, 17, 26, 31.

Carex subbracteata Mack. (Small-bracted Sedge) – Perennial herb. Common in coastal prairie. May hybridize with *Carex harfordii*. *Kenny 185B* (OBI 158525). Study Regions: 6–21, 27–30.

Carex tumulicola Mack. (Foothill Sedge) – Perennial herb. Occasional in mixed evergreen forest. *Kenny 470* (OBI 148060). Study Regions: 1, 2, 6, 11, 17, 23.

Cyperus eragrostis Lam. (Umbrella Sedge) – Perennial herb. Common in riparian areas. *Richardson 66* (OBI 14650). Study Regions: 2, 5, 10, 13, 17–21, 24, 26–28, 30–32.

Isolepis carinata Hook. & Arn. ex Torr. (Dwarf Club Rush) – Annual herb. Occasional in wet areas in coastal prairie. *Kenny 415* (OBI 158487). Study Region: 11.

Isolepis cernua (Vahl) Roem. & Schult. (Low Club Rush) – Annual to perennial herb. Common in wet areas in coastal prairie. *Kenny 559* (OBI 179973). Study Regions: 6, 11, 13, 17, 19, 21, 27, 28.

Schoenoplectus americanus (Pers.) Volkart ex Schinz & R.Keller (Olney's Three-square Bulrush) – Perennial herb. Common in the Scott Creek marsh. *Buchloh 25* (OBI 146863). Study Region: 31.

Schoenoplectus californicus (C.A.Mey.) Soják (Southern Bulrush) – Perennial herb. Common in the Scott Creek marsh. *Buchloh 33* (OBI 146921). Study Regions: 26, 31.

Scirpus microcarpus J.Presl & C.Presl (Small-fruited Bulrush) – Perennial herb. Common in riparian areas. *Richardson 251* (OBI 149304). Study Regions: 5, 13, 14, 23, 26, 31.

Iridaceae

Iris douglasiana Herb. (Douglas's Iris) – Perennial herb. Occasional in mixed evergreen forest. *Richardson 378* (OBI 186109). Study Regions: 1, 3, 5, 24, 25, 32.

Iris fernaldii R.C.Foster (Fernald's Iris) – Perennial herb. Occasional in mixed evergreen forest. *Kenny 205* (OBI 148050). Study Region: 23.

****Romulea rosea*** (L.) Eckl. var. ***australis*** (Ewart) M.P.deVos (Rosy Sand Crocus) – Geophyte. Occasional in coastal prairie. Observed in 2019, but not collected. Study Regions: 20, 28.

Sisyrinchium bellum S.Watson (Western Blue-eyed-grass) – Perennial herb. Common in coastal prairie. *Kenny 260* (OBI 171075). Study Regions: 4, 9–11, 17, 19, 21, 25, 27, 28, 32.

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Juncaceae

[R]***Juncus acuminatus*** Michx. (Sharp-fruited Rush) – Perennial herb. Uncommon, only occurring on a sandbank in Scott Creek. Historically present in study region 5, though not encountered during study. *West s.n.* (UCSC). Study Region: 5.

Juncus balticus Willd. subsp. ***ater*** (Rydb.) Snogerup (Baltic Rush) – Perennial herb. Common in wet areas in coastal prairie. *Richardson 105* (OBI 155722). Study Regions: 11, 17.

Juncus bufonius L. var. ***occidentalis*** F.J. Herm. (Western Toad Rush) – Annual herb. Common in wet areas in coastal prairie. Additional surveys needed to establish distribution. *Richardson 100* (OBI 155578). Study Regions: 8, 10, 11, 17, 19–21, 25, 27, 28, 31.

Juncus effusus L. subsp. ***pacificus*** (Fernald & Wiegand) Piper & Beattie (Pacific Rush) – Perennial herb. Occasional in freshwater marshes in coastal prairie. *Kenny 865* (OBI 166679). Study Regions: 3, 17, 24, 26, 31.

Juncus hesperius (Piper) Lint (Coast Rush) – Perennial herb. Common in wet areas in coastal prairie. Seeds have a mucilaginous coating that causes them to clump together when wet. *Kenny 861* (OBI 158502). Study Regions: 5, 6, 8, 10, 13, 18, 20, 21, 24, 26–28, 30.

Juncus lescurii Bol. (San Francisco Rush) – Perennial herb. Occasional in wet areas in coastal prairie. Common in the Scott Creek marsh. *Kenny 688* (OBI 167523). Study Regions: 13, 17, 30, 31.

Juncus occidentalis (Coville) Wiegand (Western Rush) – Perennial herb. Common in coastal prairie and in wet openings in mixed evergreen forest. *Richardson 98* (OBI 155746). Study Regions: 4, 6–8, 10, 11, 19–21, 32.

Juncus patens E.Mey. (Spreading Rush) – Perennial herb. Common in wet areas in coastal prairie. *Doiron 2* (OBI 155787). Study Regions: 1, 2, 4–12, 14, 17–21, 23–28, 30–32.

Juncus phaeocephalus Engelm. var. ***phaeocephalus*** (Brown-headed Rush) – Perennial herb. Common in wet areas in coastal prairie. *Kenny 645* (OBI 161363). Study Regions: 6, 8, 10, 11, 19–21, 27, 28, 30.

Juncus xiphioides E.Mey. (Iris-leaved Rush) – Perennial herb. Uncommon in the Scott Creek riparian corridor. *Kenny 936A* (OBI 166670). Study Regions: 5, 26.

[R]***Luzula subsessilis*** (S.Watson) Buchenau (Short-stalked Wood Rush) – Perennial herb. Uncommon in mixed evergreen forest. *Kenny 438* (OBI 158511). Study Regions: 1, 3, 5, 9, 10, 20, 25.

Juncaginaceae

Triglochin scilloides (Poir.) Mering & Kadereit (Flowering-quillwort) – Perennial herb. Uncommon, only occurring in a wet depression in coastal prairie. *Kenny 644* (OBI 161428). Study Region: 6.

Liliaceae

Calochortus albus (Benth.) Douglas ex Benth. (Fairy Lantern) – Geophyte. Occaisional on cliff faces in coastal prairie and in mixed evergreen forest. *Kenny 611* (OBI 161426). Study Regions: 10, 12, 23.

Clintonia andrewsiana Torr. (Red Clintonia) – Perennial herb. Uncommon in redwood forest. *Richardson 60* (OBI 151485). Study Regions: 3, 24, 25.

Fritillaria affinis (Schult. & Schult. f.) Sealy (Checker Lily) – Geophyte. Occasional in redwood forest. *Kenny 105* (OBI 143570). Study Regions: 18, 23, 24, 33.

Prosartes hookeri Torr. (Hooker's Fairy Bells) – Perennial herb. Occasional in redwood forest. *Richardson 377* (OBI 179971). Study Regions: 3, 5, 24, 25.

Scoliopus bigelovii Torr. (Fetid Adder's Tongue) – Perennial herb. Uncommon in redwood forest. *Kenny 768* (OBI 161859). Study Region: 24.

Melanthiaceae

Toxicoscordion fremontii (Torr.) Rydb. (Fremont's Star Lily) – Geophyte. Uncommon in openings in mixed evergreen forest. *Kenny 122* (OBI 161420). Study Regions: 1, 23.

Trillium chloropetalum (Torr.) Howell (Giant Trillium) – Perennial herb. Occasional in the Scott Creek riparian corridor. Three floral morphs are found in study area, each with a distinct scent. *Wayman 301* (HSC 218837). Study Region: 26.

Trillium ovatum Pursh subsp. ***ovatum*** (Western Trillium) – Perennial herb. Occasional in mixed evergreen forest. *Kenny 128* (OBI 161410). Study Regions: 1, 3, 5, 23, 25.

[R]***Trillium albidum*** J.D.Freeman (White Trillium) – Perennial herb. Uncommon, only occurring in coastal scrub overlooking Big Willow gulch. Historically present somewhere in study regions 9–11, though not encountered during study. Documented in West (2016). Study Regions: 9–11.

Orchidaceae

Corallorhiza maculata (Raf.) Raf. var. ***occidentalis*** (Lindl.) Ames (Unspotted Coralroot) – Perennial herb. Uncommon in mixed evergreen forest. *Kenny 893* (OBI 66675). Study Regions: 5, 18.

Corallorhiza striata Lindl. (Striped Coralroot) – Perennial herb. Uncommon in mixed evergreen forest. Historically present, though not encountered during study. Flavistic form has also been observed in study area. *Buck & West 228* (JEPS 89230). Study Region: 2.

****Epipactis helleborine*** (L.) Crantz (Broad-leaved Helleborine) – Perennial herb. Common in mixed evergreen forest. *Richardson 382* (OBI 149296). Study Regions: 1, 2, 5, 10, 23, 25.

Piperia elegans (Lindl.) Rydb. subsp. ***elegans*** (Coast Piperia) – Perennial herb. Uncommon. Historically present on the Western Terrace, though not encountered during study. Documented in West (2016). *Morgan 925* (OBI 151948). Study Region: 20.

Piperia elongata Rydb. (Chaparral Orchid) – Perennial herb. Uncommon in mixed evergreen forest, only occurring on a steep slope in opening in mixed evergreen forest. *Kenny 961* (OBI 161834). Study Region: 1.

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[RR]***Piperia michaelii*** (Greene) Rydb. (Michael's Rein Orchid) (CRPR 4.2) – Perennial herb. Uncommon in northern coastal scrub. *Kenny 973* (OBI 161846). Study Regions: 26, 32.

Piperia transversa Suksd. (Flat-spurred Piperia) – Perennial herb. Uncommon in mixed evergreen forest. *Kenny 956* (OBI 161832). Study Regions: 1, 3, 5.

[R]***Piperia unalascensis*** (Spreng.) Rydb. (Alaska Rein Orchid) – Perennial herb. Uncommon, only occurring sporadically on the Western Terrace. Historically present on the Western Terrace, though not encountered during study. Specimens similar to *Morgan 1805* collected on coast near the study area are more similar to the type specimen than to specimens found elsewhere in California. *Stone 459* (JEPS81531). Study Region: 6.

Spiranthes romanzoffiana Cham. (Hooded Lady's Tresses) – Perennial herb. Uncommon in interface between coastal prairie and mixed evergreen forest. Historically present in study region 5, though not encountered during study. *Neubauer 97a* (UCSC 100008833). Study Region: 5.

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[RR]***Agrostis blasdalei*** Hitchc. (Blasdale's Bent Grass) (CRPR 1B.2) – Perennial grass. Occasional in sandy, disturbed areas on coastal bluffs and in wet depressions in coastal prairie. *Kenny 750* (OBI 166705). Study Regions: 6, 7, 11, 14, 17, 19, 20, 27.

[R]***Agrostis densiflora*** Vasey (California Bent Grass) – Perennial grass. Uncommon, occurring primarily on coast in habitat that is commonly degraded. *Kenny 751* (OBI). Study Regions: 9, 11, 14, 17.

Agrostis exarata Trin. (Spike Bent Grass) – Perennial grass. Occasional in sandy soil on coastal bluffs and in vernally wet depressions in coastal prairie. A formerly recognized variant, *A. exarata* var. *exarata,* is found just outside the study area and is locally rare in county. *Kenny 891* (OBI 173063). Study Regions: 3, 18.

Agrostis hallii Vasey (Hall's Bent Grass) – Perennial grass. Occasional in coastal prairie and in openings in mixed evergreen forest. Often appearing intermediate between *A. hallii* and *A. pallens*. *Kenny 939* (OBI 166663). Study Regions: 3, 10, 23, 24, 26, 32, 33.

Agrostis pallens Trin. (Dune Bent Grass) – Perennial grass. Occasional in coastal prairie and in openings in mixed evergreen forest. *Kenny 935B* (OBI 166667). Study Regions: 2, 20, 24, 32.

****Agrostis stolonifera*** L. (Creeping Bent Grass) – Perennial grass. Uncommon, only occurring in the Scott Creek riparian corridor. Reproductive material has never been observed. *Kenny 996* (OBI 149768). Study Region: 26.

****Aira caryophyllea*** L. (Silver Hair Grass) – Annual grass. Common in disturbed areas in coastal prairie and on roadsides throughout study area. *Kenny 511* (OBI 148065). Study Regions: 1, 5–7, 9–11, 13, 17, 19–21, 23–25, 27, 30, 32.

****Alopecurus pratensis*** L. (Meadow Foxtail) – Perennial grass. Locally abundant in one disturbed pasture. Potentially seeded. *Kenny 901* (OBI 171054). Study Regions: 2, 5.

Anthoxanthum occidentale (Buckley) Veldkamp (California Sweet Grass) – Perennial grass. Occasional in redwood forest. *Kenny 102* (OBI 161394). Study Regions: 23, 24.

****Avena barbata*** Pott ex Link (Slender Wild Oat) – Annual grass. Common in disturbed areas in coastal prairie. *Keil 20600* (OBI 171782). Study Regions: 1, 2, 6–12, 14, 17–21, 23, 25–28, 30–32.

****Brachypodium distachyon*** (L.) P.Beauv. (Purple False Brome) – Annual grass. Common in disturbed areas throughout study area. *Kenny 681* (OBI 161338). Study Regions: 2, 6, 7, 10–12, 14, 17, 19–21, 24, 26–28, 30, 32.

****Briza maxima*** L. (Rattlesnake Grass) – Annual grass. Common in disturbed areas in coastal prairie and on roadsides throughout study area. *Kenny 849* (OBI 166735). Study Regions: 1, 3, 8–12, 17, 19, 23, 25, 28.

****Briza minor*** L. (Annual Quaking Grass) – Annual grass. Common in disturbed areas in coastal prairie. *Kenny 737* (OBI 161853). Study Regions: 4, 7–12, 17–21, 25, 27, 28, 30, 32.

****Bromus diandrus*** Roth (Ripgut Grass) – Annual grass. Common in disturbed areas throughout study area. *Keil 20599* (OBI 162493). Study Regions: 1, 2, 4, 6, 9, 11, 12, 17, 19–21, 24, 25, 27, 28, 30.

****Bromus hordeaceus*** L. (Soft Chess) – Annual grass. Common in disturbed areas in coastal prairie and on roadsides throughout study area. *Keil 20605-1* (OBI 162585). Study Regions: 2, 4, 6, 7, 9–11, 17, 19–21, 23, 25, 27, 28, 30, 32.

****Bromus madritensis*** L. (Foxtail Chess) – Annual grass. Uncommon in disturbed areas near roads. *Keil 20595* (OBI 162713). Study Region: 25.

Bromus sitchensis Trin. var. ***carinatus*** (Hook. & Arn.) R.E.Brainerd & Otting (California Brome) – Perennial grass. Occasional in mixed evergreen forest. *Kenny 758C* (OBI 161863). Study Regions: 2–6, 18–21, 23–26, 28, 30, 32, 33.

[R]***Bromus sitchensis*** Trin. var. ***maritimus*** (Piper) Otting & R.E.Brainerd (Maritime Brome) – Perennial grass. Uncommon in sandy soils on coastal bluffs. *Kenny 269* (OBI 161274). Study Region: 17.

Bromus vulgaris (Hook.) Shear (Columbia Brome) – Perennial grass. Occasional in mixed evergreen forest. *Richardson 186* (OBI 162936). Study Regions: 1, 3, 5, 23, 24.

[R]***Calamagrostis nutkaensis*** (J.Presl) Steud. (Pacific Reed Grass) – Perennial grass. Uncommon in wet areas in coastal prairie. *Kenny 657* (OBI 161503). Study Regions: 10, 13, 14.

Calamagrostis rubescens Buckley (Pine Reed Grass) – Perennial grass. Occasional in mixed evergreen forest. *Richardson 303* (OBI 149486). Study Regions: 1, 5, 9, 23–25.

****Cortaderia jubata*** (Lemoine) Stapf (Jubata Grass) – Perennial grass. Uncommon in disturbed areas on coastal bluffs. *Kenny 855* (OBI 166751). Study Regions: 1, 10, 12, 17, 23, 24, 31.

****Cynodon dactylon*** (L.) Pers. (Bermuda Grass) – Perennial grass. Uncommon on roadsides. *Kenny 974* (OBI 161847). Study Region: 32.

****Cynosurus echinatus*** L. (Bristly Dogtail Grass) – Annual grass. Common in disturbed areas throughout study area.

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Kenny 714 (OBI). Study Regions: 1–4, 6, 7, 10, 11, 18–21, 24, 25, 27, 32.

****Dactylis glomerata*** L. (Orchard Grass) – Perennial grass. Occasional in disturbed areas in coastal prairie and on roadsides. *Kenny 852* (OBI 166732). Study Regions: 5, 19, 25.

Danthonia californica Bol. (California Oat Grass) – Perennial grass. Common in coastal prairie. *Kenny 754* (OBI 161864). Study Regions: 4, 6, 7, 9–11, 17, 19, 20, 27, 28, 32.

[R]***Deschampsia cespitosa*** (L.) P.Beauv. subsp. ***cespitosa*** (Tufted Hair Grass) – Perennial grass. Common in coastal prairie. *Kenny 655* (OBI 161368). Study Regions: 1, 7, 8, 10, 17.

[R]***Deschampsia cespitosa*** (L.) P.Beauv. subsp. ***holciformis*** (J.Presl) W.E.Lawr. (California Hair Grass) – Perennial grass. Common in coastal prairie. *Kenny 648* (OBI 161440). Study Regions: 1, 6, 7.

Deschampsia elongata (Hook.) Munro (Slender Hair Grass) – Perennial grass. Occasional in mixed evergreen forest. *Richardson 163* (OBI 156357). Study Regions: 5, 11, 18, 24–26.

Distichlis spicata (L.) Greene (Salt Grass) – Perennial grass. Common in the Scott Creek marsh. *Richardson 247* (OBI 156447). Study Region: 31.

****Ehrharta erecta*** Lam. (Panic Veldt Grass) – Perennial grass. Occasional along roasides. *Kenny 857* (OBI 166749). Study Regions: 3, 5, 23, 25, 26, 33.

[RR]***Elymus californicus*** (Boland. ex Thurb.) Gould (California Bottle-brush Grass) (CRPR 4.3) – Perennial grass. Uncommon in the Scott Creek riparian corridor. *Richardson 132* (OBI 156606). Study Regions: 5, 20.

Elymus glaucus Buckley subsp. ***glaucus*** (Blue or Western Wild Rye) – Perennial grass. Occasional in coastal prairie and mixed evergreen forest. *Kenny 992* (OBI 153827). Study Regions: 2, 4–6, 8, 10, 11, 18, 20, 24, 26, 32, 33.

Elymus glaucus Buckley subsp. ***virescens*** (Piper) Gould (Virginia Wild Rye) – Perennial grass. Uncommon in coastal prairie. *Richardson 146* (OBI 156809). Study Region: 9.

****Festuca arundinacea*** Schreb. (Tall Fescue) – Perennial grass. Uncommon on coastal bluffs. *Kenny 720* (OBI 161349). Study Region: 17.

****Festuca bromoides*** L. (Brome Fescue) – Annual grass. Occasional in disturbed areas in northern coastal scrub and coastal prairie. *Keil 20602-1* (OBI 177209). Study Region: 33.

Festuca californica Vasey (California Fescue) – Perennial grass. Occasional in mixed evergreen forest. *Kenny 491A* (OBI). Study Regions: 1, 18, 23, 33.

[R]***Festuca elmeri*** Scribn. & Merr. (Elmer's Fescue) – Perennial grass. Occasional in mixed evergreen forest. *Richardson 188* (OBI 177270). Study Regions: 5, 23, 24.

[R]***Festuca idahoensis*** Elmer (Idaho Fescue) – Perennial grass. Uncommon on coastal bluffs. *Kenny 979* (OBI 171050). Study Region: 17.

[R]***Festuca microstachys*** Nutt. (Hair Fescue) – Annual grass. Uncommon in northern coastal scrub and in openings in mixed evergreen forest. Historically present, though not encountered during study. Documented in West (2016) as occurring in Study Region 1. Study Region: 1.

****Festuca myuros*** L. (Rattail Sixweeks Grass) – Annual grass. Common in disturbed areas in coastal prairie and on roadsides throughout study area. *Kenny 851* (OBI 166733). Study Regions: 1, 4, 6, 7, 9, 10, 12, 17, 19–21, 25, 27, 28, 30, 32.

Festuca occidentalis Hook. (Western Fescue) – Perennial grass. Occasional in mixed evergreen forest. *Kenny 614* (OBI 173062). Study Regions: 5, 23, 24.

****Festuca perennis*** (L.) Columbus & J.P.Sm. (Rye Grass) – Perennial grass. Common in disturbed areas in coastal prairie. *Richardson 75* (OBI 177624). Study Regions: 1, 2, 4–8, 10, 11, 14, 17–21, 25–28, 30–32.

[R]***Festuca subulata*** Trin. (Bearded Fescue) – Perennial grass. Uncommon in mixed evergreen forest. Population is extreme disjunct from the majority of the species' range. *Kenny 641* (OBI 161360). Study Region: 5

[R]***Festuca subuliflora*** Scribn. (Crinkle-awn Fescue) – Perennial grass. Uncommon in mixed evergreen forest. *Kenny 862* (OBI 149746). Study Regions: 5, 24, 25.

****Gastridium phleoides*** (Nees & Meyen) C.E.Hubb. (Nit Grass) – Annual grass. Common in disturbed areas throughout study area. *Kenny 712* (OBI 148056). Study Regions: 4, 11, 19, 24, 28, 32.

****Glyceria declinata*** Bréb. (Fowl Manna Grass) – Perennial grass. Occasional in margins of stock ponds in coastal prairie. *Kenny 548* (OBI 149355). Study Regions: 10, 20, 27, 28.

****Holcus lanatus*** L. (Common Velvet Grass) – Perennial grass. Common in moist, disturbed areas in coastal prairie. *Richardson 280* (OBI). Study Regions: 1–8, 10, 11, 14, 18–21, 24–28, 30, 31, 33.

Hordeum brachyantherum Nevski subsp. ***brachyantherum*** (Northern Barley) – Perennial grass. Occasional in coastal prairie. *Kenny 529* (OBI 148064). Study Regions: 6, 8, 10, 11, 17–19, 21, 27.

****Hordeum marinum*** Huds. subsp. ***gussoneanum*** (Parl.) Thell. (Mediterranean Barley) – Annual grass. Common in disturbed areas in coastal prairie. *Kenny 853* (OBI 166753). Study Regions: 6–8, 10, 11, 18–21, 27, 28, 30, 32.

****Hordeum murinum*** L. (Wall Barley) – Annual grass. Common in disturbed areas in coastal prairie. *Richardson 39* (OBI 180033). Study Regions: 1, 5, 6, 10, 11, 17–21, 23, 25–28, 30–32.

Koeleria macrantha (Ledeb.) Schult. (June Grass) – Perennial grass. Uncommon, only observed in one disturbed area in coastal prairie. *Kenny 673* (OBI 161509). Study Region: 9.

Melica californica Scribn. (California Melic) – Perennial grass. Occasional in northern coastal scrub and mixed evergreen forest. *Kenny 449* (OBI 166693). Study Regions: 4, 26.

Melica harfordii Boland. (Harford's Melic) – Perennial herb. Uncommon in hard chaparral margins. Historically present, though not encountered during study. Documented in West (2016) as occurring in study region 23. Study Region: 23.

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Melica imperfecta Trin. (Little California Melic) – Perennial grass. Occasional in northern coastal scrub and mixed evergreen forest. *Kenny 618* (OBI 161433). Study Regions: 1, 20, 23, 24, 28, 32.

Melica subulata (Griseb.) Scribn. (Alaskan Oniongrass) – Perennial grass. Occasional in mixed evergreen forest. *Richardson 127* (OBI). Study Regions: 5, 23.

Melica torreyana Scribn. (Torrey's Melic) – Perennial grass. Occasional in mixed evergreen forest. *Kenny 290* (OBI 148252). Study Regions: 1, 3–5, 18, 23–25, 28.

****Parapholis incurva*** (L.) C.E.Hubb. (Sickle Grass) – Annual grass. Uncommon in coastal prairie. Historically present, though not encountered during study. Personal observation by James A. West. Study Region: NA.

****Pennisetum clandestinum*** Hochst. ex Chiov. (Kikuyu Grass) – Perennial grass. Uncommon in dirt roads. Synonym for *Cenchrus clandestinus* (Hochst. ex Chiov.) Morrone. *Kenny 943* (OBI 166661). Study Regions: 31, 32.

****Phalaris aquatica*** L. (Harding Grass) – Perennial grass. Occasional in disturbed riparian areas in coastal prairie. *Richardson 68* (OBI 174247). Study Regions: 18, 24, 27.

Phalaris californica Hook. & Arn. (California Canary Grass) – Perennial grass. Occasional in wet areas in northern coastal scrub and coastal prairie. *Kenny 522* (OBI 179969). Study Regions: 5, 9, 10, 13, 25, 26, 31.

****Poa annua*** L. (annual blue grass) – Annual Grass. Occasional in disturbed areas on roadsides. *Kenny 646* (OBI 161364). Study Regions: 6, 25, 28.

Poa howellii Vasey & Scribn. (Howell's Bluegrass) – Perennial grass. Uncommon, only occurring on roadcut in mixed evergreen forest. *Kenny 462* (OBI 148059). Study Region: 3.

****Poa pratensis*** L. subsp. ***pratensis*** (Kentucky Blue Grass) – Perennial grass. Uncommon, only occurring on roadside. *Keil 20627* (OBI 169202). Study Region: 5.

Poa secunda J.Presl subsp. ***secunda*** (Nevada Blue Grass) – Perennial grass. Occasional in mixed evergreen forest. *Kenny 433* (OBI 148052). Study Regions: 1, 12, 17.

Poa unilateralis Scribn. ex Vasey subsp. ***unilateralis*** (San Francisco Blue Grass) – Perennial grass. Uncommon in areas of thin soil in coastal prairie. *Kenny 401* (OBI 161472). Study Regions: 9, 14, 32.

****Polypogon monspeliensis*** (L.) Desf. (Rabbitfoot Grass) – Annual grass. Common in disturbed areas throughout study area. *Kenny 981* (OBI 167670). Study Regions: 2, 5, 8, 10, 11, 18, 20, 21, 24, 26, 28, 30, 31.

Stipa lepida Hitchc. (Foothill Needle Grass) – Perennial grass. Uncommon in coastal prairie. *Kenny 486* (OBI 149357). Study Regions: 26, 32.

Stipa pulchra Hitchc. (Purple Needle Grass) – Perennial grass. Common in coastal prairie. *Keil 20621* (OBI 148720). Study Regions: 4–6, 9, 11, 12, 17–20, 25, 27, 28, 32.

Trisetum canescens Buckley (Tall False Oat) – Perennial grass. Uncommon in mixed evergreen forest. *Kenny 769* (OBI 143571). Study Regions: 23, 24.

Ruscaceae

Maianthemum racemosum (L.) Link (False Solomon's Seal) – Perennial herb. Common in mixed evergreen forest. *Richardson 379* (OBI 186151). Study Regions: 1, 5, 24, 25.

Maianthemum stellatum (L.) Link (Slim Solomon's Seal) – Perennial herb. Common in mixed evergreen forest. *Kenny 194* (OBI 161527). Study Regions: 2, 5, 24.

Themidaceae

Brodiaea terrestris Kellogg subsp. ***terrestris*** (Dwarf Brodiaea) – Geophyte. Uncommon in coastal prairie. *Kenny 441* (OBI 153819). Study Regions: 4, 9, 25, 27, 32.

Dipterostemon capitatus (Benth.) Rydb. (Blue Dicks) subsp. ***capitatus*** – Geophyte. Occasional in coastal prairie. *Kenny 434* (OBI 148274). Study Regions: 1, 9, 18, 20, 23, 25, 28.

Triteleia laxa Benth. (Ithuriel's Spear) – Geophyte. Occasional in northern coastal scrub and coastal prairie. *Kenny 649* (OBI 161366). Study Regions: 7, 27, 28, 32.

Typhaceae

****Typha angustifolia*** L. (Narrow-leaved Cattail) – Perennial herb. Common in the Scott Creek marsh. *Kenny 877* (OBI 166723). Study Region: 31.

Typha latifolia L. (Broad-leaved Cattail) – Perennial herb. Common in the Scott Creek marsh. *Kenny 999* (OBI 158574). Study Regions: 24, 26, 31.

Literature Cited

Gottlieb, L. D. 1972. A proposal for the classification of the annual species of *Stephanomeria* (Compositae). Madroño 21:463–481.

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